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Investor Presentation | Japan

Electronic Components: Renewed Focus on High-value-added MLCCs and ABF Package Substrates for AI/DC

We expect continued earnings growth for Murata, the MLCC market leader, and Ibiden, which boasts unrivaled technological competitiveness in ABF package substrates.

Key Takeaways

- OW on Murata Mfg: Top Pick. Earnings expansion and utilization rates for high-value-added MLCCs.
- OW on Ibiden: Upgraded to Overweight by two notches on Aug 4
- OW on TDK: Earnings expansion for rechargeable batteries and HDD-related products.
- OW on Hirose Electric: Earnings expansion for general industrial machinery/AI server connectors.
- OW on Niterra: Continued earnings growth for plugs, exhaust gas sensors, and SPE electrostatic chucks.

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ELECTRONIC COMPONENTS

Japan

Industry View

In-Line

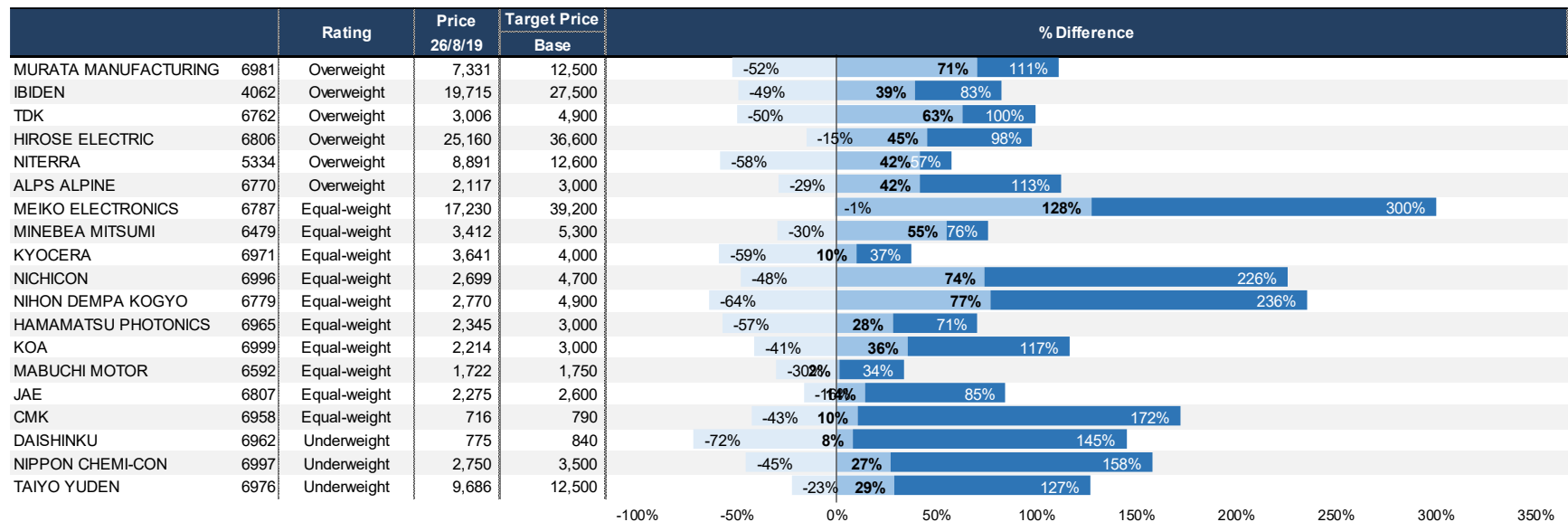
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Renewed Focus on High-value-added MLCCs and ABF Package Substrates for AI/DC

■ Risk-Reward Snapshot: Ranked in order of preference



Note: Not included Iriso Electronics (6908) data in this material as full-year F3/26 results have not yet been announced

Source: FactSet, Morgan Stanley Research

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iPhone Sell-in Forecasts / iPhone Camera Actuators by Model

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Meiko Electronics: Sales Guidance of Major Sites in VietNam/Mid-term Forecasts

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HDD Head Market

HDD Motor Market

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Electronic Components: Valuation Comparison and Earnings Forecast Comparison

		Rating	Price	Mkt Capt	Net F.Asset	EV	P/E					Ex. Cash P/E					P/B					ROE				
			26/8/19	(JPY bn)	(JPY bn)	(JPY bn)	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e	F26e	F27e	F28e	F29e	F30e
IBIDEN	4062	Overweight	19,715	5,533.8	135.5	5,398.4	63.2	42.2	31.1	24.7	21.2	58.7	39.2	28.9	22.9	19.7	9.1	7.6	6.2	5.0	4.1	15.6%	19.8%	22.1%	22.5%	21.3%
NITERRA	5334	Overweight	8,891	1,765.5	26.8	1,738.7	14.6	12.9	11.0	9.5	8.5	14.7	13.4	11.7	10.4	9.3	2.1	1.9	1.7	1.5	1.4	14.9%	15.4%	16.3%	16.9%	17.2%
MINEBEA MITSUMI	6479	Equal-weight	3,412	1,381.6	-227.5	1,609.1	16.3	13.5	12.1	11.0	10.0	19.2	15.9	14.2	12.9	11.7	1.4	1.3	1.2	1.1	1.0	9.0%	10.1%	10.5%	10.6%	10.7%
MABUCHI MOTOR	6592	Equal-weight	1,722	443.6	146.8	296.8	18.0	16.0	15.5	14.1	11.4	12.6	11.8	11.4	10.4	8.4	1.2	1.3	1.3	1.4	1.4	7.2%	7.9%	8.4%	9.5%	11.9%
TDK	6762	Overweight	3,006	5,704.5	525.9	5,178.5	23.3	20.8	18.2	16.2	14.5	21.1	18.8	16.5	14.7	13.1	2.4	2.2	2.1	1.9	1.7	10.8%	11.2%	11.8%	12.1%	12.4%
ALPS ALPINE	6770	Overweight	2,117	415.0	129.0	286.0	13.3	8.4	7.3	6.4	5.6	9.2	5.8	5.0	4.4	3.9	0.9	0.8	0.8	0.7	0.6	6.8%	10.1%	10.9%	11.4%	11.9%
NIHON DEMP KOGYO	6779	Equal-weight	2,770	60.9	-16.4	77.3	22.7	19.2	15.1	12.2	9.5	27.6	23.4	18.4	14.9	11.5	1.9	1.7	1.6	1.5	1.3	8.5%	9.4%	11.0%	12.5%	14.4%
MEIKO ELECTRONICS	6787	Equal-weight	17,230	443.4	-92.6	535.9	16.2	12.9	9.8	8.3	7.1	19.6	15.6	11.9	10.0	8.6	2.9	2.4	2.0	1.7	1.4	18.9%	20.5%	22.3%	21.9%	21.2%
HIROSE ELECTRIC	6806	Overweight	25,160	797.6	225.5	572.1	22.6	19.6	17.3	15.3	13.3	15.8	14.2	12.7	11.5	10.3	2.1	2.0	1.9	1.8	1.7	9.6%	10.6%	11.5%	12.3%	13.1%
JAE	6807	Equal-weight	2,275	156.9	10.1	146.8	22.9	21.9	20.7	19.7	18.9	21.9	21.0	19.8	18.8	18.1	1.0	1.0	1.0	1.0	0.9	4.6%	4.7%	4.9%	5.0%	5.0%
CMK	6958	Equal-weight	716	48.3	-22.8	71.1	30.0	23.2	19.6	16.5	13.4	41.8	32.3	27.3	22.9	18.7	0.6	0.6	0.6	0.6	0.6	2.1%	2.7%	3.2%	3.7%	4.5%
DAISHINKU	6962	Underweight	775	24.6	-17.2	41.9	-	41.1	22.4	16.4	12.3	-	69.8	38.1	27.9	20.9	0.7	0.7	0.7	0.7	0.7	-0.8%	1.7%	3.1%	4.2%	5.5%
HAMAMATSU PHOTONICS	6965	Equal-weight	2,345	724.7	31.0	693.7	38.0	33.7	27.8	21.1	18.4	38.5	35.2	29.6	22.9	20.4	2.2	2.2	2.2	2.1	2.1	5.7%	6.5%	7.9%	10.3%	11.4%
KYOCERA	6971	Equal-weight	3,641	4,796.2	1,797.8	2,998.4	34.1	31.1	24.0	21.5	19.6	21.7	21.4	17.5	16.2	15.2	1.6	1.6	1.5	1.5	1.4	4.3%	4.6%	5.8%	6.2%	6.6%
TAIYO YUDEN	6976	Underweight	9,686	1,266.8	-71.1	1,338.0	43.2	28.4	24.7	21.8	19.6	43.4	28.5	24.8	21.9	19.7	3.7	3.4	3.1	2.8	2.5	8.7%	12.3%	12.9%	13.3%	13.4%
MURATA MANUFACTURING	6981	Overweight	7,331	13,521.1	663.0	12,858.1	39.5	27.0	22.0	18.4	15.1	38.5	26.8	22.2	18.9	15.8	4.8	4.3	3.9	3.4	3.0	12.2%	16.7%	18.5%	19.6%	20.8%
NICHICON	6996	Equal-weight	2,699	182.9	-0.3	183.2	27.1	19.9	16.3	13.6	11.1	27.3	20.1	16.5	13.8	11.2	1.4	1.4	1.3	1.2	1.1	5.5%	7.1%	8.1%	9.1%	10.4%
NIPPON CHEMI-CON	6997	Underweight	2,750	76.9	-48.7	125.5	32.9	22.2	17.2	14.5	12.1	38.0	25.6	19.9	16.7	13.9	1.1	1.1	1.0	0.9	0.8	5.1%	7.2%	8.7%	9.6%	10.5%
KOA	6999	Equal-weight	2,214	75.2	-6.6	81.8	16.4	25.7	18.3	15.8	13.5	16.4	25.6	18.2	15.7	13.4	1.0	0.9	0.9	0.9	0.8	5.8%	3.7%	5.1%	5.6%	6.3%
Average (or total)				37,419.6		34,231.4	27.5	23.1	18.5	15.6	13.4	27.0	24.4	19.2	16.2	13.9	2.2	2.0	1.8	1.7	1.5	8.1%	9.6%	10.7%	11.4%	12.0%

(JPY bn, JPY/Shares)		Sales					OP					Net Profits					EPS					BPS
		F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e	Ce	FSe	F27e	FSe	F26e
IBIDEN	4062	567.3	550.0	537.8	707.0	678.4	133.8	127.0	114.8	197.1	174.0	92.0	84.0	81.5	137.7	124.9	312.0	299.4	286.7	466.8	440.2	2,171.4
NITERRA	5334	813.2	790.0	797.7	854.8	849.9	158.2	150.0	152.4	174.4	180.7	117.9	105.0	126.9	129.6	130.1	609.6	535.0	647.5	687.7	666.2	4,203.2
MINEBEA MITSUMI	6479	1,711.2	1,690.0	1,707.2	1,766.0	1,791.3	121.5	120.0	122.8	137.8	139.6	84.0	87.0	88.0	101.4	101.5	209.1	216.6	218.4	252.6	252.1	2,394.6
MABUCHI MOTOR	6592	218.8	225.0	218.4	233.7	232.4	27.2	26.5	25.1	29.9	27.3	23.5	22.2	21.8	25.2	22.5	95.8	92.4	89.8	107.7	93.3	1,381.8
TDK	6762	2,768.0	2,580.0	2,809.0	3,020.6	3,023.9	335.8	295.0	321.9	369.1	374.8	245.0	225.0	248.5	274.8	292.5	129.1	118.5	130.4	144.8	153.6	1,244.4
ALPS ALPINE	6770	1,064.5	1,045.0	1,048.4	1,157.6	1,085.9	49.1	48.5	43.5	68.7	49.8	31.1	30.0	28.6	48.9	34.4	159.2	150.4	145.3	250.7	174.9	2,388.8
NIHON DEMP KOGYO	6779	61.9	62.5	61.0	66.6	65.8	4.6	5.2	4.2	5.3	5.5	2.8	3.2	2.7	3.3	3.5	122.1	139.6	117.8	143.9	150.5	1,484.2
MEIKO ELECTRONICS	6787	323.8	320.0	323.7	412.9	400.0	38.6	38.0	38.1	48.7	51.3	27.3	27.0	27.3	34.3	36.4	1,063.6	1,052.0	1,064.4	1,336.4	1,416.4	5,932.4
HIROSE ELECTRIC	6806	234.6	250.0	240.3	254.6	260.1	47.3	50.0	52.2	53.4	61.2	36.2	35.0	37.9	40.4	44.0	1,114.9	1,069.6	1,158.6	1,280.5	1,358.3	11,831.8
JAE	6807	235.1	240.0	237.7	236.9	246.5	8.9	9.5	11.9	9.6	15.1	6.7	6.0	7.8	7.0	9.8	99.3	89.0	115.3	103.8	145.3	2,168.2
CMK	6958	103.6	110.0	104.8	106.2	109.6	3.2	5.0	3.5	3.9	5.0	1.7	3.0	1.9	2.2	2.6	23.8	42.1	26.0	30.9	36.5	1,151.0
DAISHINKU	6962	40.9	41.0	41.1	43.4	43.4	1.0	1.4	1.7	1.5	2.4	-0.3	0.1	0.2	0.6	0.9	-9.4	3.1	7.3	18.9	28.1	1,120.7
HAMAMATSU PHOTONICS	6965	232.8	212.8	212.1	245.8	231.4	22.5	18.0	16.2	25.4	20.4	18.0	14.0	14.2	19.7	16.9	61.7	46.9	47.3	69.5	57.5	1,067.4
KYOCERA	6971	1,951.5	2,080.0	2,054.9	1,977.2	2,138.3	132.6	160.0	147.2	148.5	171.4	138.1	160.0	148.2	140.0	160.2	106.8	122.0	114.9	117.1	129.4	2,254.0
TAIYO YUDEN	6976	411.3	424.0	400.9	475.1	457.6	45.2	45.0	41.7	70.2	73.6	30.8	29.0	28.4	46.9	50.4	224.0	219.0	219.2	340.9	389.3	2,639.5
MURATA MANUFACTURING	6981	2,044.2	2,110.0	2,079.3	2,449.7	2,435.0	431.9	430.0	440.4	624.0	628.6	334.4	338.0	350.1	480.2	503.7	185.6	185.7	192.6	271.9	278.7	1,540.9
NICHICON	6996	187.0	185.0	188.9	197.2	205.0	9.4	8.7	10.9	12.4	13.7	6.7	6.7	8.3	9.1	10.5	99.8	99.8	123.0	135.5	155.1	1,872.4
NIPPON CHEMI-CON	6997	154.0	160.0	157.0	158.3	167.5	7.0	8.0	8.1	9.1	10.7	3.3	4.0	4.1	4.9	5.9	83.5	142.2	148.2	124.0	207.9	2,415.6
KOA	6999	78.9	87.3	82.4	83.6	87.0	3.2	5.4	4.7	5.2	6.8	5.0	7.7	6.2	3.2	4.8	134.6	208.4	167.3	86.2	128.0	2,296.4
Average (or total)		13,202.5	13,162.6	13,302.5	14,447.2	14,509.1	1,581.0	1,551.2	1,561.1	1,994.2	2,011.7	1,204.2	1,186.9	1,232.5	1,509.4	1,555.3	4,825.3	4,831.6	5,020.1	5,969.6	6,261.0	

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; Diluted EPS for our forecasts and basic EPS for company and consensus estimates; e = Morgan Stanley Research estimates; FSe = FactSet consensus; Ce = Company estimates; Net Financial Asset is based on current FY end actuals (Net Financial Asset = Cash & cash equivalents + Short-term marketable securities + Investments in securities – Interest-bearing debt)

Source: FactSet, Company data, Morgan Stanley Research

Sales Trends (Quarterly, Semi-annual, Annual)

(JPY bn)	F25 1Q	2Q	3Q	4Q	F26 1Qe	2Qe	3Qe	4Qe	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce 1He	2He	F26Ce	F26e FSE	F27e FSE	F28e FSE	+1Q FSE		
Sales																											
USD/JPY:Average Rate	144.5	147.5	154.0	156.9	159.5	155.0	155.0	155.0	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0	155.0									
IBIDEN	4062	97.5	98.0	103.1	118.2	117.8	122.5	132.1	137.6	401.1	417.5	370.5	369.4	416.2	509.8	614.5	730.2	877.6	976.4	230.0	270.0	500.0	516.2	646.2	796.4	116.9	
NITERRA	5334	169.9	181.2	174.9	178.0	200.1	202.2	205.4	205.6	491.7	562.6	614.5	653.0	731.2	813.2	854.8	919.7	990.6	1,068.3	391.0	399.0	790.0	786.3	841.3	959.4	191.4	
MINEBEA MITSUMI	6479	366.9	411.4	453.9	385.1	417.1	440.7	436.7	416.6	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4	1,711.2	1,766.0	1,839.2	1,910.5	1,985.6	846.5	843.5	1,690.0	1,697.7	1,780.1	1,852.8	407.7	
MABUCHI MOTOR	6592	47.0	48.0	52.1	53.4	50.4	55.5	57.6	55.3	134.6	156.7	178.7	196.2	200.4	218.8	233.7	249.0	265.8	283.8	104.6	108.4	213.0	210.5	220.3	232.6	53.8	
TDK	6762	535.8	647.6	675.2	556.6	651.8	703.2	719.3	693.7	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,768.0	3,020.6	3,295.6	3,600.9	3,938.7	-	-	2,580.0	2,689.3	2,879.0	3,109.5	633.9	
ALPS ALPINE	6770	238.9	266.8	255.5	220.6	251.5	271.0	276.0	266.0	802.9	933.1	964.1	990.4	1,019.5	1,064.5	1,157.6	1,222.2	1,300.1	1,385.1	508.0	537.0	1,045.0	1,042.9	1,083.2	1,117.5	243.3	
NIHON DEMP A KOGYO	6779	12.7	14.0	13.2	13.5	14.8	15.3	15.6	16.2	45.4	52.5	50.3	53.1	54.6	61.9	66.6	72.4	79.7	89.1	29.5	31.1	60.6	61.9	66.6	72.4	14.8	
MEIKO ELECTRONICS	6787	53.1	58.4	60.5	62.6	73.7	78.2	85.2	86.7	151.3	167.3	179.5	206.8	240.6	323.8	412.9	482.1	539.7	595.1	-	-	320.0	321.2	397.5	469.0	70.1	
HIROSE ELECTRIC	6806	49.0	53.1	54.5	49.0	57.2	58.9	59.5	59.0	163.7	183.2	165.5	189.4	211.3	234.6	254.6	275.4	298.2	323.4	115.0	115.0	230.0	231.2	249.2	267.3	55.1	
JAE	6807	51.6	58.8	56.3	58.1	55.6	61.9	57.0	60.6	225.1	235.9	225.8	221.6	227.9	235.1	236.9	245.1	248.0	251.3	120.0	120.0	240.0	237.2	246.3	257.2	57.4	
CMK	6958	22.8	24.4	25.8	25.6	23.8	25.6	26.7	27.5	81.5	83.8	90.6	95.5	100.2	103.6	106.2	109.2	112.5	116.2	-	-	104.0	104.8	109.6	114.6	24.0	
DAISHINKU	6962	9.4	10.2	10.0	10.0	10.2	10.6	10.1	10.0	41.3	38.4	39.3	38.6	39.6	40.9	43.4	46.2	49.1	52.3	-	-	41.0	41.1	43.6	46.2	10.2	
HAMAMATSU PHOTONICS	6965	50.6	56.2	48.7	56.6	51.9	60.6	58.2	62.1	169.0	208.8	221.4	204.0	212.1	232.8	245.8	265.7	287.6	311.9	112.5	119.5	232.0	231.2	245.8	262.7	57.6	
KYOCERA	6971	478.0	513.3	530.6	465.3	468.0	492.0	495.5	496.0	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2	1,951.5	1,977.2	2,102.9	2,236.8	2,379.3	-	-	1,940.0	2,019.5	2,101.0	2,219.3	468.5	
TAIYO YUDEN	6976	84.8	92.8	88.5	86.3	94.2	101.2	107.1	108.8	349.6	319.5	322.6	341.4	355.3	411.3	475.1	523.8	573.9	624.8	-	-	384.0	396.7	450.7	520.2	93.2	
MURATA MANUFACTURING	6981	416.2	486.6	467.5	412.2	466.2	535.5	532.0	510.5	1,812.5	1,686.8	1,640.2	1,743.4	1,830.9	2,044.2	2,449.7	2,830.5	3,292.2	3,860.7	960.0	1,000.0	1,960.0	2,026.8	2,356.4	2,802.7	471.0	
NICHICON	6996	39.6	41.1	43.5	47.0	42.5	44.4	49.4	50.7	142.2	184.7	181.6	175.8	169.7	187.0	197.2	207.1	220.1	237.4	-	-	185.0	189.9	205.4	222.0	45.0	
NIPPON CHEMI-CON	6997	30.9	33.4	35.9	33.5	36.2	38.6	39.1	40.1	140.3	161.9	150.7	122.7	136.8	154.0	158.3	165.0	173.4	184.1	77.0	83.0	160.0	155.0	164.8	171.0	36.2	
KOA	6999	16.8	17.8	18.5	17.7	19.2	19.9	19.6	20.2	65.0	75.1	64.8	64.1	72.3	78.9	83.6	88.6	94.0	100.1	38.6	38.8	77.4	78.4	82.7	87.3	19.5	
Total	2,771.4	3,113.2	3,168.3	2,849.3	3,102.1	3,337.7	3,382.0	3,323.2	10,082.4	10,966.2	10,970.4	11,407.4	12,257.9	13,145.0	14,354.7	15,669.8	17,150.7	18,763.6		-	-	12,752.0	13,037.7	14,169.6	15,580.2	3,069.6	
(JPY bn)																											
Sales																											
USD/JPY:Average Rate	110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	155.0	155.0
IBIDEN	4062	205.4	212.0	220.5	247.6	319.0	398.6	413.5	309.3	274.2	305.0	300.9	285.9	310.3	318.1	314.1	266.5	300.4	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	509.8
NITERRA	5334	224.3	221.4	228.2	228.8	241.2	284.9	344.9	345.6	292.1	243.9	269.2	284.7	302.8	329.8	347.6	383.3	372.9	409.9	425.0	426.1	427.5	491.7	562.6	614.5	653.0	731.2
MINEBEA MITSUMI	6479	287.0	279.3	272.2	268.6	294.4	318.4	331.0	334.4	256.2	228.4	269.1	251.4	282.4	371.5	500.7	609.8	638.9	879.1	884.7	978.4	988.4	1,124.1	1,292.2	1,402.1	1,522.7	1,664.4
MABUCHI MOTOR	6592	116.2	105.1	116.4	105.7	99.3	93.9	100.5	107.6	92.6	70.4	82.8	78.9	85.3	108.4	122.5	143.1	140.7	146.9	143.1	131.8	116.4	134.6	156.7	178.7	196.2	200.4
TDK	6762	689.9	575.0	608.9	658.9	657.9	795.2	862.0	866.3	727.4	808.9	875.7	814.5	851.6	984.5	1,082.6	1,152.3	1,178.3	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8
ALPS ALPINE	6770	573.1	540.3	601.8	619.7	643.6	709.6	708.1	692.7	539.0	493.6	550.7	526.5	546.4	684.4	748.6	774.0	753.3	858.3	851.3	810.6	718.0	802.9	933.1	964.1	990.4	1,019.5
NIHON DEMP A KOGYO	6779	74.1	49.3	49.6	55.9	60.7	63.6	73.3	78.1	59.2	52.6	54.9	50.8	50.6	50.8	47.7	44.9	43.8	44.0	42.5	39.5	39.2	45.4	52.5	50.3	53.1	54.6
MEIKO ELECTRONICS	6787	23.0	17.2	21.6	29.2	40.2	50.3	65.3	78.5	75.8	69.8	74.7	63.0	60.7	79.2	90.9	95.3	95.9	108.5	118.9	115.5	119.3	151.3	167.3	179.5	206.8	240.6
HIROSE ELECTRIC	6806	83.5	61.1	67.9	79.0	89.6	104.4	115.5	118.0	89.9	84.5	92.4	94.8	95.9	125.0	125.7	120.3	115.1	125.3	124.6	121.8	133.5	163.7	183.2	165.5	189.4	211.3
JAE	6807	120.4	102.6	105.6	113.5	115.8	126.3	137.6	156.1	123.0	104.3	110.6	112.4	129.6	159.8	191.2	179.5	209.5	253.9	222.1	208.1	209.7	225.1	235.9	225.8	221.6	227.9
CMK	6958	122.3	104.5	111.7	115.4	116.0	119.2	128.2	138.5	107.6	90.1	87.0	80.6	72.7	71.0	73.7	73.1	75.4	86.9	90.2	82.6	70.0	81.5	83.8	90.6	95.5	100.2
DAISHINKU	6962	48.0	32.6	31.5	30.2	34.8	35.9	40.9	41.9	36.5	32.0	38.0	35.2	32.9	33.8	31.1	32.2	31.0	30.3	28.5	29.9	33.2	41.3	38.4	39.3	38.6	39.6
HAMAMATSU PHOTONICS	6965	51.6	62.6	60.7	60.9	70.5	73.9	87.0	96.3	98.8	72.0	91.0	101.9	98.1	102.2	112.1	120.7	121.9	130.5	144.3	145.9	140.3	169.0	208.8	221.4	204.0	212.1
KYOCERA	6971	1,285.1	1,034.6	1,069.8	1,140.8	1,180.7	1,181.5	1,283.9	1,290.4	1,128.6	1,073.8	1,266.9	1,190.9	1,280.1	1,447.4	1,526.5	1,479.6	1,422.8	1,577.0	1,623.7	1,599.1	1,526.9	1,838.9	2,025.3	2,004.2	2,014.5	2,070.2
TAIYO YUDEN	6976	190.0	152.1	153.7	163.3	172.3	186.5	221.2	238.3	185.5	195.7	210.4	183.8	192.9	208.2	227.1	240.4	230.7	244.1	274.3	282.3	300.9	349.6	319.5	322.6	341.4	355.3
MURATA MANUFACTURING	6981	584.0	394.8	395.0	414.2	424.5	490.8	566.8	631.7	523.9	530.8	618.0	584.7	681.0	846.7	1,043.5	1,210.8	1,135.5	1,371.8	1,575.0	1,534.0	1,630.2	1,812.5	1,686.8	1,640.2	1,743.4	1,830.9
NICHICON	6996	121.2	105.9	110.1	100.8	104.1	106.9	118.7	119.6	91.5	84.5	105.9	107.7	90.8	104.7	107.3	109.8	100.4	114.8	122.9	119.7	116.1	142.2	184.7	181.6	175.8	169.7
NIPPON CHEMI-CON	6997	130.0	93.7	98.2	102.3	105.7	120.9	135.1	143.2	114.6	105.9	127.8	100.3	93.0	114.0	123.4	118.4	116.3	133.4	141.0	114.6	110.8	140.3	161.9	150.7	122.7	136.8
KOA	6999	73.4	51.3	56.8	55.1	48.2	50.6	56.0	53.2	40.9	36.3	43.6	40.5	39.0	44.9	46.6	45.5	45.6	52.5</								

Operating Profit Trends (Quarterly, Semi-annual, Annual)

		F25				F26			F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce		F26Ce	F26e	F27e	F28e		+1Q						
(JPY bn)		1Q	2Q	3Q	4Q	1Qe	2Qe	3Qe	4Qe										1He	2He		FSE	FSE	FSE		FSE						
OP																																
USD/JPY:Average Rate		144.5	147.5	154.0	156.9	159.5	155.0	155.0	155.0	112.4	135.4	144.5	152.5	150.6	155.0	155.0	155.0	155.0														
IBIDEN	4062	17.6	14.9	12.0	17.5	21.6	22.9	24.6	25.6	70.8	72.4	47.6	47.6	62.0	94.7	128.5	166.6	213.3	242.6	38.0	52.0	90.0	99.7	151.5	223.7	20.2						
NITERRA	5334	33.8	38.1	38.9	27.3	39.6	40.0	41.9	36.6	75.5	89.2	107.6	129.7	138.2	158.2	174.4	200.8	227.4	254.4	72.0	78.0	150.0	154.0	174.9	206.8	30.7						
MINEBEA MITSUMI	6479	17.4	27.0	31.3	28.3	26.0	30.0	31.5	34.0	90.3	76.5	69.0	91.6	102.7	121.5	137.8	154.3	169.7	186.3	53.0	67.0	120.0	122.0	138.3	153.6	26.3						
MABUCHI MOTOR	6592	6.4	5.7	7.4	5.9	5.4	7.1	7.9	6.8	13.8	10.8	15.5	21.6	25.5	27.2	29.9	32.4	35.9	45.2	12.8	13.2	26.0	24.7	26.8	28.8	6.8						
TDK	6762	56.4	91.2	83.1	41.7	64.5	90.1	97.7	83.5	166.8	168.8	172.9	224.2	272.4	335.8	369.1	414.7	464.7	518.6	-	-	295.0	307.7	355.7	414.3	64.6						
ALPS ALPINE	6770	3.7	17.5	12.0	8.8	3.7	8.8	18.4	18.4	35.2	33.6	19.7	34.1	42.0	49.1	68.7	79.7	91.4	103.8	9.0	39.5	48.5	43.4	49.7	52.2	5.3						
NIHON DEMP KOGYO	6779	0.8	0.7	0.8	1.2	1.1	1.1	1.1	1.3	5.2	8.3	4.3	4.6	3.4	4.6	5.3	6.6	8.1	10.3	1.4	2.6	4.0	4.6	5.3	6.6	1.1						
MEIKO ELECTRONICS	6787	5.6	5.8	6.1	7.1	8.1	8.9	10.2	11.4	13.3	9.6	11.7	19.1	24.6	38.6	48.7	62.5	73.4	83.8	-	-	38.0	38.1	51.3	65.3	8.2						
HIROSE ELECTRIC	6806	9.8	10.6	12.0	10.5	11.3	12.1	12.7	11.2	40.8	46.8	34.0	42.7	43.0	47.3	53.4	59.9	67.0	75.4	23.0	23.0	46.0	49.8	58.1	66.9	11.5						
JAE	6807	1.5	3.3	1.2	3.0	0.6	3.1	1.6	3.6	18.0	17.6	14.4	15.6	8.9	8.9	9.6	10.1	10.6	11.0	3.8	5.7	9.5	13.2	16.1	18.8	2.8						
CMK	6958	0.2	0.3	0.9	1.5	0.7	0.5	1.0	1.0	3.0	2.6	3.5	3.8	2.8	3.2	3.9	4.5	5.2	6.2	-	-	3.2	3.5	5.0	7.5	0.3						
DAISHINKU	6962	0.1	0.1	0.3	0.6	0.3	0.3	0.3	0.1	5.2	4.2	2.1	0.9	1.1	1.0	1.5	2.3	2.9	3.6	-	-	1.4	1.4	2.0	2.3	0.3						
HAMAMATSU PHOTONICS	6965	4.3	6.5	1.5	3.9	2.4	7.6	4.9	7.6	34.3	57.0	56.7	32.1	16.2	22.5	25.4	30.7	40.6	45.9	10.0	10.0	20.0	20.1	24.4	30.4	5.3						
KYOCERA	6971	18.6	23.4	28.7	47.5	28.9	34.9	34.9	33.9	148.9	128.5	92.9	27.3	118.1	132.6	148.5	178.8	211.6	247.8	-	-	130.0	127.8	150.5	154.4	27.4						
TAIYO YUDEN	6976	3.1	5.9	7.5	3.5	7.0	10.5	13.5	14.3	68.2	32.0	9.1	10.5	20.0	45.2	70.2	80.2	90.5	100.2	-	-	30.0	40.2	67.2	99.7	7.3						
MURATA MANUFACTURING	6981	61.6	103.5	37.9	78.8	90.4	109.7	117.3	114.5	424.1	297.9	215.4	279.7	281.8	431.9	624.0	754.3	889.1	1,068.9	177.0	203.0	380.0	423.5	579.3	831.2	85.6						
NICHICON	6996	0.7	0.8	2.3	2.6	1.7	1.8	2.3	3.6	6.4	12.7	8.9	5.2	6.5	9.4	12.4	15.1	18.1	22.2	-	-	8.7	9.8	12.5	14.1	1.5						
NIPPON CHEMI-CON	6997	0.2	0.8	1.0	1.4	1.2	1.7	1.9	2.2	8.8	12.9	9.4	3.7	3.4	7.0	9.1	11.2	13.0	15.2	3.0	5.0	8.0	7.4	10.0	12.0	1.2						
KOA	6999	0.5	0.8	1.8	0.5	0.6	0.6	0.7	1.3	5.7	10.2	3.3	1.2	3.6	3.2	5.2	7.0	8.0	9.3	1.1	1.8	2.8	3.2	6.3	7.0	0.5						
Total		242.3	357.0	286.6	291.5	315.0	391.7	424.4	410.9	1,234.3	1,091.6	898.1	995.2	1,176.2	1,541.9	1,925.6	2,271.7	2,640.5	3,050.7	-	-	1,411.1	1,494.0	1,885.0	2,395.4	306.9						
OP																																
(JPY bn)		F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e
OP																																
USD/JPY:Average Rate		110.5	124.9	121.9	113.1	107.5	113.3	116.9	114.4	100.7	92.9	85.7	79.0	82.9	100.1	109.8	120.1	108.3	110.8	110.9	108.7	106.1	112.4	135.4	144.5	152.5	155.0	155.0	155.0	155.0	155.0	155.0
IBIDEN	4062	13.6	12.0	11.1	13.6	20.1	43.6	69.0	67.3	10.6	21.3	33.8	15.5	5.4	23.4	26.0	22.6	7.1	16.7	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	94.7	128.5	166.6	213.3	242.6
NITERRA	5334	25.5	11.8	15.8	20.7	26.1	41.5	52.4	35.0	-5.2	10.7	28.8	24.5	23.8	51.7	62.2	66.3	53.6	67.3	58.7	48.4	47.4	75.5	89.2	107.6	129.7	138.2	158.2	174.4	200.8	227.4	254.4
MINEBEA MITSUMI	6479	33.0	22.0	19.4	18.1	14.1	19.3	26.3	30.8	13.4	12.1	22.2	8.6	10.2	32.2	60.1	51.4	49.0	79.2	72.1	58.9	59.8	90.3	76.5	69.0	91.6	102.7	121.5	137.8	154.3	169.7	186.3
MABUCHI MOTOR	6592	28.6	22.1	29.1	24.6	17.3	8.1	10.7	11.7	5.7	3.0	6.6	3.1	5.1	9.3	16.9	23.0	24.2	24.1	21.2	17.5	12.9	13.8	10.8	15.5	21.6	25.5	27.2	29.9	32.4	35.9	45.2
TDK	6762	56.3	-43.7	22.1	54.3	59.8	60.5	79.6	87.2	-54.3	25.8	63.8	18.7	21.6	36.6	72.5	93.4	208.7	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	335.8	369.1	414.7	464.7	518.6
ALPS ALPINE	6770	24.6	13.2	41.8	36.2	31.1	45.5	22.1	19.9	-26.5	4.7	28.9	15.2	6.9	28.5	53.5	52.3	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	49.1	68.7	79.7	91.4	103.8
NIHON DEMP KOGYO	6779	12.6	0.5	0.1	2.5	4.5	5.3	9.4	9.6	-6.9	4.0	2.1	2.1	0.1	0.2	0.2	0.4	0.7	-9.6	0.4	-8.3	2.8	5.2	8.3	4.3	4.6	3.4	4.6	5.3	6.6	8.1	10.3
MEIKO ELECTRONICS	6787	1.3	-0.9	1.7	3.5	5.0	6.5	5.4	7.3	1.8	3.8	3.3	0.8	-0.6	0.9	-2.9	3.3	5.8	7.5	8.9	5.2	6.7	13.3	9.6	11.7	19.1	24.6	38.6	48.7	62.5	73.4	83.8
HIROSE ELECTRIC	6806	26.0	14.4	20.1	26.0	29.7	36.9	38.0	35.4	19.2	20.4	22.0	19.6	21.0	33.0	32.6	28.8	28.5	28.6	23.2	20.4	27.9	40.8	46.8	34.0	42.7	43.0	53.4	59.9	67.0	75.4	
JAE	6807	10.6	5.2	6.6	10.1	7.9	11.0	11.7	14.4	-0.8	2.9	6.3	6.5	8.6	16.5	25.9	17.9	12.1	20.6	17.3	14.2	8.7	18.0	17.6	14.4	15.6	8.9	9.9	10.1	10.6	11.0	
CMK	6958	9.3	-1.1	0.7	9.2	10.4	9.6	6.0	4.8	-2.6	-0.4	-1.0	0.8	0.4	-1.1	0.6	0.4	2.7	4.3	3.8	1.6	-1.7	3.0	2.6	3.5	3.8	2.8	3.2	3.9	4.5	5.2	6.2
DAISHINKU	6962	3.6	-3.5	-2.7	1.7	1.6	2.5	3.8	4.5	-0.4	1.2	1.4	0.3	0.4	0.8	-2.3	0.7	1.4	0.3	0.0	0.3	2.1	5.2	4.2	2.1	0.9	1.1	1.0	1.5	2.3	2.9	3.6
HAMAMATSU PHOTONICS	6965	5.9	8.7	5.9	4.1	9.7	11.7	15.7	19.9	20.1	3.7	15.8	21.8	17.6	16.8	21.7	23.6	20.5	22.8	27.3	25.4	21.8	34.3	57.0	56.7	32.1	16.2	22.5	25.4	30.7	40.6	45.9
KYOCERA	6971	207.2	51.6	83.4	109.0	101.0	103.2	135.1	152.4	43.4	63.9	155.9	97.7	76.9	120.6	93.4	92.7	104.5	95.6	94.8	100.2	70.6	148.9	128.5	92.9	27.3	118.1	132.6	148.5	178.8	211.6	247.8
TAIYO YUDEN	6976	35.4	6.2	8.2	11.0	2.1	6.7	22.0	21.3	-12.8	4.2	8.8	-8.0	5.0	11.4	13.2	23.4	12.4	20.2	35.2	37.2	40.8	68.2	32.0	9.1	10.5	20.0	45.2	70.2	80.2	90.5	100.2
MURATA MANUFACTURING	6981	174.2	51.0	59.2	74.2	69.5	89.8	113.4	115.8	-16.3	26.7	77.5	45.0	58.6	125.9	214.5	275.4	201.2	162.1	266.8	253.2	313.2	424.1	297.9	215.4	279.7	281.8	431.9	624.0	754.3	889.1	1,068.9
NICHICON	6996	10.3	5.1	7.7	4.6	6.3	6.0	6.7	4.6	-7.9	-4.5	5.1	2.1	-3.4	4.2	3.9	4.8	3.0	6.2	5.5	2.5	1.6	6.4	12.7	8.9	5.2	6.5	9.4	12.4	15.1	18.1	22.2
NIPPON CHEMI-CON	6997	13.6	-1.0	3.5	3.6	4.7	6.1	9.4	8.7	-6.9	-3.0	8.2	-2.6	-7.0	4.9	5.1	2.2	3.3	5.8	5.1	-2.9	3.0	8.8	12.9	9.4	3.7	3.4	7.0	9.1	11.2	13.0	15.2
KOA	6999	10.2	-																													

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in

Capex

	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce	
(JPY bn)																																	
Capex																																	
IBIDEN	4062	26.2	19.9	16.9	12.5	23.9	65.8	70.4	42.4	61.8	18.7	60.3	47.3	43.3	37.7	56.4	41.0	21.0	22.4	22.9	58.2	79.4	61.7	131.3	146.6	157.3	64.3	157.6	158.0	208.0	208.0	208.0	210.0
NITERRA	5334	23.5	25.5	10.8	10.4	14.0	26.9	29.3	63.2	24.2	11.0	8.8	14.0	24.0	41.0	36.4	45.3	36.3	37.6	56.6	51.0	33.4	29.6	31.3	41.2	37.5	49.0	47.4	45.0	45.0	45.0	47.3	
MINEBEA MITSUMI	6479	39.9	26.2	16.4	18.8	31.1	21.9	17.0	24.9	19.6	10.8	27.0	26.9	42.8	19.8	35.0	41.6	29.8	53.2	53.1	48.8	44.2	71.9	144.8	81.4	88.7	96.5	106.0	106.0	110.4	114.6	119.1	95.0
MABUCHI MOTOR	6592	4.6	5.1	6.3	5.7	11.6	5.5	3.6	3.7	3.8	3.7	3.2	6.5	6.8	5.8	9.6	16.2	11.2	12.8	16.3	17.9	7.7	9.6	10.6	12.8	14.1	10.3	12.6	15.0	15.0	15.0	10.3	
TDK	6762	99.5	58.8	41.5	44.9	61.0	73.9	70.4	84.3	98.4	64.4	78.6	99.7	85.6	68.6	102.5	160.7	167.6	178.6	173.6	173.4	212.4	291.3	275.7	218.6	225.3	298.6	370.0	350.0	350.0	350.0	370.0	
ALPS ALPINE	6770	31.4	31.8	34.9	47.1	47.1	50.1	45.3	43.2	47.2	18.5	23.5	34.0	31.8	26.6	31.4	41.2	47.7	76.2	52.9	42.4	40.4	53.0	50.8	55.7	51.8	61.6	75.2	60.0	60.0	60.0	75.2	
NIHON DEMP A KOGYO	6779	10.8	3.6	3.5	6.8	11.9	6.9	9.8	13.2	10.9	2.0	3.4	4.9	4.7	2.7	2.0	2.1	6.8	7.1	2.4	3.1	3.2	3.2	4.0	4.6	4.6	3.9	10.0	9.0	9.8	10.8	12.0	10.0
MEIKO ELECTRONICS	6787	3.5	3.2	2.0	4.2	10.0	10.1	16.6	11.2	13.7	7.0	17.4	9.6	6.6	6.6	4.1	3.1	3.1	9.6	15.8	11.8	10.7	14.4	17.4	19.9	29.8	-	70.0	70.0	70.0	78.4	86.4	70.0
HIROSE ELECTRIC	6806	7.5	4.3	3.5	6.6	5.5	9.0	13.0	10.2	7.3	7.2	9.0	11.3	11.4	11.3	14.6	13.3	15.2	19.6	21.3	18.2	14.6	15.2	19.6	32.8	21.2	19.9	25.8	28.0	30.3	32.8	35.6	25.0
JAE	6807	10.5	7.6	9.1	9.1	11.5	15.2	16.5	17.8	13.8	10.1	10.7	13.5	26.6	22.0	22.2	22.4	21.7	17.6	22.2	24.5	18.2	19.0	22.6	20.4	18.0	23.7	21.0	21.0	21.0	21.1	21.4	21.0
CMK	6958	9.7	10.1	8.5	2.9	4.6	6.2	12.3	14.7	8.0	2.4	1.9	1.8	3.5	2.6	5.0	3.1	4.6	7.0	7.5	8.3	3.7	6.1	6.7	16.0	18.9	9.2	8.4	9.0	9.3	9.6	9.9	6.3
DAISHINKU	6962	3.9	3.4	2.3	1.6	1.5	3.4	4.7	2.6	3.3	2.0	4.9	2.9	1.9	2.6	3.5	3.3	3.6	2.9	2.5	4.0	4.4	7.1	5.9	3.6	7.5	8.2	4.0	4.3	4.6	4.9	5.2	4.8
HAMAMATSU PHOTONICS	6965	9.7	13.7	8.3	6.2	5.8	9.8	11.7	9.7	8.6	9.4	9.2	7.3	8.6	9.2	15.5	14.3	9.3	13.6	14.2	17.4	20.3	13.0	20.4	31.2	30.9	34.8	40.2	15.0	15.0	15.0	-	
KYOCERA	6971	105.9	54.6	40.6	54.9	63.2	90.3	69.9	85.1	63.1	37.9	70.7	66.4	56.7	56.6	56.7	68.9	67.8	86.5	117.0	106.0	117.1	151.8	173.9	161.7	141.9	149.1	225.0	160.0	160.0	160.0	225.0	
TAIYO YUDEN	6976	54.0	16.8	7.4	14.4	22.5	15.8	30.2	44.6	27.0	9.4	17.5	29.1	19.7	16.9	18.8	37.4	31.6	26.5	42.6	44.1	43.9	51.6	63.3	79.9	64.2	40.4	40.0	60.0	60.0	60.0	42.0	
MURATA MANUFACTURING	6981	107.8	34.6	18.2	33.1	48.0	51.0	99.7	125.6	65.4	22.9	56.8	68.4	77.7	68.2	101.2	172.5	158.6	306.6	291.6	281.6	196.7	152.8	208.1	219.5	180.5	247.8	250.0	250.0	300.0	320.0	255.0	
NICHICON	6996	25.9	16.0	8.1	2.6	12.4	11.0	11.9	9.4	7.5	2.2	6.6	15.4	5.4	2.3	2.4	2.6	7.5	6.8	8.6	7.1	6.5	10.1	11.2	16.2	11.1	7.6	10.0	11.8	12.4	13.2	14.2	10.0
NIPPON CHEMI-CON	6997	14.6	11.0	9.4	6.1	7.6	8.4	14.6	14.6	11.9	4.0	9.6	13.5	6.0	3.1	5.2	4.4	4.6	7.5	9.6	5.6	3.5	5.9	7.7	11.2	7.6	5.9	5.5	6.1	6.3	6.6	7.1	5.7
KOA	6999	1.4	2.8	1.9	1.4	1.3	2.8	2.3	2.9	2.7	0.9	2.7	2.4	4.7	3.1	2.3	3.7	2.3	5.1	6.2	4.0	2.2	6.9	11.9	19.1	24.2	5.9	4.1	5.9	6.2	6.6	7.0	4.1
Total	590.1	349.1	249.7	289.3	394.4	484.1	549.1	623.2	498.3	244.3	421.8	474.9	467.8	406.7	524.8	697.1	650.3	897.2	936.7	927.6	862.5	974.1	1,217.2	1,192.3	1,135.1	1,136.7	1,481.8	1,384.0	1,443.3	1,511.7	1,551.0	-	
% of sales																																	
IBIDEN	4062	13%	9%	8%	6%	10%	21%	18%	10%	20%	7%	20%	16%	15%	12%	18%	13%	8%	7%	8%	20%	25%	15%	31%	40%	43%	15%	28%	22%	24%	20%	18%	50%
NITERRA	5334	10%	12%	5%	5%	6%	9%	8%	18%	8%	5%	3%	5%	8%	12%	10%	12%	10%	9%	13%	12%	8%	6%	7%	6%	7%	6%	5%	5%	5%	4%	6%	
MINEBEA MITSUMI	6479	14%	9%	6%	7%	11%	7%	5%	7%	8%	5%	10%	11%	15%	5%	7%	7%	5%	6%	6%	5%	4%	6%	11%	6%	6%	6%	6%	6%	6%	6%	6%	
MABUCHI MOTOR	6592	4%	5%	5%	5%	12%	6%	4%	3%	4%	5%	4%	8%	8%	5%	8%	11%	8%	9%	11%	14%	7%	7%	7%	7%	7%	5%	6%	6%	6%	5%	5%	
TDK	6762	14%	10%	7%	7%	9%	9%	8%	10%	14%	8%	9%	12%	10%	7%	9%	14%	14%	14%	13%	13%	14%	15%	13%	10%	10%	12%	13%	12%	11%	10%	9%	15%
ALPS ALPINE	6770	5%	6%	6%	8%	7%	7%	6%	6%	9%	4%	4%	6%	6%	4%	4%	5%	6%	9%	6%	5%	6%	7%	5%	6%	5%	6%	7%	5%	5%	4%	7%	
NIHON DEMP A KOGYO	6779	15%	7%	7%	12%	20%	11%	13%	17%	18%	4%	6%	10%	9%	5%	4%	5%	15%	16%	6%	8%	8%	7%	8%	9%	7%	16%	14%	14%	14%	13%	18%	
MEIKO ELECTRONICS	6787	-	-	-	14%	25%	20%	25%	14%	18%	10%	23%	15%	11%	8%	5%	3%	3%	9%	13%	10%	9%	10%	10%	11%	14%	-	22%	17%	15%	15%	29%	
HIROSE ELECTRIC	6806	9%	7%	5%	8%	6%	9%	11%	9%	8%	9%	10%	12%	12%	9%	12%	11%	13%	16%	17%	15%	11%	9%	11%	20%	11%	9%	11%	11%	11%	11%	12%	
JAE	6807	9%	7%	9%	8%	10%	12%	12%	11%	11%	10%	10%	12%	21%	14%	12%	12%	10%	7%	10%	12%	9%	8%	10%	9%	8%	10%	9%	9%	9%	9%	9%	
CMK	6958	8%	10%	8%	2%	4%	5%	10%	11%	7%	3%	2%	2%	5%	4%	7%	4%	6%	8%	8%	10%	5%	8%	8%	18%	20%	9%	8%	8%	9%	9%	6%	
DAISHINKU	6962	8%	10%	7%	5%	4%	10%	11%	6%	9%	6%	13%	8%	6%	8%	11%	10%	12%	10%	9%	14%	13%	17%	15%	9%	19%	21%	10%	10%	10%	10%	12%	
HAMAMATSU PHOTONICS	6965	19%	22%	14%	10%	8%	13%	13%	10%	9%	13%	10%	7%	9%	9%	14%	12%	8%	10%	10%	12%	15%	8%	10%	14%	15%	16%	17%	6%	6%	5%	5%	-
KYOCERA	6971	8%	5%	4%	5%	5%	8%	5%	7%	6%	4%	6%	6%	4%	4%	5%	5%	5%	7%	7%	7%	8%	9%	8%	7%	7%	7%	12%	8%	8%	7%	7%	11%
TAIYO YUDEN	6976	28%	11%	5%	9%	13%	8%	14%	19%	15%	5%	8%	16%	10%	8%	8%	16%	14%	11%	16%	16%	15%	15%	20%	25%	19%	11%	10%	13%	11%	10%	10%	12%
MURATA MANUFACTURING	6981	18%	9%	5%	8%	11%	10%	18%	20%	12%	4%	9%	12%	11%	8%	10%	14%	22%	19%	18%	12%	8%	12%	13%	10%	14%	12%	10%	9%	9%	8%	14%	
NICHICON	6996	21%	15%	7%	3%	12%	10%	10%	8%	8%	3%	6%	14%	6%	2%	2%	2%	7%	6%	6%	6%	6%	7%	6%	9%	6%	4%	5%	6%	6%	6%	6%	6%
NIPPON CHEMI-CON	6997	11%	12%	10%	6%	7%	7%	11%	10%	10%	4%	8%	13%	6%	3%	4%	4%	4%	6%	7%	5%	3%	4%	5%	7%	6%	4%	4%	4%	4%	4%	4%	4%
KOA	6999	2%	5%	3%	2%	3%	6%	4%	6%	7%	3%	6%	6%	12%	7%	5%	8%	5%	10%	11%	8%	4%	11%	16%	29%	38%	8%	5%	7%	7%	7%	6%	-
Total	12%	8%	6%	6%	8%	9%	10%	10%	10%	5%	8%	9%	9%	7%	8%	10%	9%	11%	11%	11%	10%	10%	11%	11%	10%	10%	9%	11%	10%	9%	9%	8%	-

Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; e = Morgan Stanley Research estimates; Ce = Company estimates
Source: Company data, FactSet, Morgan Stanley Research

Depreciation

	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	F26Ce	
(JPY bn)																																	
Depreciation																																	
IBIDEN	4062	14.6	17.5	16.6	13.8	14.4	22.3	31.5	39.2	43.5	38.8	40.7	42.4	42.7	35.7	39.4	44.1	33.1	24.6	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	81.4	95.5	110.5	130.5	130.5	81.0
NITERRA	5334	18.1	20.0	18.5	15.9	14.5	18.0	18.9	25.5	31.8	18.8	16.6	15.4	12.8	11.6	13.3	16.8	19.6	20.9	24.0	32.0	34.7	37.9	39.2	39.8	39.7	45.0	42.1	43.0	43.0	43.0	42.1	
MINEBEA MITSUMI	6479	22.1	24.4	22.8	21.7	22.5	24.0	24.6	26.4	24.0	21.1	20.8	19.6	20.8	23.7	28.8	34.9	28.2	31.6	36.4	46.2	48.6	45.2	53.1	58.4	66.2	70.4	75.0	77.4	80.6	83.7	87.0	75.0
MABUCHI MOTOR	6592	6.4	6.6	6.3	5.6	5.2	5.7	5.4	4.9	4.4	4.0	3.6	3.3	3.6	4.6	5.1	6.6	6.4	7.4	8.0	8.6	8.9	10.0	11.9	12.8	14.2	13.5	13.9	14.0	14.0	14.0	13.9	
TDK	6762	63.6	61.9	57.8	51.2	52.8	58.5	65.3	71.3	89.6	83.8	77.6	80.2	77.9	83.1	80.2	83.2	87.5	92.2	106.6	125.0	140.3	177.0	206.3	190.5	196.2	204.2	240.0	260.0	280.0	280.0	240.0	
ALPS ALPINE	6770	31.7	32.5	30.0	32.7	40.9	36.9	39.2	37.8	29.9	21.2	18.7	19.8	21.6	24.5	28.0	30.7	33.1	36.0	44.2	46.1	41.3	45.7	46.8	41.5	35.1	34.0	44.8	42.0	42.0	42.0	44.8	
NIHON DEMP A KOGYO	6779	5.7	5.9	5.9	6.2	6.3	7.1	7.5	7.8	9.0	3.1	3.2	3.0	3.4	3.9	3.8	3.6	3.6	4.1	3.5	3.7	3.1	3.0	3.1	3.3	3.4	3.9	4.4	4.9	5.4	6.2	7.0	4.4
MEIKO ELECTRONICS	6787	1.0	1.0	1.1	1.4	2.1	3.2	4.7	6.0	7.1	6.8	6.5	6.4	6.3	6.2	7.2	6.5	5.5	5.8	6.3	6.6	7.4	8.0	9.8	11.2	12.5	13.7	17.5	34.8	42.0	49.0	56.7	-
HIROSE ELECTRIC	6806	5.1	5.3	4.3	5.3	6.2	6.9	8.1	10.2	10.7	8.3	7.5	8.9	8.7	11.4	11.4	11.6	8.1	10.8	13.1	15.8	16.8	16.4	16.7	16.8	17.9	19.5	21.3	23.1	25.0	27.1	29.3	18.0
JAE	6807	6.0	6.4	6.0	6.2	6.6	7.7	8.4	11.2	13.0	10.6	10.0	11.6	14.7	18.9	19.9	17.7	20.3	20.8	19.6	21.1	21.2	20.4	20.6	20.6	20.9	19.7	22.0	21.7	21.4	21.3	21.3	22.0
CMK	6958	10.3	10.1	9.5	8.2	6.8	6.3	6.2	8.1	9.2	6.8	6.3	5.3	4.1	4.7	5.0	4.9	4.1	4.1	4.3	4.7	4.5	4.2	4.8	5.3	5.9	6.5	8.5	8.9	9.3	9.8	10.2	8.5
DAISHINKU	6962	4.0	3.9	3.4	3.1	2.8	2.7	3.1	3.4	3.6	3.2	3.7	3.9	3.6	3.5	3.6	2.5	2.7	2.9	2.9	2.7	3.0	3.6	4.1	4.0	4.1	4.3	5.2	5.4	5.6	5.9	6.1	5.2
HAMAMATSU PHOTONICS	6965	5.2	6.6	7.1	7.1	7.1	6.7	7.1	8.0	9.2	9.7	8.7	8.1	8.7	8.7	8.0	9.5	10.5	10.0	11.3	12.1	12.7	13.6	13.5	14.5	16.7	18.9	23.5	27.7	26.7	25.4	24.1	-
KYOCERA	6971	74.2	88.5	75.3	70.3	65.9	73.2	82.2	87.0	97.6	72.8	71.5	73.1	73.6	76.2	73.0	76.6	77.4	82.8	63.9	92.7	109.1	129.0	149.6	157.0	160.9	158.2	120.0	125.0	120.0	120.0	120.0	
TAIYO YUDEN	6976	17.1	20.7	20.3	18.3	18.0	17.1	16.4	23.3	27.9	23.9	19.3	19.3	19.8	20.8	21.8	23.8	24.9	25.6	26.5	27.0	29.3	31.3	34.9	39.4	46.3	49.1	48.0	52.0	52.0	52.0	48.0	
MURATA MANUFACTURING	6981	54.0	56.5	50.8	44.6	42.4	45.1	49.8	65.1	81.0	69.9	61.8	61.0	72.3	76.9	84.9	99.1	113.5	141.6	124.4	140.3	143.1	155.6	161.3	175.9	173.3	178.2	178.0	190.0	200.0	220.0	240.0	182.0
NICHICON	6996	13.0	14.6	13.2	9.7	8.8	8.9	9.4	10.5	11.1	8.4	7.7	9.2	7.6	5.1	4.3	4.4	3.4	3.1	4.5	5.3	5.2	6.1	7.5	6.9	8.5	8.1	8.5	9.0	9.9	10.7	11.6	8.5
NIPPON CHEMI-CON	6997	8.8	8.6	8.8	8.2	7.2	7.5	8.8	11.1	11.6	9.5	9.1	9.2	9.5	8.6	7.9	7.6	6.7	6.6	7.0	7.7	6.0	6.3	6.3	6.8	7.1	7.3	7.0	6.8	6.8	6.7	6.8	6.6
KOA	6999	2.0	2.5	2.8	2.2	1.9	2.0	2.3	2.4	2.6	2.4	2.0	1.9	1.9	2.1	2.2	2.4	2.3	2.5	2.9	3.4	3.6	3.5	4.1	4.9	5.9	7.1	7.2	7.3	7.4	7.6	7.7	7.2
Total	363.0	393.6	360.6	331.8	332.5	360.0	400.8	459.4	516.7	423.3	395.5	401.6	413.5	430.1	447.8	486.3	491.0	533.4	534.6	625.2	674.3	769.4	848.5	855.6	889.1	927.9	968.3	1,048.5	1,101.6	1,154.9	1,189.3	-	
% of sales																																	
IBIDEN	4062	7%	8%	8%	6%	6%	7%	8%	9%	14%	14%	13%	14%	15%	12%	12%	14%	12%	8%	9%	8%	11%	13%	13%	12%	15%	16%	14%	14%	13%	13%	11%	19%
NITERRA	5334	8%	9%	8%	8%	7%	6%	5%	7%	11%	8%	6%	5%	4%	4%	4%	5%	5%	6%	7%	8%	8%	7%	6%	6%	6%	5%	5%	5%	4%	4%	6%	
MINEBEA MITSUMI	6479	8%	9%	8%	8%	8%	8%	7%	8%	9%	8%	8%	7%	6%	6%	6%	4%	4%	4%	5%	5%	5%	4%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
MABUCHI MOTOR	6592	6%	6%	5%	5%	5%	6%	5%	5%	5%	6%	4%	4%	4%	4%	5%	5%	6%	7%	8%	7%	8%	7%	7%	7%	7%	6%	6%	6%	5%	5%	7%	
TDK	6762	9%	11%	9%	9%	8%	8%	7%	8%	8%	12%	10%	9%	10%	9%	8%	7%	7%	7%	8%	9%	9%	9%	9%	9%	9%	8%	9%	8%	8%	7%	10%	
ALPS ALPINE	6770	6%	6%	5%	5%	6%	5%	6%	5%	6%	4%	3%	4%	4%	4%	4%	4%	4%	4%	5%	6%	6%	6%	5%	4%	4%	3%	4%	3%	3%	4%		
NIHON DEMP A KOGYO	6779	8%	12%	12%	11%	10%	11%	10%	10%	15%	6%	6%	6%	7%	8%	8%	8%	8%	9%	8%	9%	8%	7%	6%	7%	7%	7%	7%	7%	8%	8%	8%	
MEIKO ELECTRONICS	6787	4%	6%	5%	5%	5%	6%	7%	8%	9%	10%	9%	10%	10%	8%	8%	7%	6%	5%	6%	6%	6%	5%	6%	6%	6%	5%	8%	9%	9%	10%	-	
HIROSE ELECTRIC	6806	6%	9%	6%	7%	7%	7%	7%	9%	12%	10%	8%	9%	9%	9%	10%	7%	9%	11%	13%	13%	10%	9%	10%	9%	9%	9%	9%	9%	9%	9%	9%	
JAE	6807	5%	6%	6%	5%	6%	6%	6%	7%	11%	10%	9%	10%	11%	12%	10%	10%	10%	8%	9%	10%	10%	9%	9%	9%	9%	9%	9%	9%	9%	8%	10%	
CMK	6958	8%	10%	8%	7%	6%	5%	5%	6%	9%	8%	7%	7%	6%	7%	7%	5%	5%	5%	6%	6%	5%	6%	6%	6%	7%	8%	8%	9%	9%	9%	8%	
DAISHINKU	6962	8%	12%	11%	10%	8%	8%	7%	8%	10%	10%	10%	11%	11%	10%	12%	8%	9%	10%	10%	9%	9%	9%	11%	10%	11%	11%	13%	12%	12%	12%	12%	13%
HAMAMATSU PHOTONICS	6965	10%	11%	12%	12%	10%	9%	8%	8%	9%	14%	10%	8%	9%	8%	7%	8%	9%	8%	8%	8%	9%	8%	6%	7%	8%	9%	10%	11%	10%	9%	8%	-
KYOCERA	6971	6%	9%	7%	6%	6%	6%	6%	7%	9%	7%	6%	6%	6%	5%	5%	5%	5%	5%	4%	6%	7%	7%	7%	8%	8%	8%	6%	6%	5%	5%	6%	
TAIYO YUDEN	6976	9%	14%	13%	11%	10%	9%	8%	10%	15%	12%	9%	10%	10%	10%	10%	11%	10%	10%	10%	10%	9%	11%	12%	14%	14%	12%	11%	10%	9%	8%	14%	
MURATA MANUFACTURING	6981	9%	14%	13%	11%	10%	9%	9%	10%	15%	13%	10%	10%	10%	11%	9%	8%	8%	10%	10%	8%	9%	9%	10%	11%	10%	10%	9%	8%	7%	7%	6%	10%
NICHICON	6996	11%	14%	12%	10%	8%	8%	8%	9%	12%	10%	7%	9%	8%	5%	4%	3%	3%	4%	4%	5%	4%	4%	4%	4%	4%	5%	5%	5%	5%	5%	5%	
NIPPON CHEMI-CON	6997	7%	9%	9%	8%	7%	6%	7%	8%	10%	9%	7%	9%	10%	8%	6%	6%	6%	5%	5%	7%	5%	4%	4%	4%	4%	4%	4%	4%	4%	4%	5%	
KOA	6999	3%	5%	5%	4%	4%	4%	4%	5%	6%	7%	5%	5%	5%	5%	5%	5%	5%	5%	5%	7%	7%	5%	5%	8%	9%	10%	9%	8%	8%	8%	10%	
Total	7.3%	9.4%	8.3%	7%	7%	7%	7%	7%	8%	11%	9%	7%	8%	8%	7%	7%	7%	7%	6%	7%	8%	8%	8%	8%	8%	8%	7%	7%	7%	7%	6%	-	

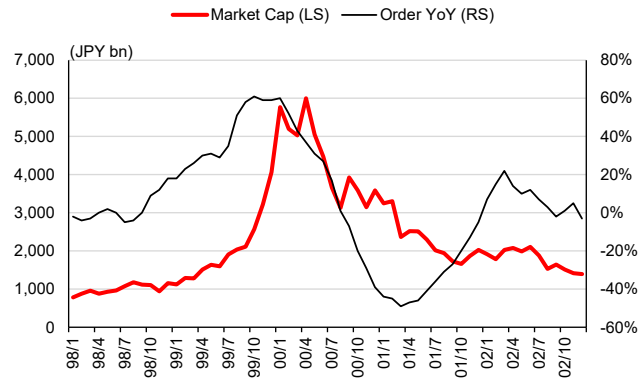
Note: For Mabuchi Motor, fiscal year ends in December; For Hamamatsu Photonics, fiscal year ends in Sep and one year behind the other companies shown; Fiscal year ends in Mar for all others; e = Morgan Stanley Research estimates; Ce = Company estimates
Source: Company data, FactSet, Morgan Stanley Research

Inventory and Inventory Days

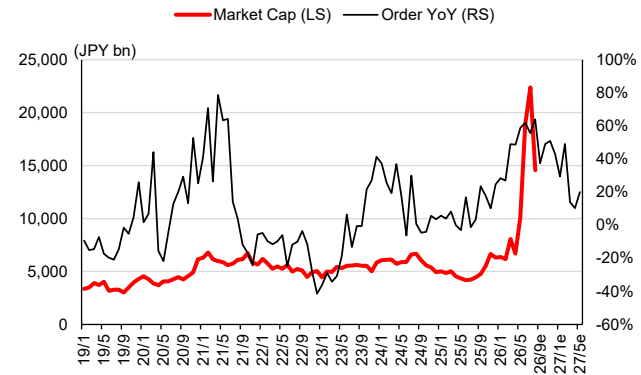
	F00	F01	F02	F03	F04	F05	F06	F07	F08	F09	F10	F11	F12	F13	F14	F15	F16	F17	F18	F19	F20	F21	F22	F23	F24	F25	F26e	F27e	F28e	F29e	F30e	
(JPY bn)																																
Inventory																																
IBIDEN	4062	20.3	16.7	15.4	20.0	20.9	32.0	37.4	42.2	31.8	25.1	29.8	35.8	38.7	38.6	38.0	37.1	33.6	40.0	48.8	45.3	57.0	74.6	79.2	65.3	67.2	71.0	67.7	79.4	90.8	105.6	114.4
NITERRA	5334	51.7	51.1	46.0	42.2	45.4	51.1	65.4	76.5	52.7	59.7	62.1	62.1	67.8	71.1	78.4	82.1	86.9	95.0	113.1	116.7	126.8	156.8	192.3	189.6	183.9	217.7	232.5	228.3	236.1	245.1	255.5
MINEBEA MITSUMI	6479	52.8	49.9	43.2	41.5	47.0	48.9	45.9	42.4	38.7	35.9	44.0	49.0	58.2	63.7	92.2	103.0	120.4	150.1	141.4	169.8	171.4	219.3	263.1	294.9	350.9	391.3	397.7	407.9	423.0	437.9	453.5
MABUCHI MOTOR	6592	26.5	23.5	17.6	14.7	16.4	17.1	17.8	18.0	17.1	13.5	16.1	18.6	20.5	24.2	29.0	31.0	26.7	33.0	35.7	33.9	32.5	50.9	61.4	60.4	59.8	57.8	62.4	62.7	62.5	61.9	59.2
TDK	6762	116.4	91.1	73.9	77.3	74.9	89.0	89.8	88.8	105.7	105.1	121.7	137.2	137.9	136.4	151.0	157.1	154.5	207.5	226.9	236.5	288.9	437.0	443.0	406.1	410.0	585.4	464.8	485.0	501.9	541.7	557.2
ALPS ALPINE	6770	93.1	79.8	82.3	78.6	85.0	83.7	77.4	72.5	52.1	48.1	56.6	69.1	74.3	73.1	85.4	87.5	89.6	96.1	95.8	100.5	95.8	133.4	165.4	163.7	138.6	138.3	142.2	152.6	160.0	169.1	179.1
NIHON DENPA KOGYO	6779	11.6	6.8	6.9	6.6	7.3	8.6	8.5	9.8	4.8	9.4	9.0	11.7	11.3	11.0	12.5	10.6	12.2	13.7	14.4	12.9	10.1	11.8	12.8	11.1	10.5	12.4	12.7	13.3	14.1	15.1	16.4
MEIKO ELECTRONICS	6787	3.1	2.8	4.1	4.6	4.3	5.9	8.8	9.8	8.6	8.9	11.3	9.1	10.5	11.6	13.7	11.6	12.5	14.3	16.0	19.0	22.2	29.8	32.6	35.6	38.5	49.5	64.0	78.2	87.4	93.4	99.7
HIROSE ELECTRIC	6806	7.6	5.0	4.7	5.7	5.5	6.6	6.7	6.6	5.6	6.3	7.3	7.2	7.9	8.4	9.1	8.2	8.9	11.7	13.9	14.5	16.9	21.7	27.7	24.4	25.1	30.2	33.5	36.4	39.4	42.6	46.2
JAE	6807	17.6	14.0	13.8	12.5	11.7	12.4	12.4	12.6	10.3	9.1	9.9	9.8	11.3	12.4	12.7	12.9	16.1	18.9	18.0	20.3	21.1	29.0	28.6	28.3	29.2	32.8	33.2	32.8	33.3	33.0	32.7
CMK	6958	9.9	10.3	9.9	9.6	8.5	9.5	10.3	12.4	7.9	9.6	9.0	8.0	6.7	7.9	9.2	8.5	8.5	9.8	10.3	10.6	10.3	16.1	17.3	17.3	18.6	18.4	18.8	19.0	19.2	19.5	19.8
DAISHINKU	6962	14.0	12.4	6.2	6.7	7.3	8.6	9.2	8.7	7.2	7.4	8.1	8.0	8.3	9.1	9.8	8.4	9.4	10.7	12.8	13.1	14.8	16.5	17.2	16.4	18.0	24.6	26.3	27.0	27.7	28.4	29.1
HAMAMATSU PHOTONICS	6965	13.0	13.0	12.8	12.8	13.3	13.7	15.4	17.8	20.2	20.0	20.7	22.3	24.4	25.1	26.9	29.5	30.0	31.8	34.5	37.5	43.5	44.4	59.1	75.6	80.6	77.0	79.4	78.0	78.0	79.3	83.5
KYOCERA	6971	252.8	205.8	183.2	197.2	213.4	190.6	209.2	205.2	199.6	177.4	232.9	270.3	296.5	335.8	354.5	327.9	331.2	364.9	343.9	344.3	345.4	452.5	539.4	540.2	521.8	522.0	511.5	501.9	540.5	556.9	552.4
TAIYO YUDEN	6976	36.9	28.3	25.1	23.5	25.5	27.3	32.2	40.5	30.3	28.6	32.5	30.5	35.9	37.6	44.1	48.7	48.8	52.1	60.9	62.6	67.9	96.8	104.8	102.3	110.0	125.8	114.5	123.9	132.4	158.1	164.7
MURATA MANUFACTURING	6981	92.0	59.2	53.3	54.8	59.0	66.4	86.1	114.6	94.1	89.2	110.5	132.0	160.9	170.6	186.3	217.5	211.4	290.3	349.3	334.4	361.3	464.7	575.0	513.0	482.8	520.5	541.3	612.9	697.2	796.4	909.2
NICHICON	6996	21.1	18.4	18.9	16.0	17.3	18.2	20.6	20.1	16.0	14.5	17.6	16.9	17.6	17.7	17.5	17.8	18.8	22.1	21.7	21.6	31.1	37.2	36.7	33.2	32.3	35.1	35.9	36.6	37.7	39.3	
NIPPON CHEMI-CON	6997	23.1	18.2	18.9	18.8	18.7	21.9	23.5	24.4	22.4	19.5	21.0	27.1	21.8	20.9	22.7	20.8	20.9	23.0	27.9	26.5	25.5	31.6	33.0	29.2	33.1	31.2	34.7	35.3	36.3	37.7	39.5
KOA	6999	9.4	7.5	5.7	5.4	5.1	5.4	5.8	5.6	3.8	3.7	4.6	4.4	4.9	4.6	5.2	4.9	5.0	5.9	7.6	7.3	7.8	10.8	13.9	14.0	13.4	13.5	14.7	15.1	15.5	16.0	16.5
Total		873.0	713.8	641.8	648.6	686.5	717.0	782.5	828.5	728.9	690.9	824.5	930.9	1,014.7	1,079.7	1,198.6	1,224.8	1,244.6	1,487.5	1,593.4	1,627.3	1,740.8	2,328.8	2,703.1	2,624.1	2,625.3	2,951.9	2,887.2	3,025.8	3,231.7	3,475.4	3,667.8
Days in inventory (days)																																
IBIDEN	4062	38.6	38.2	33.6	36.1	38.1	41.0	44.7	49.6	54.2	49.8	45.1	50.8	58.0	59.5	59.0	57.5	61.3	57.3	69.6	75.3	78.4	85.4	96.8	98.4	94.4	88.6	85.0	75.0	74.0	71.6	72.1
NITERRA	5334	99.9	106.5	99.6	93.9	89.9	86.6	85.6	98.7	93.6	106.4	113.7	106.2	103.3	114.0	122.1	118.7	123.3	123.7	135.1	143.5	149.5	154.3	174.7	179.5	172.6	162.8	161.2	157.9	149.0	143.8	138.6
MINEBEA MITSUMI	6479	89.6	90.9	83.5	76.1	69.6	70.0	67.2	63.5	75.1	77.7	72.1	85.5	87.0	77.8	74.7	73.2	79.5	70.4	74.1	70.0	75.9	78.5	82.2	87.0	94.1	98.8	103.3	102.9	102.3	102.4	102.4
MABUCHI MOTOR	6592	122.7	137.7	107.6	93.0	89.6	91.9	88.5	84.8	93.1	107.8	90.0	105.8	111.1	99.9	111.7	110.0	111.4	109.4	126.9	139.9	148.5	155.1	171.9	165.3	153.1	152.6	142.4	137.6	128.2	118.5	110.4
TDK	6762	74.0	81.5	65.5	58.2	57.4	51.1	52.4	51.3	58.6	62.3	62.9	74.5	74.3	65.5	65.4	67.7	66.4	71.0	80.5	88.1	91.8	99.0	100.6	103.2	98.2	105.5	102.0	83.5	78.9	72.5	69.3
ALPS ALPINE	6770	64.6	69.9	61.9	58.4	56.4	53.0	49.3	46.6	47.0	44.1	43.3	53.0	57.1	48.2	48.7	50.7	53.7	50.6	50.8	53.4	59.3	63.0	71.2	75.3	67.7	60.3	59.5	58.3	59.0	58.7	58.7
NIHON DENPA KOGYO	6779	72.9	82.9	62.0	56.3	55.6	61.4	59.6	58.9	49.1	68.1	78.8	97.1	102.4	99.3	110.9	117.0	120.2	125.3	144.7	139.8	129.1	121.6	124.5	121.4	106.3	107.5	106.8	102.9	100.4	98.0	95.9
MEIKO ELECTRONICS	6787	55.1	69.5	73.3	71.3	52.2	47.5	49.6	52.1	50.0	54.0	58.2	67.9	67.1	58.4	55.5	56.2	55.2	54.3	55.9	64.6	73.8	76.5	80.6	83.6	80.9	84.5	82.6	80.9	81.8	80.3	78.2
HIROSE ELECTRIC	6806	59.5	65.6	48.2	48.4	46.3	44.1	41.7	37.9	41.9	45.4	47.0	47.0	49.7	43.1	46.3	47.9	50.9	55.7	66.9	73.8	74.6	79.4	94.9	102.2	86.7	82.5	83.5	85.3	86.2	87.0	88.0
JAE	6807	76.2	73.6	63.4	57.4	50.5	46.5	43.4	38.1	39.8	41.4	39.6	40.3	37.0	35.0	31.8	33.5	30.1	30.3	37.0	40.8	42.0	49.9	54.2	55.8	58.4	59.1	60.9	60.6	58.7	58.2	57.1
CMK	6958	36.7	39.0	37.0	37.3	34.7	33.1	32.3	34.0	37.1	39.1	42.4	43.2	41.7	41.5	48.0	49.9	47.7	45.2	47.6	53.1	60.4	69.5	85.6	82.9	81.9	79.2	77.1	76.7	75.9	75.0	74.4
DAISHINKU	6962	125.0	159.5	119.4	100.5	93.1	107.1	107.7	107.6	96.8	106.2	93.0	102.0	111.5	119.1	129.0	131.6	139.1	155.7	192.0	201.2	204.5	196.6	229.6	207.4	214.6	256.2	297.9	294.9	288.6	279.8	271.0
HAMAMATSU PHOTONICS	6965	144.4	157.6	152.4	150.9	139.8	141.2	128.3	136.5	149.8	186.8	163.0	160.2	178.2	177.1	177.5	178.6	178.6	171.7	171.7	182.7	206.1	187.4	195.9	242.2	284.7	259.8	236.0	221.7	199.9	188.1	175.3
KYOCERA	6971	83.8	105.3	89.1	80.7	87.6	87.9	81.0	85.6	88.3	87.3	84.2	105.5	108.6	108.0	110.8	113.9	114.6	105.8	111.5	108.5	112.4	109.9	124.0	135.8	133.2	130.2	143.3	140.2	131.0	130.9	130.4
TAIYO YUDEN	6976	96.9	104.2	85.2	73.2	64.1	64.7	65.8	73.9	79.0	67.6	67.8	73.2	78.2	83.5	85.5	95.7	99.2	101.1	107.4	114.5	112.4	133.8	158.5	147.0	143.6	157.4	145.7	128.7	124.7	113.7	116.5
MURATA MANUFACTURING	6981	96.1	108.9	83.8	76.6	81.2	77.3	82.9	94.4	95.7	87.4	87.8	106																			

Market Cap/Order Monthly Trend for Murata Mfg.

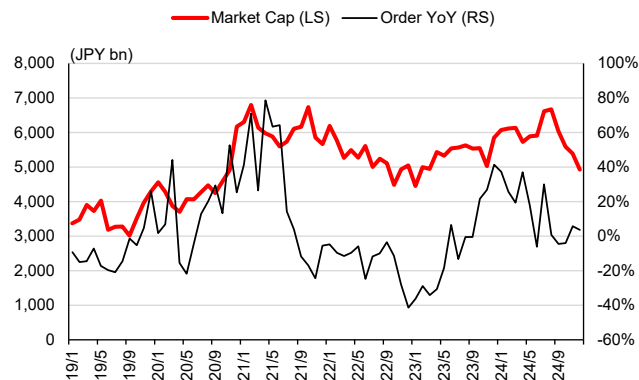
■ For dot-com bubble era (1998/1~2003/3)



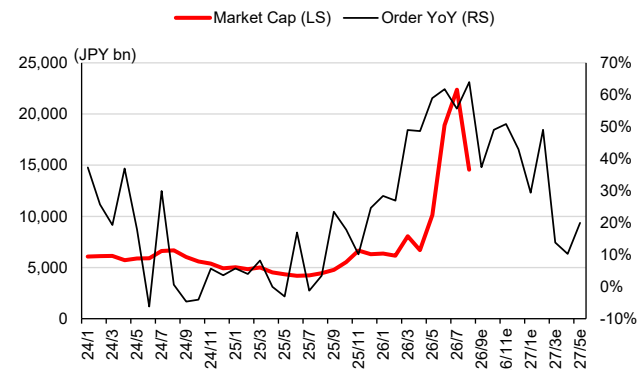
■ From 2019



■ For COVID-19 era (2019/1~2024/12)



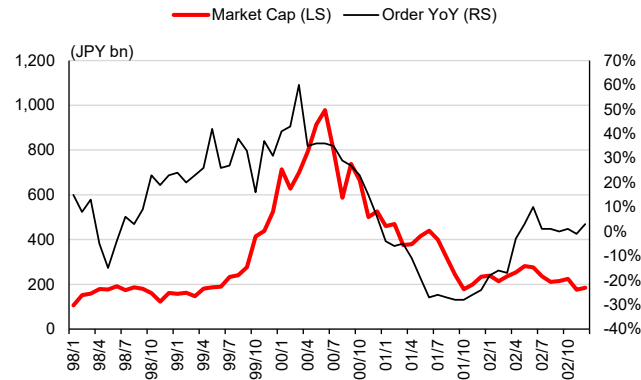
■ After COVID-19 era



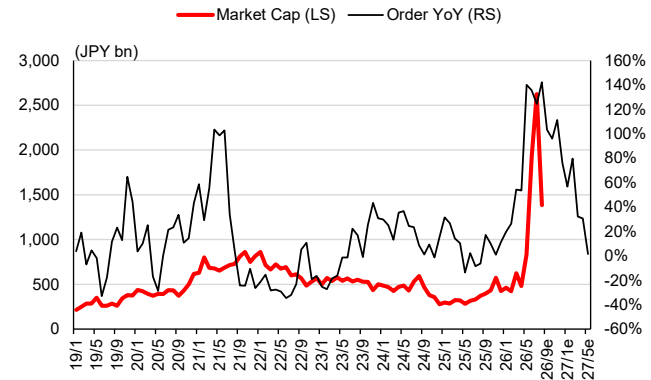
Note: For July 26 and thereafter, we estimate the same order level as June 26
Source: FactSet, Company data, Morgan Stanley Research

Market Cap/Order Monthly Trend for Taiyo Yuden

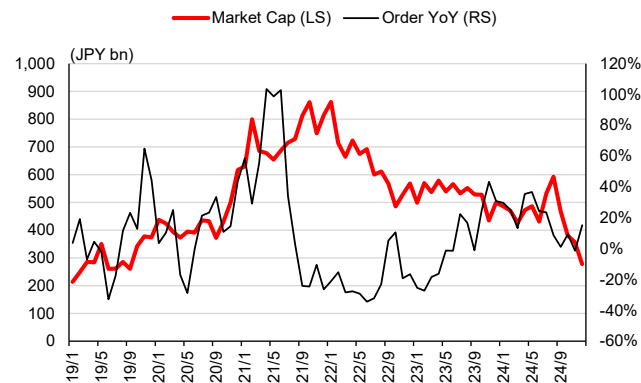
■ For dot-com bubble era (1998/1~2003/3)



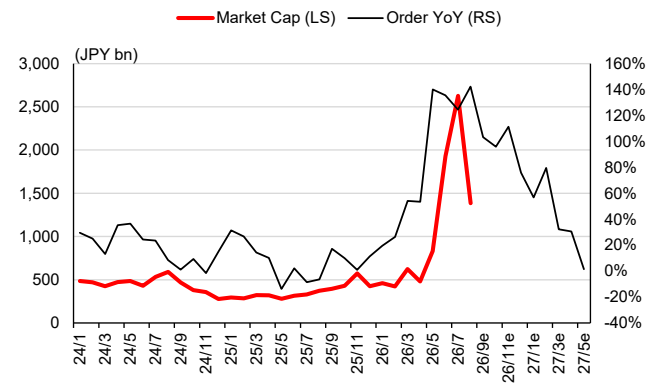
■ From 2019



■ For COVID-19 era (2019/4~2024/12)



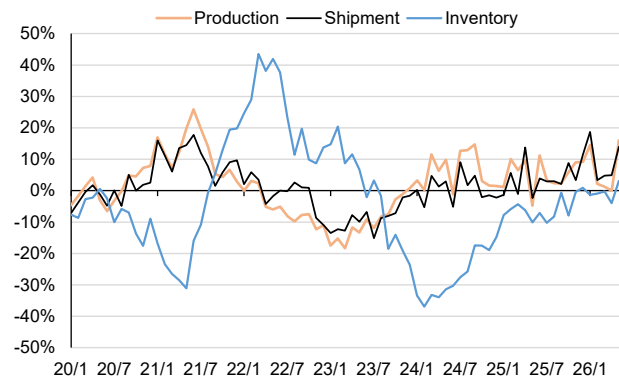
■ After COVID-19 era



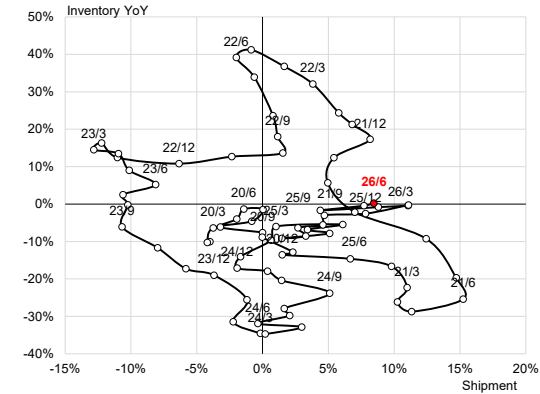
Note: Order data after Jan 19 are capacitor only; For July 26 and thereafter, we estimate the same order level as June 26
Source: FactSet, Company data, Morgan Stanley Research

Production & Inventory

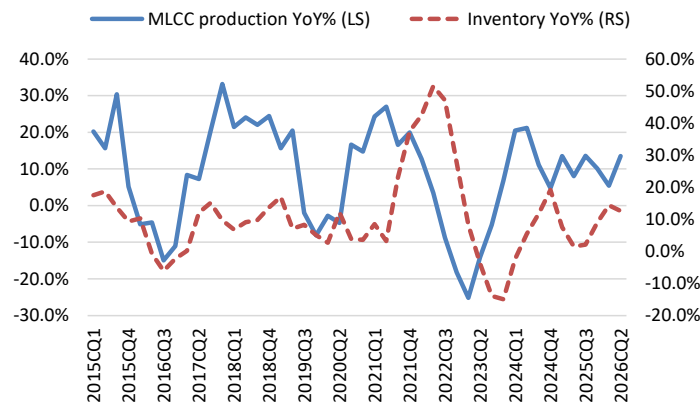
Industrial production: Production, shipments, inventories for Electronic parts and devices* (1)



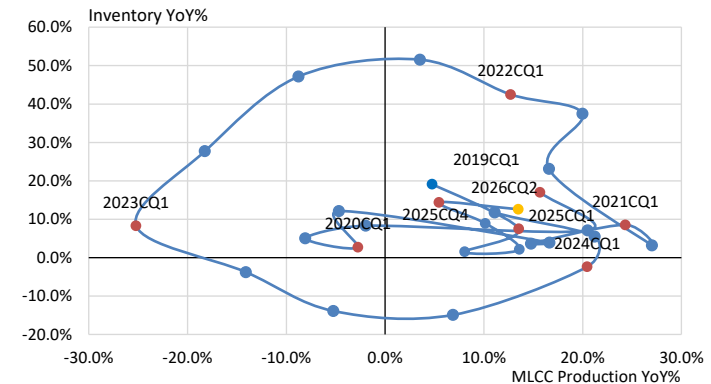
Industrial production: Production, shipments, inventories for Electronic parts and devices* (2)



Taiyo Yuden: MLCC production & inventory (1)



Taiyo Yuden: MLCC Production & inventory (2)

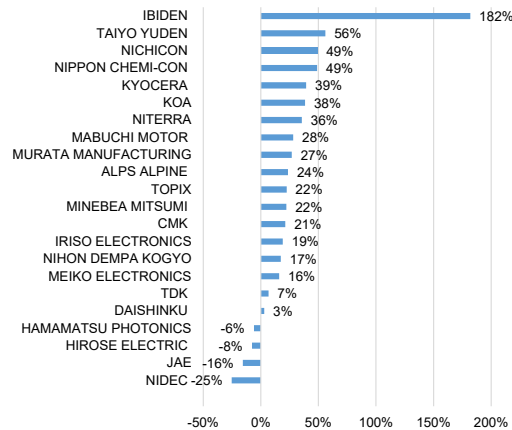


Note: *3-month average
Source: METI, Company data, Morgan Stanley Research

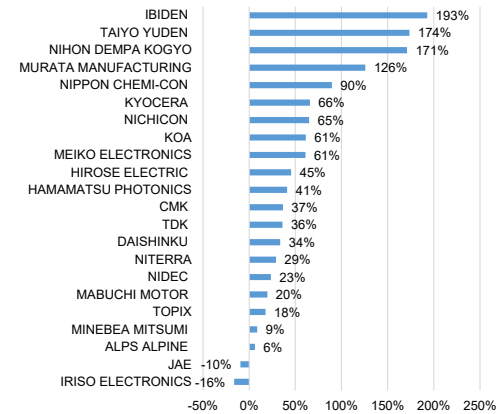
Stock Price Performance

Rank	2020	2021	2022	2023	2024	2025	2026 Based on 8/19 Closing Price
1	IBIDEN 85%	NIHON DEMP A KOGYO 184%	NITERRA 22%	CMK 80%	MEIKO ELECTRONICS 122%	IBIDEN 182%	IBIDEN 193%
2	NIDEC 73%	MEIKO ELECTRONICS 147%	KOA 17%	MEIKO ELECTRONICS 73%	TDK 54%	TAIYO YUDEN 56%	TAIYO YUDEN 174%
3	DAISHINKU 51%	DAISHINKU 135%	ALPS ALPINE 10%	IBIDEN 63%	NITERRA 52%	NICHICON 49%	NIHON DEMP A KOGYO 171%
4	TAIYO YUDEN 44%	CMK 72%	JAE 6%	TDK 55%	ALPS ALPINE 31%	NIPPON CHEMI-CON 49%	MURATA MANUFACTURING 126%
5	MURATA MANUFACTURING 38%	MINEBEA MITSUMI 59%	MABUCHI MOTOR -2%	JAE 53%	TOPIX 18%	KYOCERA 39%	NIPPON CHEMI-CON 90%
6	HAMAMATSU PHOTONICS 31%	IBIDEN 42%	IRISO ELECTRONICS -2%	MINEBEA MITSUMI 47%	HIROSE ELECTRIC 17%	KOA 38%	KYOCERA 66%
7	TDK 25%	TAIYO YUDEN 37%	NICHICON -3%	NITERRA 37%	NIDEC 0%	NITERRA 36%	NICHICON 65%
8	NIHON DEMP A KOGYO 19%	JAE 26%	TDK -3%	MURATA MANUFACTURING 36%	MABUCHI MOTOR -4%	MABUCHI MOTOR 28%	KOA 61%
9	NICHICON 14%	HAMAMATSU PHOTONICS 24%	TOPIX -5%	KYOCERA 26%	JAE -8%	MURATA MANUFACTURING 27%	MEIKO ELECTRONICS 61%
10	HIROSE ELECTRIC 12%	HIROSE ELECTRIC 24%	KYOCERA -9%	MABUCHI MOTOR 25%	MINEBEA MITSUMI -11%	ALPS ALPINE 24%	HIROSE ELECTRIC 45%
11	MABUCHI MOTOR 8%	NITERRA 14%	NIPPON CHEMI-CON -13%	TOPIX 25%	MURATA MANUFACTURING -14%	TOPIX 22%	HAMAMATSU PHOTONICS 41%
12	KOA 8%	KYOCERA 14%	NIHON DEMP A KOGYO -13%	DAISHINKU 24%	NICHICON -15%	MINEBEA MITSUMI 22%	CMK 37%
13	TOPIX 5%	TOPIX 10%	HAMAMATSU PHOTONICS -14%	NICHICON 6%	KYOCERA -23%	CMK 21%	TDK 36%
14	IRISO ELECTRONICS -8%	KOA 9%	HIROSE ELECTRIC -14%	ALPS ALPINE 3%	NIPPON CHEMI-CON -26%	IRISO ELECTRONICS 19%	DAISHINKU 34%
15	MINEBEA MITSUMI -10%	NIDEC 4%	MURATA MANUFACTURING -28%	TAIYO YUDEN -3%	IRISO ELECTRONICS -27%	NIHON DEMP A KOGYO 17%	NITERRA 29%
16	NIPPON CHEMI-CON -13%	MURATA MANUFACTURING -2%	IBIDEN -30%	HIROSE ELECTRIC -4%	NIHON DEMP A KOGYO -31%	MEIKO ELECTRONICS 16%	NIDEC 23%
17	KYOCERA -15%	NIPPON CHEMI-CON -2%	CMK -34%	HAMAMATSU PHOTONICS -8%	DAISHINKU -35%	TDK 7%	MABUCHI MOTOR 20%
18	NITERRA -17%	IRISO ELECTRONICS -3%	MINEBEA MITSUMI -40%	IRISO ELECTRONICS -13%	KOA -36%	DAISHINKU 3%	TOPIX 18%
19	MEIKO ELECTRONICS -22%	NICHICON -3%	TAIYO YUDEN -42%	NIPPON CHEMI-CON -16%	IBIDEN -39%	HAMAMATSU PHOTONICS -6%	MINEBEA MITSUMI 9%
20	JAE -29%	TDK -13%	NIDEC -49%	NIDEC -17%	HAMAMATSU PHOTONICS -39%	HIROSE ELECTRIC -8%	ALPS ALPINE 6%
21	CMK -40%	MABUCHI MOTOR -15%	MEIKO ELECTRONICS -49%	KOA -18%	TAIYO YUDEN -39%	JAE -16%	JAE -10%
22	ALPS ALPINE -46%	ALPS ALPINE -20%	DAISHINKU -55%	NIHON DEMP A KOGYO -19%	CMK -48%	NIDEC -25%	IRISO ELECTRONICS -16%

■ Stock price performance (CY2025)



■ Stock price performance (CY2026 YTD)



Source: FactSet

FX Sensitivities of Electronic Components Makers/Changes in the Competitiveness of Core Products Under a Weak Yen Environment

		F26e	FX assumption (by company)		FX sensitivity (JPY bn/year) USD/JPY	Assumption ¥150/USD		Assumption ¥160/USD	
		OP (JPY bn)	USD/JPY	USD/JPY		OP (JPY bn)	Change (%)	OP (JPY bn)	Change (%)
IBIDEN	4062	133.8	158.0	152.0	0.8	127.4	-5%	135.4	1%
NITERRA	5334	158.2	155.0	150.0	1.0	153.2	-3%	163.2	3%
MINEBEA MITSUMI	6479	121.5	155.0	156.0	1.5	114.0	-6%	129.0	6%
MABUCHI MOTOR	6592	27.2	155.0	156.6	0.3	25.7	-6%	28.7	6%
TDK	6762	335.8	155.0	150.0	2.0	325.8	-3%	345.8	3%
ALPS ALPINE	6770	49.1	155.0	150.0	1.2	43.1	-12%	55.1	12%
NIHON DEMPAC KOGYO	6779	4.6	155.0	157.5	0.1	4.2	-9%	5.0	9%
MEIKO ELECTRONICS	6787	38.6	155.0	150.0	0.1	38.1	-1%	39.1	1%
HIROSE ELECTRIC	6806	47.3	155.0	155.0	0.3	45.7	-3%	49.0	3%
JAE	6807	8.9	155.0	153.0	0.3	7.4	-17%	10.4	17%
CMK	6958	3.2	155.0	158.0	0.0	3.2	0%	3.2	0%
DAISHINKU	6962	1.0	155.0	150.0	0.1	0.8	-25%	1.3	25%
HAMAMATSU PHOTONICS	6965	22.5	155.0	158.3	0.3	21.0	-7%	24.0	7%
KYOCERA	6971	132.6	155.0	155.0	1.0	127.6	-4%	137.6	4%
TAIYO YUDEN	6976	45.2	155.0	159.7	0.9	40.7	-10%	49.7	10%
MURATA MANUFACTURING	6981	431.9	155.0	156.1	4.5	409.4	-5%	454.4	5%
NICHICON	6996	9.4	155.0	150.0	0.1	8.9	-6%	10.0	6%
NIPPON CHEMI-CON	6997	7.0	155.0	153.0	0.3	5.7	-19%	8.3	19%
KOA	6999	3.2	150.0	159.5	0.1	3.2	0%	4.4	38%
Total		1,581.0			14.9	1,504.9	-5%	1,653.4	5%

		Changes in the competitiveness of core products under a weak-yen environment	
IBIDEN	4062	↑	Flip-chip package costs rising vs. peers in S. Korea, Taiwan, Austria; profits highly likely to grow
MINEBEA MITSUMI	6479	→	Potential sales decline with over 70% of sales \$-denominated, but expecting THB/USD, CNY/USD rates to have big impact on profits
MABUCHI MOTOR	6592	→	Forecasting profit growth, but expecting no major shift in competitiveness for mainstay products given high overseas production weighting
TDK	6762	↑	Cost competitiveness for passive components may improve versus foreign peers, but limited impact for rechargeable batteries, HDD heads given lack of ¥-denominated costs
ALPS ALPINE	6770	→	Forecasting profit growth, but expecting no major shift in product competitiveness given limited rivalry with foreign firms for mainstay products
NIHON DEMPAC KOGYO	6779	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials
MEIKO ELECTRONICS	6787	→	Expecting no major change in product competitiveness given high foreign production exposure, some use of imported materials
HIROSE ELECTRIC	6806	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan
JAE	6807	→	Expecting no major change in product competitiveness as US micro connector peers also undertaking production development/mass production in Japan
CMK	6958	→	Expecting no major shift in competitiveness given high sales exposure to domestic customers
DAISHINKU	6962	↑	Expecting increase in competitiveness on high foreign sales exposure, limited use of imported materials
HAMAMATSU PHOTONICS	6965	↑	Yen weakness positive for earnings as 95% of production in Japan & 75% of sales abroad, although still certain level of ¥-denominated sales overseas
KYOCERA	6971	→	Forecasting growth in ¥-converted profits, but expecting no major shift in competitiveness of mainstay products
TAIYO YUDEN	6976	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure
MURATA MANUFACTURING	6981	↑	Cost competitiveness of mainstay MLCCs rising vs. peers in S. Korea & Taiwan; potential growth in ¥-converted profits on high foreign sales exposure
NICHICON	6996	→	Positive on aluminum electrolytic capacitor earnings, but worries over pressure on NECST product earnings given rising costs of imported parts/materials
NIPPON CHEMI-CON	6997	↑	Cost competitiveness of mainstay aluminum electrolytic capacitors rising vs. peers in China; potential growth in ¥-converted profits on high foreign sales exposure
KOA	6999	→	Expecting substantial profit growth but no major change in product competitiveness as domestic peers with high market share also getting same benefits

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

Sales and Production by Region

■ Sales by region (FY24)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	29.3%	12.9%	7.3%	-	50.5%
5334	NITERRA	18.9%	25.3%	19.3%	10.5%	26.0%
6479	MINEBEA MITSUMI	24.4%	27.2%	11.3%	17.8%	19.4%
6592	MABUCHI MOTOR	10.3%	18.1%	22.9%	0.0%	48.7%
6594	NIDEC	15.4%	22.8%	22.8%	21.1%	17.8%
6762	TDK	7.9%	6.4%	7.9%	54.1%	23.7%
6770	ALPS ALPINE	11.2%	18.0%	-	22.1%	48.7%
6779	NIHON DEMPA KOGYO	15.4%	13.7%	8.4%	35.3%	27.2%
6787	MEIKO ELECTRONICS	36.1%	0.0%	1.8%	16.7%	33.3%
6806	HIROSE ELECTRIC	15.7%	-	-	39.6%	44.7%
6807	JAE	36.6%	1.0%	-	37.0%	15.1%
6908	IRISO ELECTRONICS	16.0%	9.1%	15.5%	36.7%	22.6%
6958	CMK	58.2%	-	11.2%	18.2%	12.4%
6962	DAISHINKU	11.7%	5.3%	11.1%	32.5%	39.5%
6965	HAMAMATSU PHOTONICS	22.6%	24.2%	23.4%	19.1%	10.7%
6971	KYOCERA	29.0%	21.0%	20.2%	11.8%	18.0%
6976	TAIYO YUDEN	7.0%	-	-	32.2%	47.4%
6981	MURATA MANUFACTURING	7.4%	13.9%	9.2%	47.7%	21.7%
6996	NICHICON	50.6%	7.9%	5.7%	26.0%	9.9%
6997	NIPPON CHEMI-CON	21.4%	11.4%	10.7%	32.2%	24.4%
6999	KOA	31.3%	17.1%	18.9%	-	32.7%
Total		17.7%	16.9%	13.0%	27.6%	24.9%

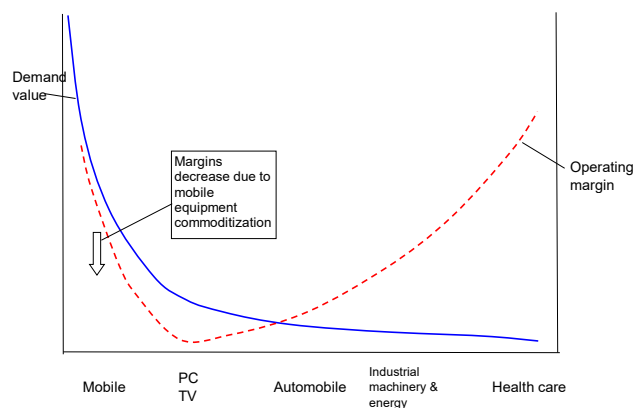
■ Production by region (FY25e)

Code	Company Name	Japan	US/America	Europe	China	Others
4062	IBIDEN	75.0%	4.0%	3.0%	2.0%	16.0%
6479	MINEBEA MITSUMI	22.0%	26.0%	11.0%	19.0%	22.0%
6592	MABUCHI MOTOR	0.0%	7.0%	3.0%	35.0%	55.0%
6762	TDK	12.5%	5.0%	10.0%	60.0%	12.5%
6770	ALPS ALPINE	30.0%	10.0%	5.0%	25.0%	30.0%
6779	NIHON DEMPA KOGYO	70.0%	0.0%	0.0%	20.0%	10.0%
6787	MEIKO ELECTRONICS	10.0%	0.0%	0.0%	45.0%	45.0%
6806	HIROSE ELECTRIC	50.0%	0.0%	0.0%	10.0%	40.0%
6807	JAE	60.0%	5.0%	0.0%	20.0%	15.0%
6908	IRISO ELECTRONICS	13.0%	0.0%	0.0%	49.0%	38.0%
6958	CMK	20.0%	0.0%	0.0%	35.0%	45.0%
6962	DAISHINKU	60.0%	0.0%	0.0%	30.0%	10.0%
6965	HAMAMATSU PHOTONICS	80.0%	5.0%	10.0%	5.0%	0.0%
6971	KYOCERA	60.0%	10.0%	10.0%	10.0%	10.0%
6976	TAIYO YUDEN	35.0%	0.0%	0.0%	20.0%	45.0%
6981	MURATA MANUFACTURING	70.0%	3.0%	3.0%	15.0%	9.0%
6996	NICHICON	75.0%	0.0%	0.0%	15.0%	10.0%
6997	NIPPON CHEMI-CON	65.0%	5.0%	0.0%	15.0%	15.0%
6999	KOA	70.0%	0.0%	0.0%	5.0%	25.0%
Total		39.9%	8.6%	7.9%	25.1%	18.6%

Note: Production data based on Morgan Stanley Research estimation
Source: Company data, Morgan Stanley Research

Preparing for Smartphone Price Compression and Slowdown in Technological Innovation

■ Potential for smartphone commoditization to knock down margins, but also greater earnings opportunities from computerization of autos and other areas



■ Electronic component firms: Sales breakdown by applications (F25)

	Sales in F25 (JPY bn)					
	Smartphone	Computers	AV/Office	Automotive	Industrial	Others
Ibiden	416.2	7.3	268.6	0.0	55.9	16.6
Nittera	731.2	6.7	0.0	0.0	587.5	40.7
Minebea Mitsumi	1,664.4	217.9	141.1	303.1	567.8	99.9
Mabuchi Motor	200.4	0.0	0.0	18.3	154.5	9.2
TDK	2,504.8	1,007.5	465.0	79.4	459.4	402.8
Alps Alpine	1,019.5	295.6	41.1	56.3	555.1	24.7
NDK	54.6	9.8	0.3	0.3	29.9	5.8
Meiko Electronics	240.6	29.4	16.5	0.0	97.5	4.4
Hirose Electric	211.3	44.3	8.9	6.8	54.3	62.0
JAE	227.9	55.5	18.0	18.0	95.1	24.1
CMK	100.2	0.0	0.4	0.5	89.5	0.0
Daishinku	39.6	10.2	5.2	4.0	14.8	4.2
Hamamatsu Photonics	204.0	0.0	0.0	1.0	5.6	66.3
Kyocera	2,070.2	139.4	154.8	491.9	303.7	414.0
Taiyo Yuden	355.3	85.3	64.0	28.4	106.6	71.1
Murata Mfg.	1,830.9	653.0	310.4	56.0	474.5	250.3
Nichicon	169.7	0.0	15.3	17.0	40.7	61.1
Nippon Chemi-Con	136.8	0.0	48.1	6.9	40.1	28.2
KOA	72.3	0.0	8.0	4.0	37.4	8.5
Total	12,249.8	2,561.8	1,565.8	1,092.1	3,770.0	1,593.9

	Smartphone	Computers	AV/Office	Automotive	Industrial	Others
Ibiden	100%	2%	65%	0%	13%	4%
Nittera	100%	1%	0%	0%	80%	6%
Minebea Mitsumi	100%	13%	8%	18%	34%	6%
Mabuchi Motor	100%	0%	0%	9%	77%	5%
TDK	100%	40%	19%	3%	18%	16%
Alps Alpine	100%	29%	4%	6%	54%	2%
NDK	100%	18%	1%	1%	55%	11%
Meiko Electronics	100%	12%	7%	0%	41%	2%
Hirose Electric	100%	21%	4%	3%	26%	29%
JAE	100%	24%	8%	8%	42%	11%
CMK	100%	0%	0%	1%	89%	0%
Daishinku	100%	26%	13%	10%	37%	11%
Hamamatsu Photonics	100%	0%	0%	1%	3%	33%
Kyocera	100%	7%	7%	24%	15%	20%
Taiyo Yuden	100%	24%	18%	8%	30%	20%
Murata Mfg.	100%	36%	17%	3%	26%	14%
Nichicon	100%	0%	9%	10%	24%	36%
Nippon Chemi-Con	100%	0%	35%	5%	29%	21%
KOA	100%	0%	11%	6%	52%	12%
Total	100%	21%	13%	9%	31%	13%

Source: Company data, Morgan Stanley Research

Memory Price Outlook

	1Q26	2Q26	3Q26E - Old	3Q26E - New	4Q26E - Old	4Q26E - New
PC DRAM	DDR4: up 120-125% DDR5: up 110-115% Blended ASP: up 110-115%	DDR4: up 35-40% DDR5: up 45-50% Blended ASP: up 45-50%	up 3-8%	DDR4: up 15-20% DDR5: up 15-20% Blended ASP: up 15-20%	up 0-5%	up 3-8%
Server DRAM	DDR4: up 95-100% DDR5: up 93-98% Blended ASP: up 93-98%	DDR4: up 50-55% DDR5: up 53-58% Blended ASP: up 53-58%	up 10-15%	DDR4: up 13-18% DDR5: up 13-18% Blended ASP: up 13-18%	up 3-8%	up 3-8%
Mobile DRAM	LPDDR4X: up 58-63% LPDDR5(x): up 58-63%	LPDDR4X: up 73-78% LPDDR5(x): up 80-85%	up 3-8%	LPDDR4X: up 8-13% LPDDR5(x): up 8-13%	up 0-5%	up 0-5%
Graphics DRAM	GDDR6: up 95-100% GDDR7: up 95-100%	GDDR6: up 30-35% GDDR7: up 40-45%	GDDR6: up 3-8% GDDR7: up 10-15%	GDDR6: up 15-20% GDDR7: up 15-20%	GDDR6: up 0-5% GDDR7: up 3-8%	GDDR6: up 0-5% GDDR7: up 3-8%
Consumer DRAM	DDR3: up 75-80% DDR4: up 75-80%	DDR3: up 55-60% DDR4: up 55-60%	up 10-15%	DDR3: up 35-40% DDR4: up 28-33%	up 3-8%	up 3-8%
Total DRAM	Conventional DRAM: up 93-98% HBM Blended: up 83-88%	Conventional DRAM: up 58-63% HBM Blended: 53-58%	Conventional DRAM: up 8-13%	Conventional DRAM: up 13-18% HBM Blended: 8-13%	Conventional DRAM: up 3-8%	Conventional DRAM: up 3-8%
	1Q26	2Q26	3Q26E - Old	3Q26E - New	4Q26E - Old	4Q26E - New
eMMC UFS	up 95-100%	up 75-80%	up 5-10%	up 5-10%	mostly flat	mostly flat
Enterprise SSD	up 75-80%	up 48-53%	up 13-18%	up 18-23%	up 3-8%	up 3-8%
Client SSD	up 105-110%	up 70-75%	up 5-10%	up 13-18%	mostly flat	mostly flat
3D NAND Wafers (TLC & QLC)	up 75-80%	up 28-33%	up 5-10%	up 5-10%	mostly flat	mostly flat
Total NAND Flash	up 85-90%	up 55-60%	up 8-13%	up 10-15%	up 0-5%	up 0-5%

Note: E = Morgan Stanley Research estimates based on published note on Jul 6
Source: TrendForce, Morgan Stanley Research

End-product Market and Major Components Market (Calendar Year)

	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e
(mn units)																
Server	8.7	9.0	9.2	9.7	9.6	10.2	11.8	11.8	12.7	13.5	14.9	12.4	14.5	16.4	20.1	22.1
YoY % Change	0%	3%	3%	5%	-2%	7%	16%	0%	7%	7%	10%	-17%	17%	13%	23%	10%
PC	346.1	311.8	304.7	272.1	256.2	261.3	260.2	267.5	306.6	353.6	293.8	253.2	261.6	288.5	250.2	241.5
YoY % Change	-4%	-10%	-2%	-11%	-6%	2%	0%	3%	15%	15%	-17%	-14%	3%	10%	-13%	-3%
Desktop	146.0	134.2	131.3	110.7	100.8	100.8	96.0	95.9	85.8	95.8	85.7	72.2	73.0	83.6	73.2	71.5
YoY % Change	-4%	-8%	-2%	-16%	-9%	0%	-5%	0%	-11%	12%	-11%	-16%	1%	15%	-12%	-2%
Notebook	200.1	177.5	173.4	161.4	155.4	160.5	164.1	171.6	220.8	257.7	208.1	181.0	188.6	204.9	176.9	170.0
YoY % Change	-4%	-11%	-2%	-7%	-4%	3%	2%	5%	29%	17%	-19%	-13%	4%	9%	-14%	-4%
Smartphone	726.7	1,018.7	1,301.5	1,437.6	1,469.5	1,465.3	1,402.5	1,372.6	1,281.2	1,359.8	1,206.2	1,165.5	1,237.5	1,264.0	1,094.0	1,132.6
YoY % Change	47%	40%	28%	10%	2%	0%	-4%	-2%	-7%	6%	-11%	-3%	6%	2%	-13%	4%
5G smartphone	0.0	0.0	0.0	0.0	0.0	0.0	0.0	16.1	255.3	559.5	636.8	703.4	814.6	893.1	829.6	901.3
YoY % Change	-	-	-	-	-	-	-	-	1485%	119%	14%	10%	16%	10%	-5%	9%
iPhone	135.9	153.5	192.7	231.5	215.4	215.8	204.9	199.3	203.4	235.8	226.7	235.6	234.3	251.7	252.8	267.0
YoY % Change	46%	13%	26%	20%	-7%	0%	-5%	-3%	2%	16%	-4%	4%	-1%	7%	0%	6%
HDD	578.0	551.9	564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2
YoY % Change	-7%	-5%	2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34%	-29%	1%	0%	5%	-4%
Enterprise	33.2	31.8	32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.6	5.8	4.4	3.2	2.7	1.9
NL	28.1	35.5	36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	76.4	78.6
3.5"ATA	216.2	203.2	202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	31.5	27.5
2.5"	298.7	280.7	291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	19.8	17.3
Global Light vehicle sales	75.0	78.2	81.4	83.5	92.3	94.4	93.9	89.9	77.2	80.3	79.0	86.7	88.7	91.1	91.4	93.2
YoY % Change	5%	4%	4%	3%	11%	2%	-1%	-4%	-14%	4%	-2%	10%	2%	3%	0%	2%
ICE				83.5	91.8	93.7	92.6	88.5	75.1	75.2	71.0	76.2	77.0	76.2	75.4	75.0
YoY % Change					10%	2%	-1%	-4%	-15%	0%	-6%	7%	1%	-1%	-1%	-1%
EV				-	0.5	0.7	1.3	1.4	2.1	5.1	8.0	10.5	11.7	14.9	16.0	18.2
YoY % Change						40%	86%	8%	50%	143%	57%	31%	11%	27%	7%	14%

Note: e = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

End-product Market and Major Components Market (Quarterly)

	CY2021			CY2022			CY2023			CY2024			CY2025			CY2026						
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q
Server	2.9	3.4	3.5	3.7	3.4	3.8	3.9	3.8	3.0	3.1	3.1	3.2	3.2	3.7	3.7	4.0	3.9	4.2	4.2	4.4	4.1	5.5
YoY % Change	9%	3%	8%	8%	14%	13%	12%	1%	-10%	-19%	-22%	-16%	7%	18%	20%	26%	22%	15%	15%	10%	3%	30%
Sequential % Change	-15%	15%	4%	7%	-10%	14%	3%	-4%	-20%	2%	0%	3%	2%	13%	1%	8%	-1%	7%	0%	4%	-7%	34%
PC	85.7	85.7	88.3	93.8	80.6	71.6	74.4	67.2	57.3	61.4	68.7	65.9	59.5	63.4	69.1	69.6	64.0	69.7	76.5	78.3	65.6	62.3
YoY % Change	58%	14%	5%	1%	-6%	-16%	-16%	-28%	-29%	-14%	-8%	-2%	4%	3%	1%	6%	8%	10%	11%	12%	3%	-11%
Sequential % Change	-8%	0%	3%	6%	-14%	-11%	4%	-10%	-15%	7%	12%	-4%	-10%	7%	9%	1%	-8%	9%	10%	2%	-16%	-5%
Desktop	22.3	23.6	24.3	25.6	22.7	21.5	21.8	19.7	17.5	17.4	18.6	18.7	17.3	17.3	18.8	19.6	18.5	19.5	22.3	23.4	18.2	17.5
YoY % Change	14%	19%	10%	5%	2%	-9%	-10%	-23%	-23%	-19%	-15%	-5%	-1%	0%	1%	5%	7%	12%	19%	19%	-2%	-10%
Sequential % Change	-9%	6%	3%	5%	-11%	-5%	2%	-10%	-11%	-1%	7%	0%	-7%	0%	8%	4%	-6%	5%	15%	5%	-22%	-4%
Notebook (incl. Netbook)	63.5	62.1	64.0	68.1	57.9	50.1	52.6	47.5	39.8	44.0	50.0	47.2	42.2	46.0	50.3	50.0	45.5	50.3	54.2	54.9	47.4	44.9
YoY % Change	83%	12%	3%	0%	-9%	-19%	-18%	-30%	-31%	-12%	-5%	-1%	6%	5%	1%	6%	8%	9%	8%	10%	4%	-11%
Sequential % Change	-7%	-2%	3%	6%	-15%	-13%	5%	-10%	-16%	11%	14%	-6%	-11%	9%	9%	-1%	-9%	10%	8%	1%	-14%	-5%
Smartphone	344.7	313.4	334.2	367.6	314.5	287.6	303.2	300.6	268.4	267.9	303.7	324.0	300.4	292.3	314.7	328.8	302.6	302.6	302.6	302.6	302.6	302.6
YoY % Change	25%	13%	-6%	-2%	-9%	-8%	-9%	-18%	-15%	-7%	0%	8%	12%	9%	4%	1%	1%	4%	-4%	-8%	0%	0%
Sequential % Change	-8%	-9%	7%	10%	-14%	-9%	5%	-1%	-11%	0%	13%	7%	-7%	-3%	8%	4%	-8%	0%	0%	0%	0%	0%
5G smartphone	125.1	118.1	136.5	179.8	155.5	146.3	161.6	173.4	160.5	149.3	182.6	211.1	187.9	186.0	211.4	229.3	213.6	203.9	228.0	247.6	219.0	198.7
YoY % Change	491%	139%	110%	50%	24%	24%	18%	-4%	3%	2%	13%	22%	17%	25%	16%	9%	14%	10%	8%	8%	3%	-3%
Sequential % Change	4%	-6%	16%	32%	-14%	-6%	10%	7%	-7%	-7%	22%	16%	-11%	-1%	14%	8%	-7%	-5%	12%	9%	-12%	-9%
iPhone	55.3	44.4	51.1	85.0	56.5	45.4	52.3	72.5	55.8	44.8	54.4	80.6	53.0	46.1	57.4	77.9	59.1	48.4	59.6	84.7	61.8	57.5
YoY % Change	50%	18%	23%	-3%	2%	2%	2%	-15%	-1%	-1%	4%	11%	-5%	3%	6%	-3%	12%	5%	4%	9%	4%	19%
Sequential % Change	-37%	-20%	15%	66%	-34%	-20%	15%	39%	-23%	-20%	21%	48%	-34%	-13%	25%	36%	-24%	-18%	23%	42%	-27%	-7%
HDD	64.1	67.6	66.3	61.2	52.8	44.6	38.5	36.3	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0	33.0
YoY % Change	-5%	15%	4%	-13%	-18%	-34%	-42%	-41%	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%
Sequential % Change	-9%	5%	-2%	-8%	-14%	-15%	-14%	-6%	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%
Enterprise	3.2	3.5	3.5	3.3	2.7	2.4	1.6	1.7	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8	0.6
NL	16.0	19.3	19.7	18.5	18.7	18.0	14.2	10.6	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.6	16.5	17.3	19.2	18.7	19.1
3.5"ATA	22.2	23.6	23.2	20.8	16.5	14.1	13.0	12.8	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3	8.7
2.5"	22.7	21.2	19.9	18.7	14.9	10.0	9.7	11.3	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3	4.7
HDD/PC	0.7	0.8	0.8	0.7	0.7	0.6	0.5	0.5	0.6	0.5	0.4	0.4	0.5	0.5	0.5	0.5	0.4	0.4	0.4	0.4	0.5	0.5

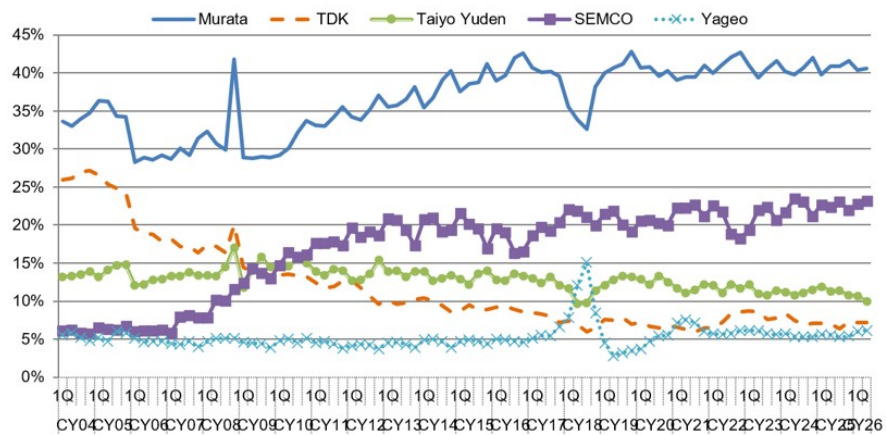
Source: Company data, Morgan Stanley Research

MLCC Market Data

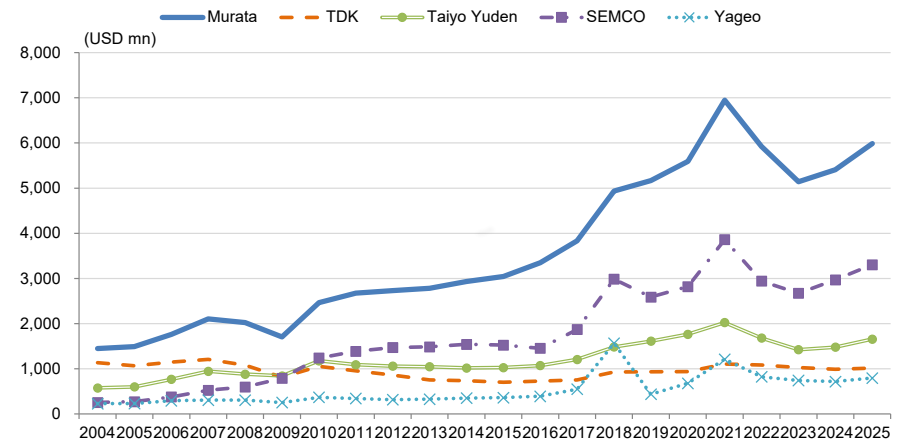
■ MLCC market forecasts

Base Case	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e	2029e	2030e	2031e
Total worldwide shipment quantity (Bil pcs)	1,504	1,480	1,640	2,050	2,029	2,167	2,293	2,605	2,735	3,080	3,480	3,828	3,254	3,695	4,446	3,572	3,187	3,386	3,773	4,037	4,360	4,709	5,086	5,493	5,932
YoY (%)	14.9%	-1.6%	10.8%	24.9%	-1.0%	6.8%	5.8%	13.6%	5.0%	12.6%	13.0%	10.0%	-15.0%	13.5%	20.3%	-19.7%	-10.8%	6.2%	11.4%	7.0%	8.0%	8.0%	8.0%	8.0%	8.0%
CAGR (1991-)	14.9%	13.8%	13.7%	14.2%	13.4%	13.1%	12.8%	12.8%	12.5%	12.5%	12.5%	12.4%	11.3%	11.4%	11.6%	10.5%	9.7%	9.6%	9.7%	9.6%	9.6%	9.5%	9.5%	9.4%	9.4%
ASP in worldwide shipment (Cent)	0.47	0.42	0.36	0.38	0.39	0.36	0.33	0.30	0.29	0.26	0.27	0.37	0.39	0.38	0.39	0.40	0.40	0.39	0.39	0.46	0.51	0.55	0.57	0.59	0.60
YoY (%)	-0.2%	-10.1%	-14.0%	6.8%	1.2%	-7.7%	-7.2%	-10.6%	-4.1%	-7.2%	3.8%	34.3%	4.6%	-2.8%	4.7%	2.0%	-1.0%	-1.0%	-1.0%	17.7%	11.0%	7.5%	4.0%	3.0%	3.0%
CAGR (1991-)	-8.6%	-8.7%	-9.0%	-8.2%	-7.8%	-7.8%	-7.7%	-7.9%	-7.7%	-7.7%	-7.3%	-6.0%	-5.6%	-5.5%	-5.2%	-5.0%	-4.9%	-4.8%	-4.6%	-4.1%	-3.7%	-3.4%	-3.2%	-3.0%	-2.9%
Total worldwide shipment value (Bil \$)	7.00	6.19	5.90	7.87	7.89	7.77	7.63	7.75	7.80	8.15	9.56	14.13	12.56	13.86	17.47	14.32	12.64	13.30	14.67	18.48	22.16	25.73	28.89	32.14	35.76
YoY (%)	14.6%	-11.6%	-4.7%	33.4%	0.2%	-1.5%	-1.8%	1.6%	0.7%	4.5%	17.3%	47.7%	-11.1%	10.3%	26.0%	-18.1%	-11.7%	5.2%	10.3%	26.0%	19.9%	16.1%	12.3%	11.2%	11.2%
CAGR (1991-)	5.0%	4.0%	3.5%	4.8%	4.6%	4.3%	4.0%	3.9%	3.8%	3.8%	4.3%	5.7%	5.0%	5.2%	5.8%	5.0%	4.4%	4.4%	4.6%	5.1%	5.5%	5.8%	6.0%	6.1%	6.2%

■ Global share of the top 5 ceramic capacitor firms



■ MLCC sales



Note: e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Capacitor Market: 016008 MLCCs are 75% Smaller than 0201 MLCCs

■ MLCC size comparison

	4532	3216	2012	1608	1005	0603	0402	0201	016008
Length	4.5mm	3.2mm	2.0mm	1.6mm	1.0mm	0.6mm	0.4mm	0.25mm	0.16mm
Width	3.2mm	1.6mm	1.25mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Thickness	2.0mm	0.85mm	0.6mm	0.8mm	0.5mm	0.3mm	0.2mm	0.125mm	0.08mm
Volume	28.8mm ³	4.35mm ³	1.5mm ³	1.024mm ³	0.25mm ³	0.054mm ³	0.016mm ³	0.004mm ³	0.001mm ³
Volume ratio	7,373	1,114	384	262	64.0	13.8	4.1	1.0	0.26

■ Number of MLCCs per vehicle

(units: pcs)	GAS	HV	BEV	PHEV
Power Train	300-500	1,000-1,600	2,000-2,500	2,500-3,000
ADAS(Lv2-5)	1,500-5,000			
Safety	300-1,000			
Infotainment	500-2,500			
Other	500-2,500			

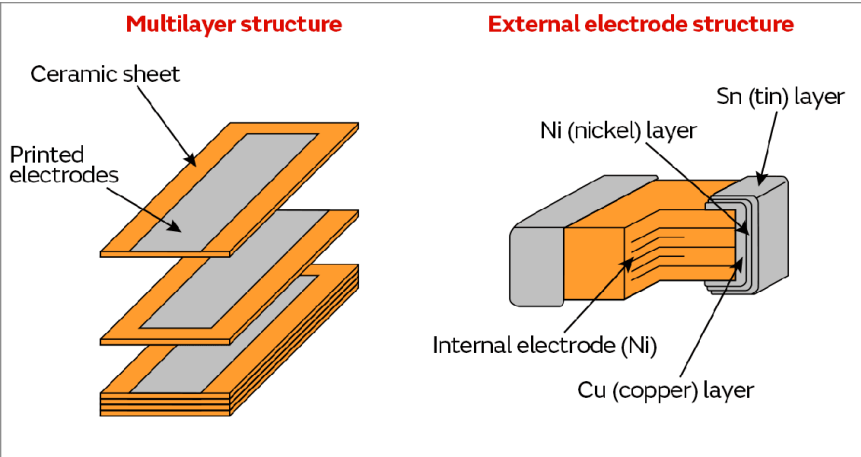
■ MLCC maximum capacitance by size

<=1000	220						
<=100		100	100	47			
<=10					10		
<=1						1	0.1
(mm)	3216	2012	1608	1005	0603	0402	0201
Murata Mfg	○	○	○	○	○	○	○

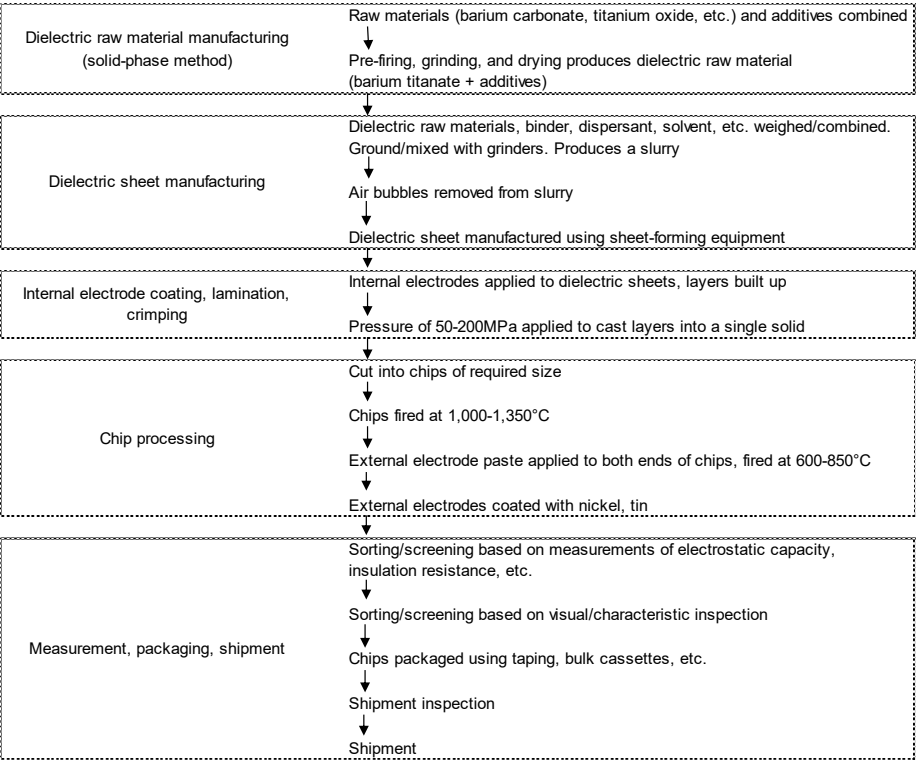
Source: Company data, Morgan Stanley Research

MLCC Basics

MLCC composition



MLCC production process

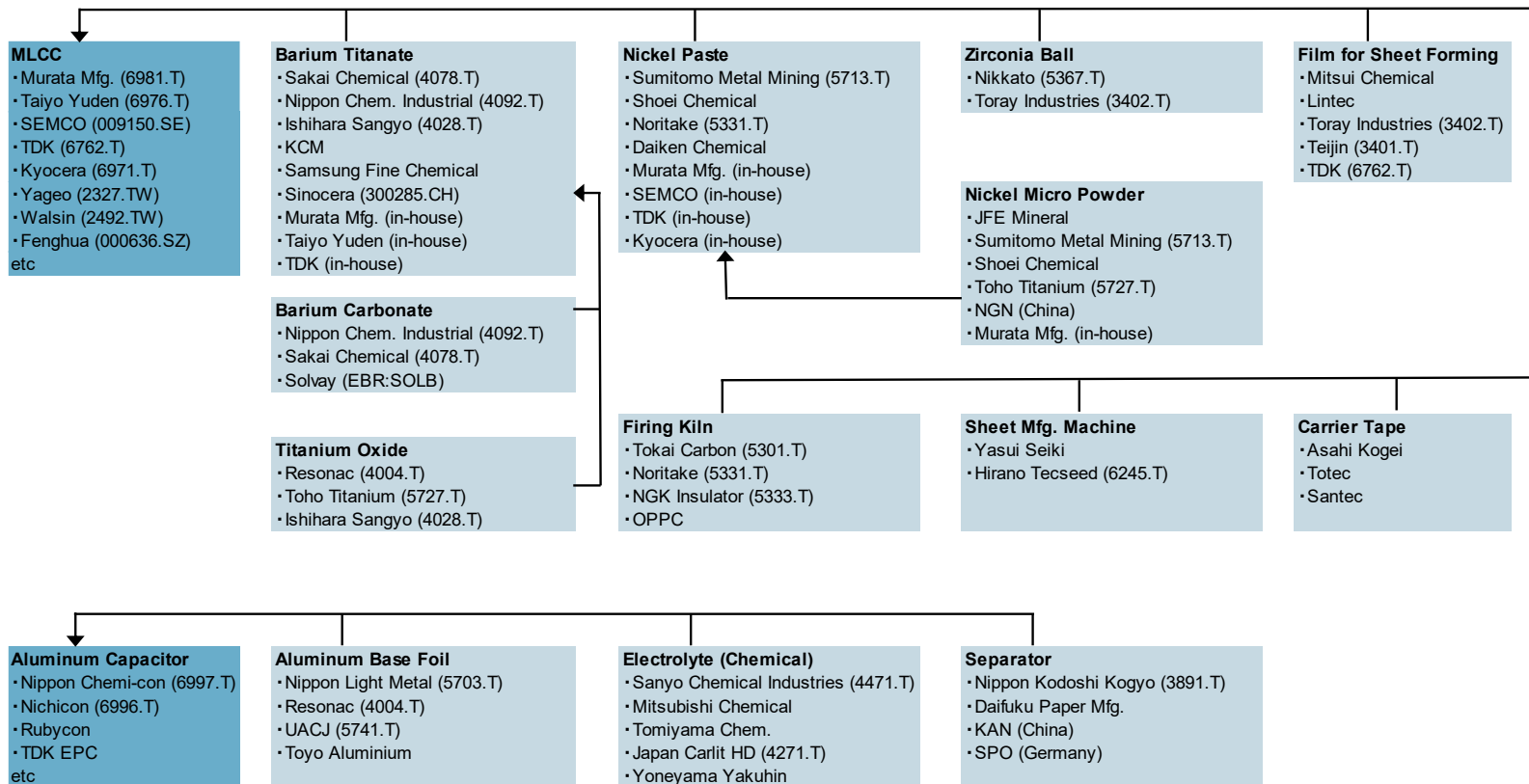


Source: Murata Mfg., Morgan Stanley Research

MLCC Sales, Market Share, OP

	CY23				CY24				CY25				CY26			2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q																		
Total (USD mn)	2,940	3,131	3,354	3,221	3,171	3,276	3,518	3,334	3,321	3,682	3,937	3,731	3,825	4,367	5,917	7,868	7,885	7,769	7,626	7,731	7,802	8,167	9,564	14,129	12,562	13,861	17,311	14,316	12,645	13,299	14,672	
Major 5	2,567	2,724	2,918	2,802	2,759	2,850	3,061	2,900	2,889	3,203	3,426	3,246	3,328	3,799	4,411	6,308	6,462	6,444	6,405	6,586	6,660	7,000	8,215	11,903	10,744	11,798	15,013	12,467	11,010	11,571	12,764	
Murata	1,205	1,233	1,362	1,340	1,273	1,305	1,432	1,399	1,320	1,504	1,611	1,553	1,545	1,773	1,708	2,467	2,677	2,730	2,784	2,933	3,043	3,349	3,834	4,934	5,168	5,591	6,945	5,921	5,140	5,409	5,988	
TDK	256	268	256	251	265	247	241	236	234	256	253	271	277	314	816	1,053	957	864	757	738	701	730	753	931	934	939	1,105	1,095	1,031	988	1,015	
Taiyo Yuden	357	343	359	367	354	354	390	383	395	416	447	401	408	433	843	1,177	1,093	1,061	1,047	1,020	1,028	1,073	1,209	1,484	1,613	1,767	2,026	1,683	1,425	1,481	1,659	
SEMCO	569	689	750	664	685	770	811	705	754	823	907	821	869	1,010	791	1,241	1,389	1,471	1,488	1,542	1,525	1,455	1,871	2,987	2,590	2,820	3,860	2,943	2,672	2,972	3,305	
Yageo	180	191	190	180	181	175	187	177	185	204	207	200	229	269	254	370	345	318	329	353	363	393	549	1,567	439	681	1,077	825	742	720	797	
Other	373	407	436	419	412	426	457	433	432	479	512	485	497	568	1,505	1,548	1,423	1,326	1,222	1,145	1,142	1,162	1,438	2,400	1,815	2,063	2,298	1,848	1,635	1,729	1,907	
Market Share																																
Major 5	87.3%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	74.6%	80.2%	81.9%	82.9%	84.0%	85.2%	85.4%	85.7%	85.9%	84.2%	85.5%	85.1%	86.7%	87.1%	87.1%	87.0%	87.0%	
Murata	41.0%	39.4%	40.6%	41.7%	40.2%	39.8%	40.7%	42.0%	39.8%	40.8%	40.9%	41.6%	40.4%	40.6%	28.9%	31.4%	33.9%	35.1%	36.5%	37.9%	39.0%	41.0%	40.1%	34.9%	41.1%	40.3%	40.1%	41.4%	40.6%	40.7%	40.8%	
TDK	8.7%	8.6%	7.6%	7.8%	8.4%	7.5%	6.8%	7.1%	7.1%	7.0%	6.4%	7.3%	7.2%	7.2%	13.8%	13.4%	12.1%	11.1%	9.9%	9.5%	9.0%	8.9%	7.9%	6.6%	7.4%	6.8%	6.4%	7.6%	8.1%	7.4%	6.9%	
Taiyo Yuden	12.1%	10.9%	10.7%	11.4%	11.2%	10.8%	11.1%	11.5%	11.9%	11.3%	11.4%	10.7%	10.7%	9.9%	14.2%	15.0%	13.9%	13.7%	13.7%	13.2%	13.2%	13.1%	12.6%	10.5%	12.8%	12.7%	11.7%	11.8%	11.3%	11.1%	11.3%	
SEMCO	19.4%	22.0%	22.4%	20.7%	21.6%	23.5%	23.1%	21.2%	22.7%	22.3%	23.0%	22.0%	22.7%	23.1%	13.4%	15.8%	17.6%	18.9%	19.5%	19.9%	19.5%	17.8%	19.6%	21.1%	20.6%	20.3%	22.3%	20.6%	21.1%	22.3%	22.5%	
Yageo	6.1%	6.1%	5.7%	5.3%	5.7%	5.3%	5.3%	5.3%	5.3%	5.5%	5.3%	5.4%	6.0%	6.2%	4.3%	4.7%	4.4%	4.1%	4.3%	4.6%	4.7%	4.8%	5.7%	11.1%	3.5%	4.9%	6.2%	5.8%	5.9%	5.4%	5.4%	
Other	12.7%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	13.0%	25.4%	19.7%	18.1%	17.1%	16.0%	14.8%	14.6%	14.2%	15.0%	17.0%	14.4%	14.9%	13.3%	12.9%	12.9%	13.0%	13.0%	
Market Share in Major 5																																
Murata	47.0%	45.3%	46.7%	47.8%	46.2%	45.8%	46.8%	48.2%	45.7%	46.9%	47.0%	47.8%	46.4%	46.7%	38.7%	39.1%	41.4%	42.4%	43.5%	44.5%	45.7%	47.8%	46.7%	41.5%	48.1%	47.4%	46.3%	47.5%	46.7%	46.8%	46.9%	
TDK	10.0%	9.9%	8.8%	9.0%	9.6%	8.7%	7.9%	8.1%	8.1%	8.0%	7.4%	8.4%	8.3%	8.3%	18.5%	16.7%	14.8%	13.4%	11.8%	11.2%	10.5%	10.4%	9.2%	7.8%	8.7%	8.0%	7.4%	8.8%	9.4%	8.5%	7.9%	
Taiyo Yuden	13.9%	12.6%	12.3%	13.1%	12.8%	12.4%	12.7%	13.2%	13.7%	13.0%	13.1%	12.3%	12.3%	11.4%	19.1%	18.7%	16.9%	16.5%	16.4%	15.5%	15.4%	15.3%	14.7%	12.5%	15.0%	15.0%	13.5%	13.5%	12.9%	12.8%	13.0%	
SEMCO	22.2%	25.3%	25.7%	23.7%	24.8%	27.0%	26.5%	24.3%	26.1%	25.7%	26.5%	25.3%	26.1%	26.6%	17.9%	19.7%	21.5%	22.8%	23.2%	23.4%	22.9%	20.8%	22.8%	25.1%	24.1%	23.9%	25.7%	23.6%	24.3%	25.7%	25.9%	
Yageo	7.0%	7.0%	6.5%	6.4%	6.6%	6.1%	6.1%	6.1%	6.4%	6.4%	6.1%	6.2%	6.9%	7.1%	5.7%	5.9%	5.3%	4.9%	5.1%	5.4%	5.5%	5.6%	6.7%	13.2%	4.1%	5.8%	7.2%	6.6%	6.7%	6.2%	6.2%	
YoY Growth	-29.4%	-17.1%	0.3%	6.2%	7.9%	4.7%	6.1%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%	-28.9%	-17.2%	-0.1%	5.9%	7.5%	4.7%	4.9%	11.9%	15.2%	18.6%	18.5%	16.7%	14.8%	13.4%	11.8%	11.2%	10.5%	10.4%
Major 5	-27.6%	-20.8%	-3.0%	3.6%	5.5%	5.9%	5.1%	4.4%	3.7%	15.2%	12.5%	11.0%	17.0%	17.9%	-5.6%	-3.1%	-9.9%	-4.9%	3.7%	-8.1%	-5.8%	-6.0%	-11.5%	4.0%	4.9%	14.9%	18.1%	22.5%				
TDK	-28.9%	-18.5%	-11.9%	3.8%	-0.7%	3.3%	8.6%	4.5%	11.6%	17.6%	14.6%	4.6%	3.2%	4.0%	-28.9%	-18.5%	-11.9%	3.8%	-0.7%	3.3%	8.6%	4.5%	11.6%	17.6%	14.6%	4.6%	3.2%	4.0%				
Taiyo Yuden	-39.3%	-16.3%	19.0%	20.3%	20.4%	11.9%	8.1%	6.2%	10.0%	6.8%	11.9%	16.4%	15.3%	22.8%	-39.3%	-16.3%	19.0%	20.3%	20.4%	11.9%	8.1%	6.2%	10.0%	6.8%	11.9%	16.4%	15.3%	22.8%				
SEMCO	-23.9%	-9.9%	-1.2%	-1.9%	0.7%	-8.7%	-1.8%	-1.8%	2.2%	16.7%	11.2%	13.1%	23.7%	31.7%	-23.9%	-9.9%	-1.2%	-1.9%	0.7%	-8.7%	-1.8%	-1.8%	2.2%	16.7%	11.2%	13.1%	23.7%	31.7%				
Yageo	-32.0%	-16.8%	2.7%	8.7%	10.4%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%	-32.0%	-16.8%	2.7%	8.7%	10.4%	4.7%	4.9%	3.5%	4.7%	12.4%	11.9%	11.9%	15.2%	18.6%				
Other	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
QoQ Growth	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	6.5%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
Major 5	-6.8%	2.3%	10.5%	-1.7%	-5.0%	2.5%	9.8%	-2.3%	-5.6%	13.9%	7.2%	-3.6%	-0.5%	14.7%	-6.8%	2.3%	10.5%	-1.7%	-5.0%	2.5%	9.8%	-2.3%	-5.6%	13.9%	7.2%	-3.6%	-0.5%	14.7%				
TDK	-3.2%	5.0%	-4.7%	-1.7%	5.5%	-6.9%	-2.4%	-1.9%	-0.7%	9.4%	-1.5%	7.4%	2.1%	14.7%	-3.2%	5.0%	-4.7%	-1.7%	5.5%	-6.9%	-2.4%	-1.9%	-0.7%	9.4%	-1.5%	7.4%	2.1%	14.7%				
Taiyo Yuden	1.1%	-3.9%	4.8%	2.0%	-3.4%	0.0%	10.1%	-1.8%	3.2%	5.4%	7.3%	-10.4%	1.8%	6.2%	1.1%	-3.9%	4.8%	2.0%	-3.4%	0.0%	10.1%	-1.8%	3.2%	5.4%	7.3%	-10.4%	1.8%	6.2%				
SEMCO	3.1%	21.0%	9.0%	-11.5%	3.2%	12.4%	5.3%	-13.0%	6.9%	9.2%	10.3%	-9.5%	5.8%	16.3%	3.1%	21.0%	9.0%	-11.5%	3.2%	12.4%	5.3%	-13.0%	6.9%	9.2%	10.3%	-9.5%	5.8%	16.3%				
Yageo	-2.1%	6.3%	-0.8%	-5.1%	0.5%	-3.6%	6.7%	-5.0%	4.6%	10.0%	1.7%	-3.4%	14.4%	17.2%	-2.1%	6.3%	-0.8%	-5.1%	0.5%	-3.6%	6.7%	-5.0%	4.6%	10.0%	1.7%	-3.4%	14.4%	17.2%				
Other	-3.0%	9.0%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%	-3.0%	9.0%	7.1%	-4.0%	-1.5%	3.3%	7.4%	-5.2%	-0.4%	10.9%	6.9%	-5.2%	2.5%	14.3%				
OP																																
Murata	35.8	42.8	52.0	50.8	47.4	63.7	64.8	58.8	47.9	61.2	73.0	66.4	70.1	98.6	-8.8	31.9	22.3	19.3	50.6	72.2	115.6	73.3	105.8	176.1	190.5	196.6	312.1	270.5	181.4	234.7	248.4	
TDK	7.9	9.7	8.2	10.7	8.8	8.1	6.4	4.7	2.4	1.8	3.0	6.6	6.0	7.8	-20.3	-3.9	-9.4	-15.4	-0.2	3.4	5.9	3.3	7.4	19.0	20.4	16.1	27.6	43.3	36.5	27.9	13.8	
Taiyo Yuden	-2.9	0.0	0.8	2.0	1.3	1.6	2.4	0.5	2.9	3.5	4.5	5.8	3.1	3.8	-0.2	14.7	-0.6	0.4	9.1	6.7	15.9	6.9	15.4	33.0	41.8	37.5	55.4	43.4	-0.1	5.8	16.7	
SEMCO	7.5	12.3	10.8	7.9	8.9	14.5	17.0	14.0	11.5	14.3	18.1	15.2	22.6	28.8	12.2	19.1	12.3	8.6	11.1	8.1	20.5	11.9	26.1	46.1	44.5	48.7	96.4	58.9	38.5	54.4	59.0	
- SEMCO (KRW bn)	72.6	117.8	98.6	70.2	80.1	127.9	154.2	128.0	109.5	138.3	169.7	142.5	211.0	271.6	160.2	252.5	168.8	146.9	126.3	78.0	190.4	126.7	261.2	459.7	470.3	536.0	1,005.3	585.8	359.2	490.2	560.0	
OP (USD mn)																																
- Murata	270.8	311.1	359.4	343.5	319.5	408.5	434.0	385.9	313.7	423.2	495.1	431.1	446.6	618.4	-90.8	367.2	277.5	241.4	515.0	672.0	953.7	680.8	942.8	1,587.4	1,746.							

Supply Chains of MLCC and Aluminium Capacitors



Source: Company data, Morgan Stanley Research

MLCC Coverages: Medium-term Results and Forecasts

■ Murata Mfg. (6981)

(bn JPY, JPY/share)	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	1,811.2	1,686.8	1,640.2	1,743.4	1,830.9	2,044.2	2,449.7	2,830.5	3,292.2	3,860.7
Components	984.3	911.2	933.8	1,033.1	1,159.7	1,376.3	1,660.1	1,882.6	2,137.3	2,429.1
Capacitors	788.5	738.8	753.5	831.8	936.4	1,143.9	1,418.4	1,631.2	1,875.9	2,157.3
Inductors and EMI Filters	195.8	172.3	180.3	201.3	223.3	232.4	241.7	251.4	261.4	271.9
Devices/Modules	815.0	761.0	695.2	697.2	656.0	652.9	776.6	934.9	1,141.9	1,418.5
High Frequency Devices/Communications Module	528.2	453.6	440.1	443.6	394.8	377.9	453.5	544.2	653.0	783.6
Battery/Power supply	180.4	214.6	164.4	155.7	154.1	165.0	206.5	267.2	357.9	496.1
Functional Devices	106.4	92.8	90.7	97.8	107.1	110.0	116.6	123.6	131.0	138.9
Others	11.9	14.7	11.2	13.1	15.2	15.0	13.0	13.0	13.0	13.0
OP	424.1	297.9	215.4	279.7	281.8	431.9	624.0	754.3	889.1	1,068.9
Components	355.5	280.1	234.2	275.2	315.1	426.4	579.2	685.2	810.6	958.1
Devices/Modules	69.7	20.6	-13.0	10.0	-26.5	11.5	50.7	75.0	84.5	116.8
Others	-1.2	-2.8	-5.7	-5.4	-6.8	-6.0	-6.0	-6.0	-6.0	-6.0
OPM	23.4%	17.7%	13.1%	16.0%	15.4%	21.1%	25.5%	26.6%	27.0%	27.7%
ROE	15.0%	10.9%	7.3%	9.1%	8.8%	12.2%	16.7%	18.5%	19.6%	20.8%
EPS	163.7	133.8	95.7	125.1	127.7	185.6	271.9	332.8	397.6	484.6
FCF	209.2	118.4	288.1	243.8	231.4	208.9	290.0	393.1	439.2	545.9

■ Taiyo Yuden (6976)

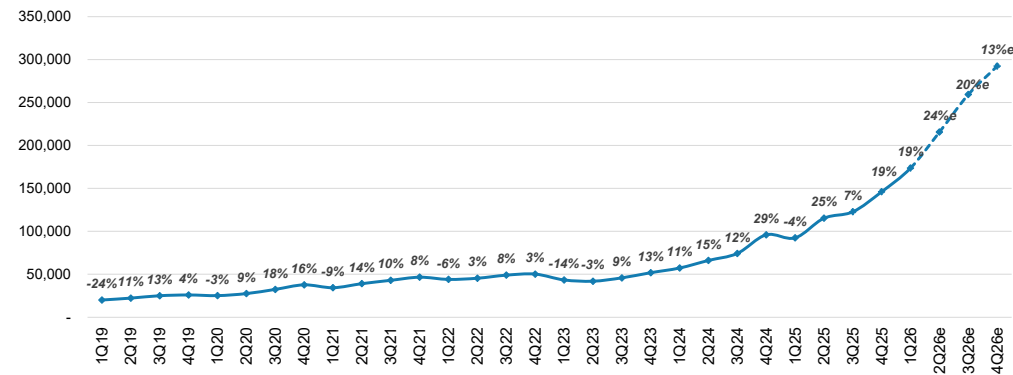
(bn JPY, JPY/share)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	282.3	300.9	349.6	319.5	322.6	341.4	355.3	411.3	475.1	523.8	573.9	624.8
Capacitor	176.5	195.2	230.4	208.1	205.8	232.1	251.8	304.8	362.7	406.2	450.9	496.0
Inductor	38.8	41.6	48.9	52.9	55.6	61.5	64.3	68.2	73.0	78.1	83.5	89.4
Complex devices	49.8	46.9	48.8	32.6	34.9	23.0	14.8	12.3	14.5	14.5	14.5	14.5
Others	17.3	17.2	21.5	25.9	26.3	24.8	24.5	26.0	25.0	25.0	25.0	25.0
OP	37.2	40.8	68.2	32.0	9.1	10.5	20.0	45.2	70.2	80.2	90.5	100.2
OPM	13.2%	13.5%	19.5%	10.0%	2.8%	3.1%	5.6%	11.0%	14.8%	15.3%	15.8%	16.0%
ROE	8.7%	12.6%	20.0%	7.5%	2.6%	0.7%	4.5%	8.7%	12.3%	12.9%	13.3%	13.4%
EPS	142.7	227.3	432.2	185.7	63.5	16.1	107.6	224.0	340.9	391.7	444.1	493.3
FCF	11.6	10.7	16.7	2.1	-31.7	-29.6	32.4	45.6	19.8	31.5	26.9	47.8

Note: MSe = Morgan Stanley Research estimates

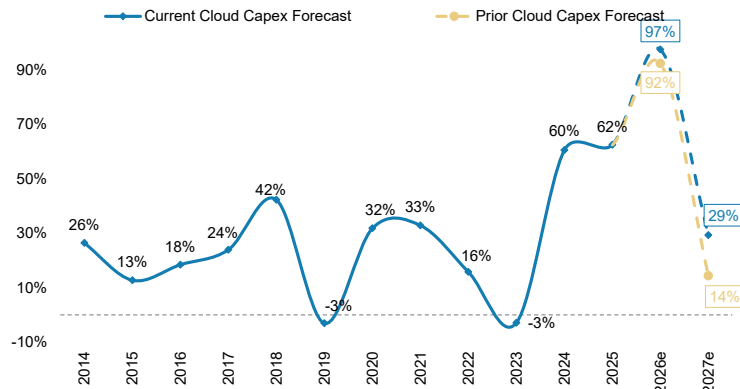
Source: Company data, Morgan Stanley Research

Capex by Hyperscalers Has Increased Significantly

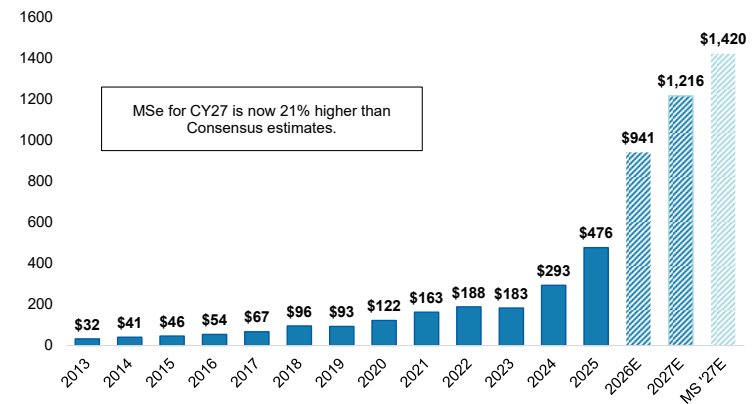
■ Total cash capex by 14 cloud suppliers and QoQ change



■ Total cash capex by 14 cloud suppliers and YoY change



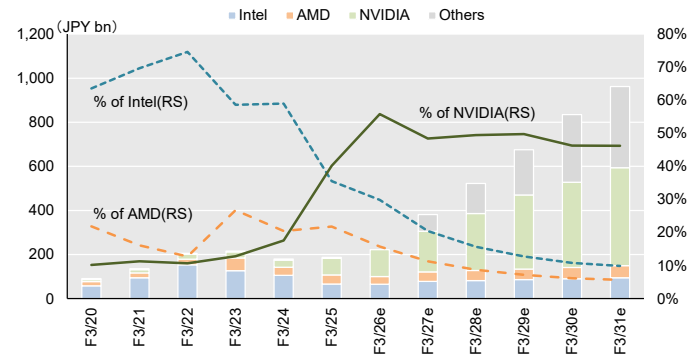
■ Total cash capex by 14 cloud suppliers



Note: E = FactSet consensus, MS'27E and e = Morgan Stanley Research estimates
 Source: Company data, FactSet, Morgan Stanley Research

We Expect Ibiden Will Maintain Nearly 100% Market Share in FC Packages for NVIDIA

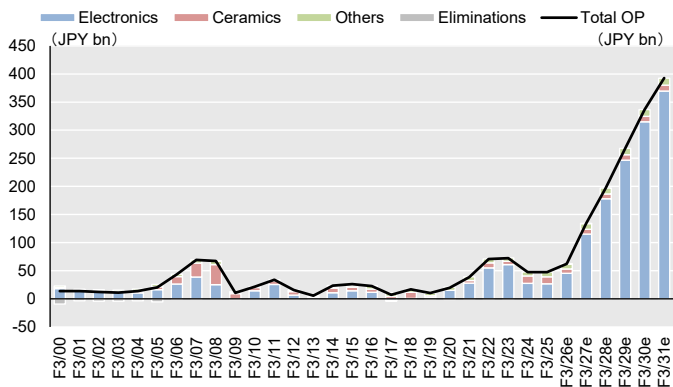
ABF package substrate sales by customers*



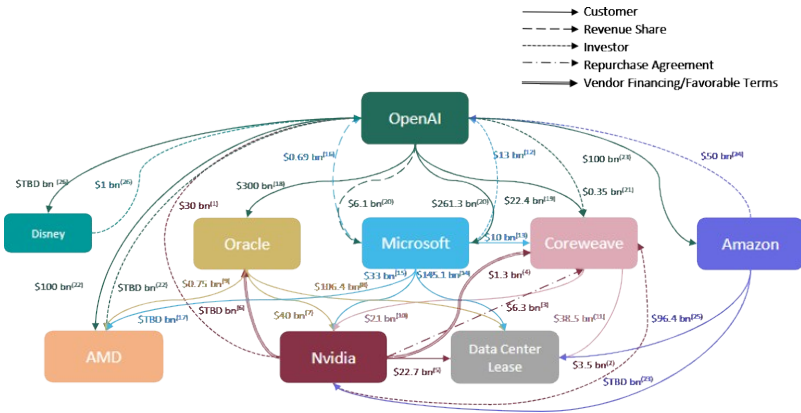
Size and # of layers for NVIDIA ABF packages

	Package Size	Surface Area	# of BU Layers	SAP Demand Size
Hopper	55 x 60mm	3,300	10	100
Blackwell	70 x 80mm	5,600	12	204
Rubin	80 x 100mm	8,000	16	388

OP by segment and total OP



AI ecosystem capital flows



Note: *Morgan Stanley Research assumption based on company data, e = Morgan Stanley Research estimates
Source: Company data, Morgan Stanley Research

Ibiden: Medium-term Results and Forecasts

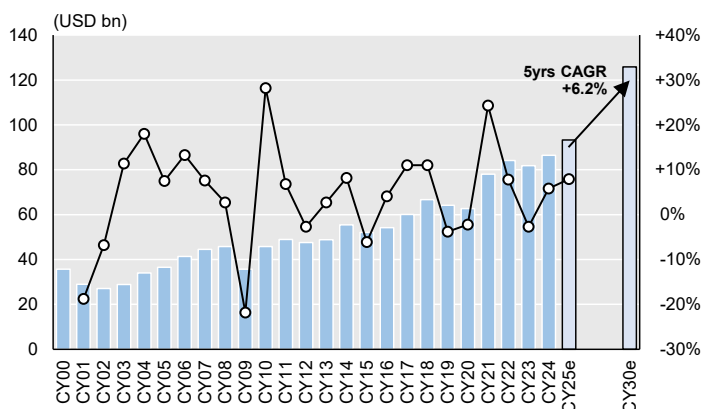
(bn JPY, JPY/share)	F3/07	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	CAGR F3/25- F3/31MSe
Sales	398.6	291.1	296.0	323.5	401.1	417.5	370.5	369.4	416.2	567.3	707.0	867.9	1,032.5	1,164.5	21.1%
Electronics	205.0	116.0	132.2	166.1	237.0	250.7	190.7	197.2	243.3	390.5	531.5	684.1	844.0	970.9	30.4%
Ceramics	96.2	102.5	88.4	87.4	90.7	89.9	96.5	84.1	82.6	86.0	81.5	87.0	88.8	90.8	1.3%
Others	97.4	72.7	75.4	70.0	73.5	76.9	83.3	88.1	90.3	90.8	94.0	96.8	99.7	102.7	2.6%
OP	69.0	10.1	19.7	38.6	70.8	72.4	47.6	47.6	62.0	133.8	197.1	267.6	336.6	392.5	42.1%
Electronics	38.6	2.5	14.9	27.8	55.1	60.6	27.3	26.8	45.2	115.0	177.9	246.8	314.8	369.6	54.8%
Ceramics	24.9	3.0	-1.0	4.6	8.7	6.1	13.4	12.2	7.6	8.7	8.5	9.5	10.0	10.5	-2.5%
Others	5.9	4.7	5.8	6.6	7.1	6.1	7.1	8.7	9.0	10.0	10.7	11.2	11.8	12.4	6.3%
Eliminations	-0.4	-0.0	-0.1	-0.4	-0.1	-0.6	-0.1	-0.2	0.2	0.0	0.0	0.0	0.0	0.0	
OPM	17.3%	3.5%	6.7%	11.9%	17.7%	17.3%	12.8%	12.9%	14.9%	23.6%	27.9%	30.8%	32.6%	33.7%	
ROE	19.0%	1.2%	4.2%	8.8%	12.1%	13.3%	6.9%	6.8%	12.2%	15.6%	19.8%	22.1%	22.5%	21.3%	
EPS	161.8	11.8	40.5	92.0	147.7	186.9	112.7	120.7	214.9	312.0	466.8	634.1	798.0	930.8	
CapEx	70.4	22.9	57.1	78.2	60.6	131.3	146.6	157.3	64.3	157.6	158.0	208.0	208.0	208.0	
Electronics	54.1	14.2	48.7	70.4	54.6	121.5	136.9	146.8	58.1	147.2	150.0	200.0	200.0	200.0	
Ceramics	9.3	5.4	6.1	3.1	3.0	3.1	5.2	7.1	1.8	1.9	3.0	3.0	3.0	3.0	
Others	6.9	3.4	2.3	4.7	3.0	6.6	4.5	3.4	4.5	8.5	5.0	5.0	5.0	5.0	
Depreciation	31.5	25.1	24.2	35.4	52.7	54.9	46.0	54.2	66.2	81.4	95.5	110.5	130.5	130.5	
Electronics	16.6	13.8	12.3	25.5	43.1	45.4	36.7	44.3	55.2	70.9	86.0	101.0	121.0	121.0	
Ceramics	12.5	9.6	8.4	6.8	6.1	5.6	5.0	5.8	6.2	5.4	5.5	5.5	5.5	5.5	
Others	2.4	3.0	3.5	3.1	3.5	3.9	4.3	4.1	4.8	5.1	4.0	4.0	4.0	4.0	
FCF	7.7	0.5	-12.0	-43.4	40.7	21.7	68.0	-45.3	54.0	3.7	49.1	59.6	127.8	174.1	

Note: MSe = Morgan Stanley Research estimates

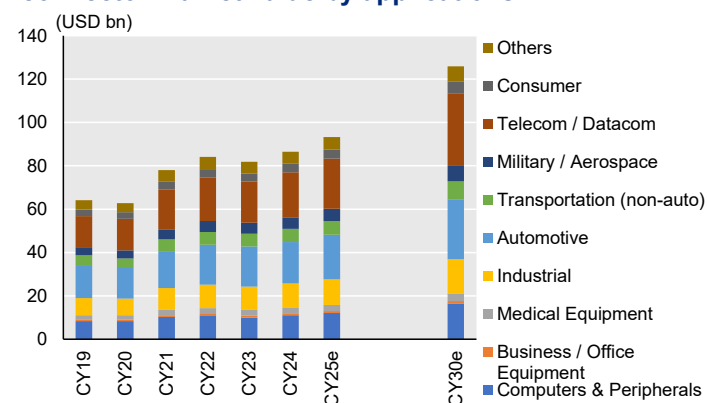
Source: Company data, Morgan Stanley Research

Connector Market

Global connector market value



Global connector market value by applications



Global connector market value CAGR by applications

	CY04-09	CY09-14	CY14-19	CY19-24	CY24-30e
Computers & Peripherals	3.0%	5.9%	-0.3%	5.7%	6.9%
Business / Office Equipment	4.1%	5.1%	-0.7%	3.1%	3.5%
Medical Equipment	8.2%	6.4%	2.6%	6.5%	4.4%
Industrial	-0.6%	9.0%	4.6%	7.0%	6.1%
Automotive	-5.7%	12.0%	4.6%	4.7%	6.4%
Transportation (non-auto)	1.4%	9.4%	7.4%	6.1%	5.4%
Military / Aerospace	4.4%	2.1%	3.4%	6.9%	5.9%
Telecom / Datacom	5.1%	10.2%	5.8%	7.9%	8.0%
Consumer	3.6%	6.4%	3.0%	4.6%	5.1%
Others	2.3%	15.5%	-6.9%	5.3%	4.3%
Market total	1.0%	9.2%	3.0%	6.1%	6.5%

Global connector sales by major suppliers

(USD bn)	CY13		(USD bn)	CY23	
	Sales	Share		Sales	Share
1 TE Connectivity	8.7	17.8%	1 TE Connectivity	12.7	15.6%
2 Amphenol	4.3	8.8%	2 Amphenol	7.0	8.5%
3 Molex	3.6	7.4%	3 Aptiv(Delphi)	4.7	5.8%
4 Aptiv(Delphi)	3.0	6.0%	4 Molex	4.6	5.7%
5 Foxconn	2.7	5.5%	5 Foxconn	2.5	3.1%
6 Yazaki	2.4	4.9%	6 Luxshare Precision	2.3	2.9%
7 JST	1.5	3.1%	7 Yazaki	2.2	2.7%
8 JAE	1.3	2.7%	8 JAE	1.7	2.1%
9 Sumitomo Wiring Systems	1.0	2.1%	9 Rosenberger	1.4	1.7%
10 Hirose Electric	1.0	2.0%	10 Hirose Electric	1.1	1.4%
Top 10	29.5	60.4%	Top 10	40.3	49.3%
Total	48.9	100.0%	Total	81.9	100.0%

Note: e = TTI estimates

Source: TTI, Company data, Morgan Stanley Research

Connector Coverage Companies: Medium-term Results and Forecasts

■ Hirose Electric (6806)

(JPY bn, JPY/share)	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28e MSe	F3/29e MSe	F3/30e MSe	F3/30 Ce	F3/31e MSe
Sales	133.5	163.7	183.2	165.5	189.4	211.3	234.6	250.0	254.6	275.4	298.2	290.0	323.4
General Industrial	40.4	56.4	69.1	50.5	46.9	62.0	73.9	83.0	80.6	87.8	95.7	84.1	104.3
Smartphone	31.7	33.9	34.0	35.5	46.4	44.3	43.0	44.5	44.2	45.6	46.9	89.9	48.3
Consumer/Mobile equipment	26.3	31.3	30.0	25.3	34.7	34.7	34.1	36.0	34.8	35.5	36.2	89.9	36.9
Automotive/Mobility related	23.8	29.2	36.0	42.9	49.2	54.3	60.0	63.0	67.2	75.3	84.3	89.9	94.5
Telecommunications infrastructure equipment	5.2	6.2	6.0	4.5	5.4	6.7	10.7	10.0	12.3	14.1	16.2	11.6	18.6
OA equipment and Others	6.1	6.7	8.1	6.8	6.8	9.3	13.0	13.5	15.5	17.1	18.8	14.5	20.7
OP	27.9	40.8	46.8	34.0	42.7	43.0	47.3	50.0	53.4	59.9	67.0	66.7	75.4
OPM	20.9%	24.9%	25.5%	20.6%	22.5%	20.4%	20.2%	20.0%	21.0%	21.8%	22.5%	23.0%	23.3%
ROE	6.3%	9.4%	10.0%	7.4%	9.0%	8.9%	9.6%	-	10.6%	11.5%	12.3%	12.0%	13.1%
EPS	549.1	885.4	1,002.1	772.4	976.3	991.9	1,114.9	1,069.6	1,280.5	1,453.9	1,648.3	-	1,885.3
FCF	15.1	33.4	52.1	12.9	12.7	36.6	22.1	-	29.5	33.3	38.6	-	41.8

■ JAE (6807)

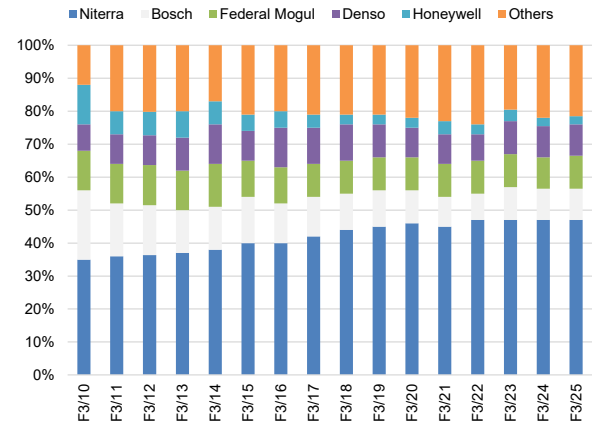
(JPY bn, JPY/share)	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28e MSe	F3/29e MSe	F3/29e Ce	F3/30 MSe	F3/31 MSe
Sales	209.7	225.1	235.9	225.8	221.6	227.9	235.1	240.0	236.9	245.1	260.0	248.0	251.3
Connector	195.8	209.0	215.9	204.9	201.8	206.9	212.1	215.0	212.9	220.1	233.0	221.8	224.0
Handhelds	99.7	88.9	86.8	75.5	66.9	65.2	61.8	-	59.1	56.5	-	54.2	52.0
Automotive	70.5	83.1	91.0	105.3	112.6	115.7	122.5	-	125.0	127.5	-	130.1	132.7
Industrial & Infrastructure	20.5	30.1	32.5	19.8	18.0	20.6	22.3	-	23.4	24.6	-	25.8	27.1
Aviation & Space (Defence)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	6.0	-	6.3	6.6
Others	5.1	6.9	5.5	4.4	4.3	5.4	5.5	-	5.5	5.5	-	5.5	5.5
Aerospace	13.3	15.3	19.1	20.1	19.3	20.5	22.5	24.5	23.5	24.5	27.0	25.7	26.8
Automotive	1.3	1.1	1.1	1.0	0.2	0.1	0.1	-	0.1	0.1	-	0.1	0.1
Industrial & Infrastructure	8.0	9.1	13.6	13.7	10.7	9.3	10.5	-	11.1	11.8	-	12.6	13.3
Aviation & Space (Defence)	4.0	5.0	4.4	5.4	8.4	11.1	11.9	-	12.3	12.6	-	13.0	13.4
Others	0.6	0.8	0.9	0.7	0.5	0.5	0.5	0.5	0.5	0.5	0.0	0.5	0.5
OP	8.7	18.0	17.6	14.4	15.6	8.9	8.9	9.5	9.6	10.1	18.0	10.6	11.0
Connector	12.3	20.8	18.1	15.8	18.0	12.0	11.5	-	12.0	12.3	-	12.7	12.9
Aerospace	0.3	1.6	4.1	3.6	2.6	2.2	2.7	-	2.9	3.1	-	3.3	3.4
Others	0.3	0.3	0.3	0.1	0.1	0.1	0.1	-	0.1	0.1	-	0.1	0.1
Adjustment	-4.2	-4.6	-4.8	-5.1	-5.0	-5.4	-5.4	-	-5.5	-5.5	-	-5.5	-5.5
OPM	4.2%	8.0%	7.4%	6.4%	7.0%	3.9%	3.8%	4.0%	4.1%	4.1%	6.9%	4.3%	4.4%
ROE	4.1%	9.6%	8.9%	8.2%	8.9%	5.1%	4.6%	-	4.7%	4.9%	8.0%	5.0%	5.0%
EPS	62.6	157.4	160.7	181.8	172.0	104.9	99.3	89.0	103.8	109.7	-	115.7	120.1
FCF	25.8	-5.8	-2.6	2.7	17.1	-7.4	7.2	-	8.3	7.2	-	8.4	8.3

Note: MSe = Morgan Stanley Research estimates, Ce = Company estimates

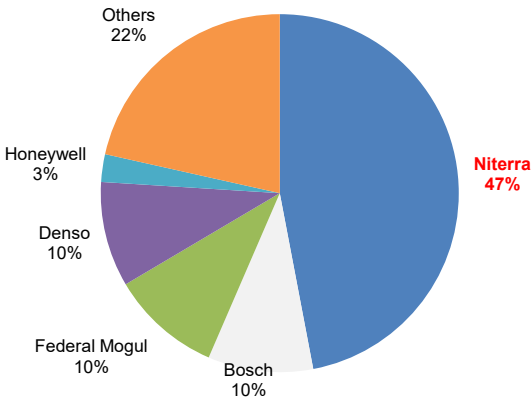
Source: Company data, Morgan Stanley Research

Global Market Share Trend and Split for Spark Plugs / Exhaust Gas Oxygen Sensors

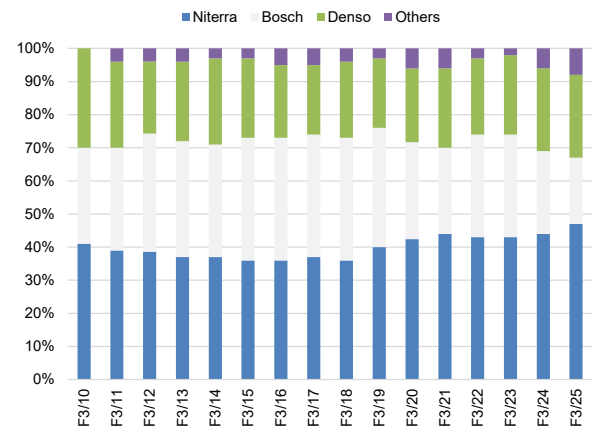
Spark plug global market share trend



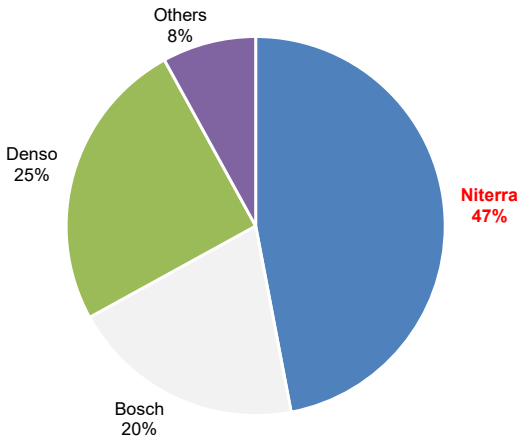
Spark plug global market share split in F3/25



Exhaust gas oxygen sensor global market share



Exhaust gas oxygen sensor global market share split in F3/25



Source: Company data, Morgan Stanley Research

Niterra: Medium-term Results and Forecasts

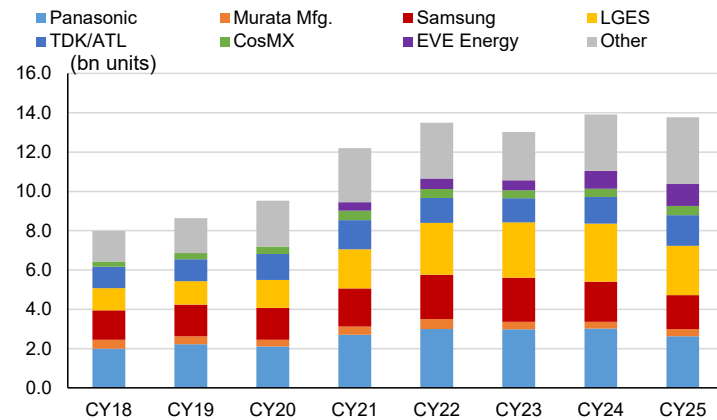
(bn JPY, JPY/share)	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	F3/22-F3/28MSe	CAGR
Sales	562.6	427.5	491.7	562.6	614.5	653.0	731.2	813.2	854.8	919.7	990.6	1,068.3		9.7%
Spark & Glow Plugs	203.5	205.3	250.0	285.1	335.3	362.6	390.7	426.2	458.4	501.9	549.9	602.6		10.6%
Automotive Sensors	140.9	133.4	137.8	158.9	170.0	180.2	196.8	222.4	226.9	233.7	240.7	247.9		8.7%
Component Solutions	36.3	80.0	95.5	110.8	96.5	103.1	130.7	154.6	163.6	178.1	194.1	211.7		9.4%
(Electrostatic Chuck)	18.5	22.9	30.0	40.4	33.2	45.0	52.2	66.0	72.6	80.6	89.5	99.3		15.9%
New Business	0.0	4.5	4.6	4.5	5.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0		-
Others/Elimination	4.1	4.4	3.9	3.3	7.4	7.1	13.0	10.0	6.0	6.0	6.0	6.0		7.5%
OP	48.4	47.4	75.5	89.2	107.6	129.7	138.2	158.2	174.4	200.8	227.4	254.4		15.0%
Spark & Glow Plugs	36.2	40.4	55.0	67.4	95.8	109.8	109.2	122.4	133.2	149.3	167.0	186.6		15.9%
Automotive Sensors	16.6	16.2	19.3	25.9	25.4	24.4	26.1	26.7	28.3	30.7	33.1	35.6		6.5%
Component Solutions	-1.1	-0.2	14.7	11.0	-0.6	-6.3	-4.6	9.0	12.9	20.4	26.8	31.7		
(Electrostatic Chuck)	-0.6	0.5	8.6	13.0	7.0	6.4	10.4	13.0	16.0	20.0	24.4	29.3		10.9%
New Business	0.0	-9.4	-13.6	-17.1	-13.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Others/Elimination	-0.6	0.4	0.1	2.0	0.2	1.8	7.4	0.0	0.0	0.4	0.4	0.4		
OPM	8.6%	11.1%	15.4%	15.9%	17.5%	19.9%	18.9%	19.5%	20.4%	21.8%	23.0%	23.8%		
ROE	8.5%	9.1%	12.5%	12.3%	13.8%	14.1%	15.7%	14.9%	15.4%	16.3%	16.9%	17.2%		
EPS	163.1	188.6	296.0	326.1	409.5	466.3	570.4	609.6	687.7	810.0	940.2	1,049.0		
FCF	6.9	20.9	82.1	31.9	26.0	98.7	-56.1	87.8	125.5	129.2	146.5	164.0		

Note: MSe = Morgan Stanley Research estimates

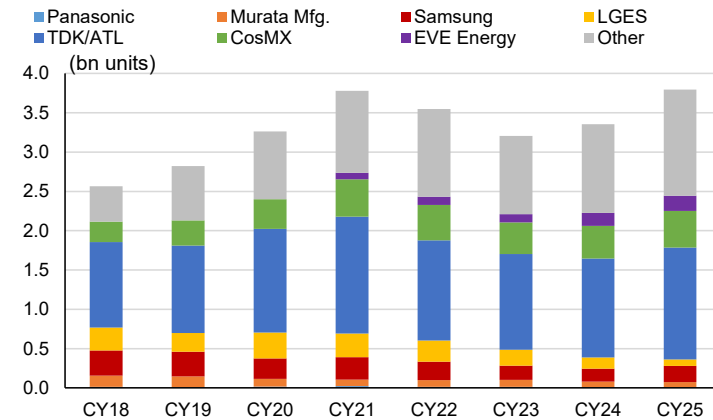
Source: Company data, Morgan Stanley Research

Small Size LiB Market Overview

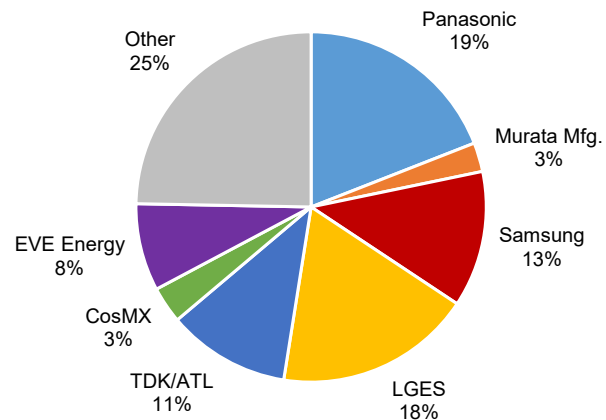
■ Small size LiB shipment volume by supplier



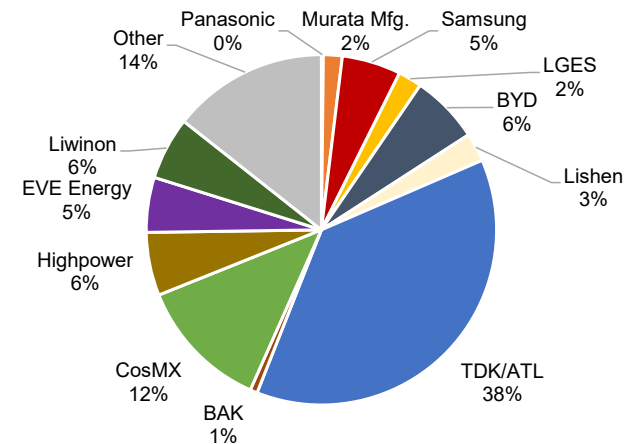
■ Laminated small size LiB shipment volume by supplier



■ Small size LiB market share by supplier (CY25)



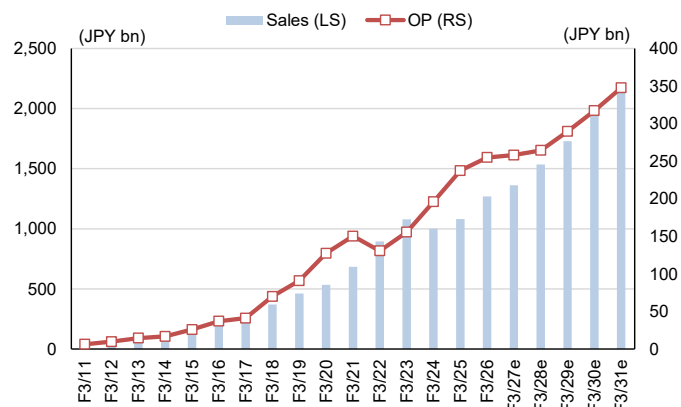
■ Laminated small size LiB market share by supplier (CY25)



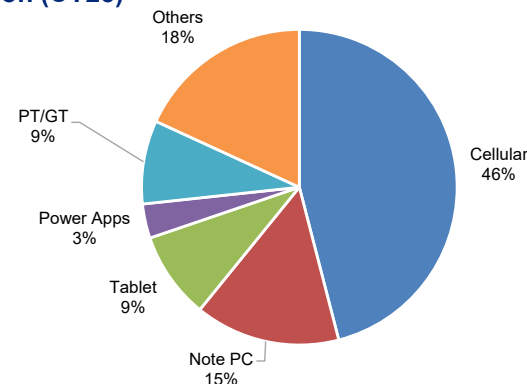
Source: TSR, Morgan Stanley Research

TDK: LiB Earnings Expansion, Major Acquisition and Transfer of Business

■ TDK: Sales and OP for rechargeable battery business



■ TDK/ATL: Small size battery shipment volume composition by application (CY25)



■ TDK: Major acquisition and transfer of business

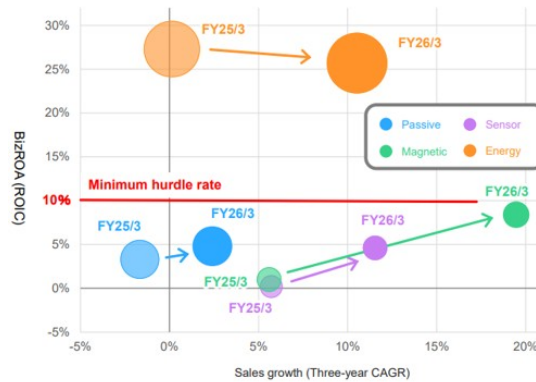
	Acquisition and Transfer of Business	Business	Acquisition/Sales value (JPY mn)
Aug-1986	Acquisition of SAE Magnetics	HDD Heads	unknown
May-1989	Acquisition of Silicon Systems	Mixed-signal Semiconductors	28,000
Jun-1996	Transfer of Silicon Systems to Texas Instruments	Mixed-signal Semiconductors	62,700
Mar-2000	Acquisition of Headway Technologies	HDD Heads	26,410
May-2005	Acquisition of Amperex Technology(ATL)	Lithium Polymer Batteries	8,700
Oct-2005	Acquisition of Lambda Power Group	Power	25,500
Aug-2007	Transfer of Recording Media business to Imation	Recording Media	31,400
Sep-2007	Acquisition of HDD head business from Alps	HDD Heads	34,400
Nov-2007	Acquisition of Magnecomp International	HDD Head Suspensions	20,900
Oct-2008	Acquisition of Epcos	Passive Components	170,600
Mar-2016	Acquisition of Micronas	MEMS Magnetic Sensors	22,500
Oct-2016	Acquisition of Hutchinson Technology	HDD Head Suspensions	14,262
Dec-2016	Acquisition of Tronics	MEMS Inertial Sensors	4,107
Feb-2017	Transfer of Epcos High Frequency Components business to Qualcomm	High Frequency Components	339,000
Mar-2017	Acquisition of ICsense	Gyroscope Sensors	2,396
May-2017	Acquisition of InvenSense	MEMS Inertial Sensors	142,758
Feb-2018	Acquisition of Chirp Microsystems	Ultrasonic 3D Sensors	3,015
May-2018	Acquisition of Faraday Semi LLC	Power Management IC	unknown
Apr-2025	Transfer of Micro Actuator Solution Business to QT Investment	Micro Actuator	unknown
Jun-2025	Acquisition of SoftEye	Smart glasses	unknown
Jun-2025	Acquisition of RF power business from QEI Corporation	RF power business	unknown
Jun-2026	Acquisition of Linergy Power	Mid-sized LiB	38,300
Jun-2026	Acquisition of Fabric8Labs	Thermal Management for DC	64,000

Note: e = Morgan Stanley Research estimates

Source: TSR, Company data, Morgan Stanley Research

TDK: Actions for Business Portfolio Improvement

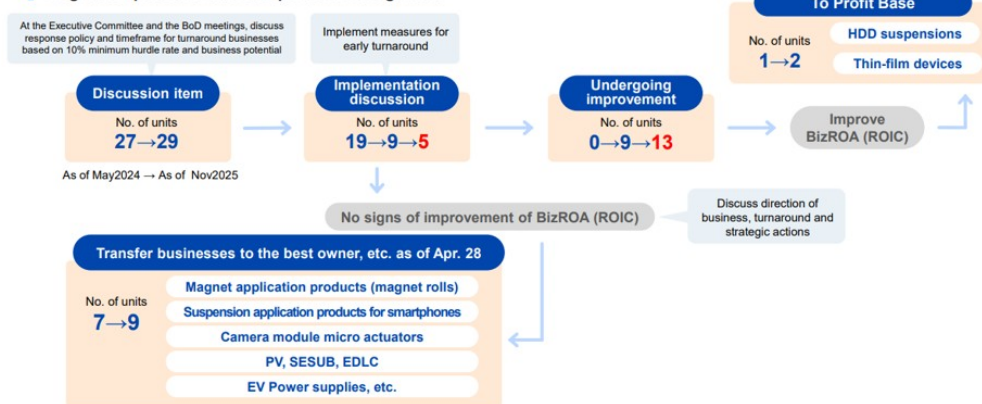
Chart for growth and capital efficiency by segment



Net sales, BizROA by segment

	(¥bn)	FY3/25	FY3/26
Net sales		2,204.8	2,504.8
Passive Components		559.6	593.2
Sensor Application Products		189.5	224.7
Magnetic Application Products		223.6	262.9
Energy Application Products		1,176.5	1,370.3
OP margin		10.2%	10.9%
BizROA (ROIC)		6.7%	7.5%
Passive Components		3.3%	4.8%
Sensor Application Products		0.2%	4.7%
Magnetic Application Products		1.0%	8.4%
Energy Application Products		27.3%	25.7%

Progress of proactive business portfolio management



Source: Company data

TDK: Medium-term Results and Forecasts

(bn JPY, JPY/share)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe
Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,768.0	3,020.6	3,295.6	3,600.9	3,938.7
Passive Components	417.6	433.4	395.5	407.1	507.8	578.8	565.6	559.6	593.2	647.4	674.9	704.0	735.2	767.9
Sensor Application Products	77.4	76.5	77.9	81.3	130.8	169.5	180.5	189.5	224.6	233.5	249.4	266.5	284.9	304.7
Magnetic Application Products	277.5	272.8	219.7	199.3	248.4	200.6	184.2	223.6	262.9	370.8	404.1	436.0	470.9	509.0
Energy Application Products	442.8	537.5	597.7	740.2	965.3	1,173.4	1,121.7	1,176.5	1,370.3	1,466.2	1,642.1	1,839.2	2,059.9	2,307.1
Other	56.4	61.6	72.3	51.1	49.7	58.6	51.8	55.6	53.8	50.0	50.0	50.0	50.0	50.0
OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	335.8	369.1	414.7	464.7	518.6
Passive Components	46.0	58.4	39.1	40.2	76.8	95.5	53.9	34.1	41.8	54.0	66.3	75.0	84.0	93.4
Sensor Application Products	-19.4	-22.1	-25.0	-24.9	-0.3	10.7	6.0	5.0	20.7	22.2	30.2	35.8	42.4	49.5
Magnetic Application Products	16.1	17.0	0.4	-2.4	4.6	-56.4	-35.6	3.4	27.0	61.5	68.2	74.4	81.1	88.1
Energy Application Products	72.9	91.0	124.1	147.4	123.2	147.4	195.7	234.4	246.7	256.1	262.3	287.6	315.2	345.6
Other	0.6	-6.7	-8.6	-16.1	-1.4	-0.4	-1.8	-4.4	-10.2	-6.0	-6.0	-6.0	-6.0	-6.0
Intersegment Elimination	-30.5	-29.8	-32.2	-32.7	-36.1	-28.0	-45.3	-48.3	-53.6	-52.0	-52.0	-52.0	-52.0	-52.0
OPM	6.7%	7.8%	7.2%	7.5%	8.8%	7.7%	8.2%	10.2%	10.9%	12.1%	12.2%	12.6%	12.9%	13.2%
ROE	7.8%	9.7%	6.7%	8.6%	11.4%	8.3%	7.9%	9.5%	9.7%	10.7%	11.2%	11.8%	12.1%	12.4%
EPS	33.4	43.3	30.4	41.8	69.3	60.2	65.7	88.0	103.0	129.1	144.8	165.2	185.5	207.6
FCF	-154.8	0.1	180.4	-8.7	-102.6	28.4	230.4	201.0	129.9	123.7	181.7	245.5	290.3	338.6

(JPY bn)	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27 MSe	F3/28 MSe	F3/29 MSe	F3/30 MSe	F3/31 MSe	CAGR F3/18-F3/25	CAGR F3/25-F3/31 MSe
Total Sales	1,271.7	1,381.8	1,363.0	1,479.0	1,902.1	2,180.8	2,103.9	2,204.8	2,504.8	2,768.0	3,020.6	3,295.6	3,600.9	3,938.7	8.2%	10.2%
Group A Sales	633.2	727.5	752.1	889.5	1,165.4	1,309.7	1,237.6	1,375.7	1,612.9	1,798.3	2,011.6	2,245.0	2,505.4	2,796.0	11.7%	12.5%
HDD Head	175.5	181.4	136.4	116.5	148.5	104.1	100.1	134.5	141.7	223.4	250.2	275.3	302.8	333.1		
HDD Suspension	60.5	56.0	54.3	57.5	66.3	49.7	43.7	56.2	81.0	91.8	98.2	105.1	112.5	120.3		
Magnetic Sensor	26.2	28.1	27.5	30.5	54.6	76.6	91.9	104.3	120.3	120.1	127.3	134.9	143.0	151.6		
LiB	371.0	461.9	533.8	685.0	896.0	1,079.3	1,001.9	1,080.7	1,269.9	1,363.0	1,535.8	1,729.7	1,947.1	2,190.9		
Group B Sales	638.6	654.3	611.0	589.5	736.7	871.1	866.3	829.1	892.0	969.6	1,008.9	1,050.6	1,095.4	1,142.7	3.8%	5.5%
Capacitors	157.0	173.3	153.9	158.2	198.1	242.5	245.0	234.3	257.5	280.9	294.9	309.6	325.1	341.4		
Inductive Devices	157.5	150.0	137.6	140.0	180.2	198.5	192.0	204.3	216.2	244.9	254.7	264.9	275.5	286.5		
Other Passive Components	103.1	110.1	104.0	109.0	129.4	137.8	128.7	121.1	119.5	121.6	125.3	129.4	134.6	139.9		
Temperature/Pressure Sensor	24.9	26.4	26.8	27.2	33.0	36.9	40.4	41.6	42.3	44.1	45.9	47.7	49.6	51.6		
MEMS Sensor	25.5	22.0	23.7	23.6	43.2	56.0	48.3	43.5	62.0	69.3	76.2	83.8	92.2	101.4		
Magnet	41.5	35.4	28.9	25.2	33.6	46.8	40.7	32.9	40.2	55.6	55.6	55.6	55.6	55.6		
Power Supply	71.8	75.6	63.9	55.2	69.3	94.0	119.7	95.8	100.4	103.2	106.3	109.5	112.8	116.2		
Others	57.2	61.6	72.3	51.1	49.8	58.6	51.6	55.6	53.8	50.0	50.0	50.0	50.0	50.0		
Total OP	85.6	107.8	97.9	111.5	166.8	168.8	172.9	224.2	272.4	335.8	369.1	414.7	464.7	518.6		
Group A OP	86.2	118.1	145.1	154.3	153.4	126.7	193.7	273.0	309.9	344.2	359.5	393.4	431.0	471.8		
Group B OP	30.0	20.0	-15.1	-10.0	53.3	70.1	24.5	-0.6	16.1	43.5	61.6	73.3	85.7	98.8		
Intersegment elimination	-30.5	-30.3	-32.2	-32.7	-39.8	-28.0	-45.3	-48.3	-53.6	-52.0	-52.0	-52.0	-52.0	-52.0		

Note: MSe = Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

iPhone Sell-in Forecasts / iPhone Camera Actuators by Model

min units	1Q22	2Q22	3Q22	4Q22	2022	1Q23	2Q23	3Q23	4Q23	2023	1Q24	2Q24	3Q24	4Q24	2024	1Q25	2Q25	3Q25	4Q25	2025	1Q26	2Q26e	3Q26e
iPhone Sell-in																							
iPhone SE3 / SE 5G / 16e / 17e	3.0	13.0	6.0	2.0	24.0	1.0	2.5	5.0	3.0	11.5	1.0	2.0	5.0	3.0	11.0	4.0	9.0	4.0	1.0	18.0	2.0	10.0	8.0
iPhone XR	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 11	1.0	1.0	0.5	0.5	3.0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 mini (5.4")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 (6.1")	2.0	1.0	1.0	0.5	4.5	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro (6.1")	1.0	0.5	0.5	0.5	2.5	0.5	-	-	-	0.5	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 12 Pro Max (6.7")	1.0	0.5	0.5	0.5	2.5	0.5	-	-	-	0.5	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 mini (5.4")	5.0	2.0	2.0	1.0	10.0	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 (6.1")	16.0	11.5	8.0	3.0	38.5	1.0	0.5	-	-	1.5	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 13 Pro (6.1")	13.0	8.0	6.0	3.5	30.5	1.0	2.0	1.0	-	4.0	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-
iPhone 13 Pro Max (6.7")	12.0	7.5	5.0	4.0	28.5	2.0	2.0	1.0	-	5.0	0.5	0.5	-	-	1.0	-	-	-	-	-	-	-	-
iPhone 14 (6.1")	-	-	-	-	-	6.5	13.0	19.5	7.0	4.0	5.0	2.0	18.0	1.0	0.5	-	1.5	-	-	-	-	-	-
iPhone 14 Plus (6.7")	-	-	-	-	-	2.0	13.5	15.5	5.0	3.0	1.0	1.0	10.0	1.0	0.5	-	1.5	-	-	-	-	-	-
iPhone 14 Pro (6.1")	-	-	-	-	-	6.0	16.0	22.0	15.0	12.0	7.0	3.0	37.0	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0
iPhone 14 Pro Max (6.7")	-	-	-	-	-	6.0	19.0	25.0	19.5	14.0	9.0	4.0	46.5	2.0	2.0	1.0	1.0	6.0	1.0	1.0	1.0	-	3.0
iPhone 15 (6.1")	-	-	-	-	-	-	-	-	-	-	6.0	12.0	18.0	6.0	5.0	4.0	1.0	16.0	1.0	0.5	0.5	0.5	2.5
iPhone 15 Plus (6.7")	-	-	-	-	-	-	-	-	-	-	5.0	10.0	15.0	4.0	3.0	1.0	1.0	9.0	1.0	0.5	0.5	0.5	2.5
iPhone 15 Pro (6.1")	-	-	-	-	-	-	-	-	-	-	12.0	16.0	24.0	12.0	10.0	8.0	2.0	22.0	2.0	2.0	2.0	1.0	1.0
iPhone 15 Pro Max (6.7")	-	-	-	-	-	-	-	-	-	-	2.0	24.0	26.0	18.0	13.0	10.5	3.0	44.5	3.0	3.0	2.0	1.0	9.0
iPhone 16 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 16 Plus (6.7")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 16 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 16 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 17 (6.1")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone Air (6.7")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 17 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 17 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 18 Pro (6.3")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone 18 Pro Max (6.9")	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
iPhone Fold	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	54.0	45.0	50.0	77.0	226.0	54.0	41.0	50.0	75.0	220.0	48.0	39.0	54.0	73.0	214.0	50.0	46.5	55.0	76.0	227.5	56.0	52.0	54.0
YoY	7%	1%	0%	-9%	-2%	0%	-9%	0%	-3%	-3%	-11%	-5%	8%	-3%	-3%	-4%	19%	2%	4%	6%	12%	12%	-2%
QoQ	-36%	-17%	11%	54%	-30%	-24%	22%	50%	-36%	-19%	38%	35%	-32%	-7%	16%	38%	-26%	-7%	4%				

	Type	Rear cam								Front cam	
		FF	OIS	VCM+OIS	OIS+FF	OIS x 2	SS+FF	OIS x 2 + FF	SS + OIS + FF	SS + periscope + FF	F Camera VCM
	number of cameras	Single	Single	Dual	Dual	Dual	Dual	Triple	Triple	Triple	Single
2022	iPhone 14 (6.1")										
	iPhone 14 Plus (6.7")										
	iPhone 14 Pro (6.1")										
	iPhone 14 Pro Max (6.7")										
2023	iPhone 15 (6.1")										
	iPhone 15 Plus (6.7")										
	iPhone 15 Pro (6.1")										
	iPhone 15 Pro Max (6.7")										
2024/25	iPhone 16 (6.1")										
	iPhone 16 Plus (6.7")										
	iPhone 16 Pro (6.3")										
	iPhone 16 Pro Max (6.9")										
	iPhone 16e (6.1")										
2025/26	iPhone 17 (6.3")										
	iPhone Air (6.5")										
	iPhone 17 Pro (6.3")										
	iPhone 17 Pro Max (6.9")										
	iPhone 17e (6.1")										

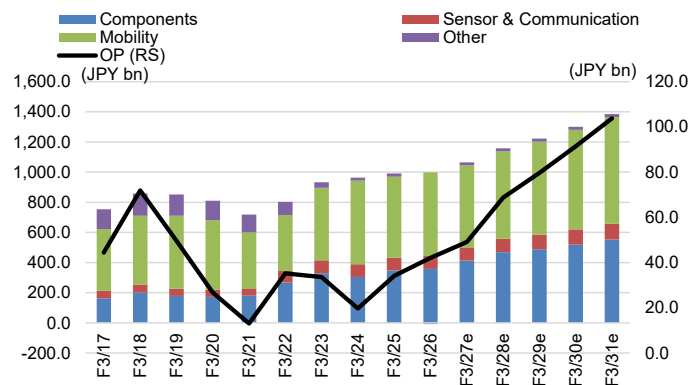
Note: e = Morgan Stanley Research estimates
Source: Morgan Stanley Research

Alps Alpine: We Expect New Smartphone Camera Actuators to Contribute to Earnings

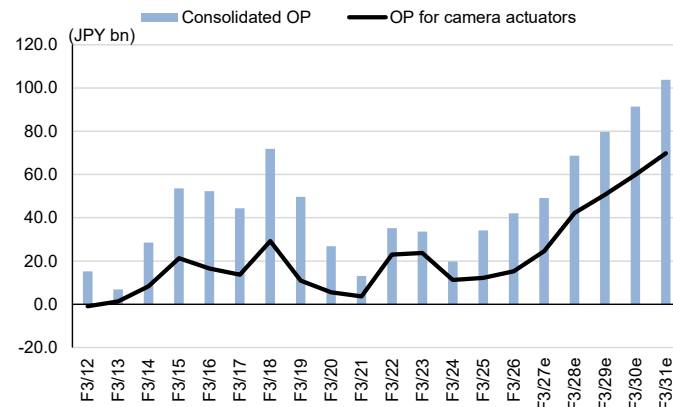
■ Alps Alpine: Medium-term results and forecasts

(bn JPY, JPY/share)	F3/17	F3/18	F3/19	F3/20	F3/21	F3/22	F3/23	F3/24	F3/25	F3/26	F3/27	F3/28	F3/29	F3/30	F3/31	F3/27	F3/28
											MSe	MSe	MSe	MSe	MSe	Ce	Ce
Sales	753.3	858.3	851.3	810.6	718.0	802.9	933.1	964.1	990.4	990.4	1,064.5	1,157.6	1,222.2	1,300.1	1,385.1	1,045.0	1,085.0
Components	162.8	200.3	178.9	173.6	180.1	268.5	329.0	305.3	348.0	358.4	413.5	468.0	489.3	519.9	552.9	398.0	415.0
Sensor & Communication	50.1	52.1	48.1	46.7	44.3	76.7	85.5	84.1	84.2	85.3	87.0	91.4	95.9	100.7	105.7	87.0	100.0
Mobility	408.3	458.6	484.4	461.1	376.3	369.4	481.4	554.4	537.2	555.1	544.0	578.2	616.9	659.5	706.4	540.0	550.0
Other	132.1	147.4	139.9	129.2	117.3	88.2	37.2	20.3	21.0	-8.3	20.0	20.0	20.0	20.0	20.0	20.0	20.0
OP	44.4	71.9	49.6	26.8	13.1	35.2	33.6	19.7	34.1	42.0	49.1	68.7	79.7	91.4	103.8	48.5	67.5
Components	20.9	41.5	19.5	14.8	10.3	32.6	38.3	20.5	30.4	30.2	37.1	51.7	60.2	69.4	79.3	28.0	35.0
Sensor & Communication	9.6	7.9	6.0	3.0	1.7	3.0	-1.6	-1.5	-3.4	-3.5	-1.2	2.0	2.5	3.0	3.5	0.5	4.0
Mobility	7.9	15.9	18.0	3.7	-4.6	-8.0	-6.6	-1.1	5.6	14.2	12.2	14.0	16.0	18.0	20.0	19.0	27.5
Other/Elimination	6.0	6.6	6.2	5.3	5.8	7.6	3.5	1.9	1.5	1.3	1.0	1.0	1.0	1.0	1.0	1.0	1.0
OPM	5.9%	8.4%	5.8%	3.3%	1.8%	4.4%	3.6%	2.0%	3.4%	4.2%	4.6%	5.9%	6.5%	7.0%	7.5%	4.6%	6.2%
ROE	14.4%	17.0%	6.6%	-1.2%	-1.1%	6.2%	2.9%	-7.6%	9.4%	6.2%	6.8%	10.1%	10.9%	11.4%	11.9%	-	10.0%
EPS	178.3	241.9	110.2	-19.5	-18.7	110.8	55.8	-145.0	184.0	134.8	159.2	250.7	290.7	333.2	378.4	-	-
FCF	3.6	3.7	5.3	44.8	1.5	-11.2	-38.8	34.1	64.1	37.5	8.9	12.0	25.6	31.1	38.5	-	-

■ Alps Alpine: Sales by segment and consolidated OP



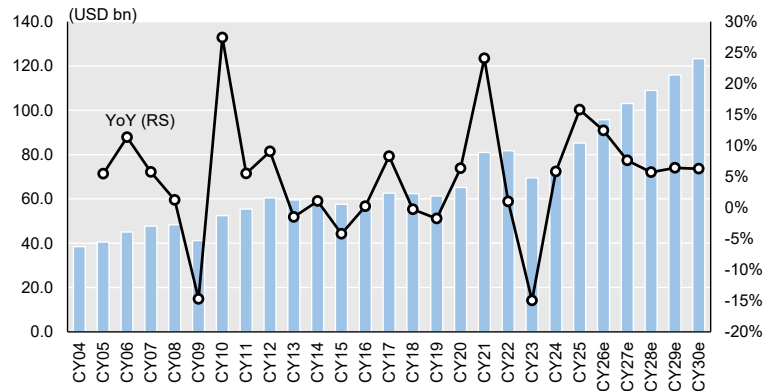
■ Alps Alpine: Consolidated OP and OP for camera actuators



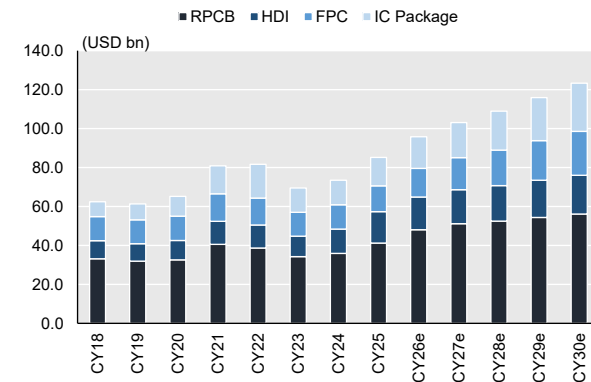
Note: MSe/e = Morgan Stanley Research estimates, Ce = Company estimates
Source: Company data, Morgan Stanley Research

PCB Market

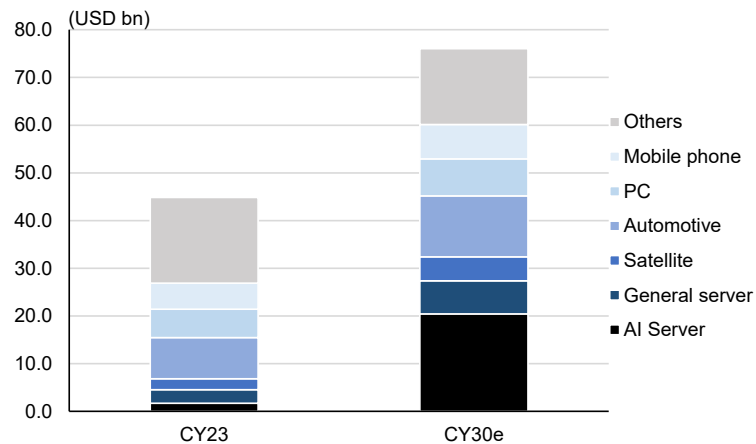
Global PCB market value



Global PCB market value by composition



Global Rigid PCB/HDI market value application



Major composition by application

Usage/Product	1-2 sided	MLB	HDI	Flexible/ Rigid-flex
General server/Computer	○	○ (4-16)	○	○
AI server	○	○ (4-40+)	○	○
Automotive	○	○ (4-8)	○	
Mobile		○ (4-12)	○	○
Consumer electronics	○	○ (4-6)	○	
5G		○ (20-26)		
Industrial	○	○ (4-16)	○	
Medical	○	○ (-20)	○	
Aerospace/Military	○	○ (4-6)		

Note: e = Morgan Stanley Research estimates
Source: Prismark, Morgan Stanley Research

Meiko Electronics: Sales Guidance of Major Sites in VietNam/Mid-term Forecasts

(JPY bn)	Major applications	F3/27 Ce	F3/28 Ce	F3/29 Ce	F3/30 Ce
Viet Nam #4	Satellites, Memory modules	50	70	75	80
Hoa Binh #1	High density HDI incl. high-end smartphones	7	20	30	50
Quang Minh #3	High density HDI incl. high-end smartphones	-	20	30	50
Yen Quang	Satellites	-	20	40	80
Hoa Binh #2	Main board for AI servers (JV with ACCL in TW)	Undisclosed, sales recorded from F3/28			

(JPY bn, JPY/share)	F3/24	F3/25	F3/26	F3/27 MSe	F3/27 Ce	F3/28 MSe	F3/28 Ce	F3/29 MSe	F3/29 Ce	F3/30 MSe	F3/31 MSe
Sales	179.5	206.8	240.6	323.8	320.0	412.9	390.0	482.1	460.0	539.7	595.1
Automotive	94.7	90.4	97.5	101.3	100.0	104.0	101.0	107.0	104.0	110.3	113.9
Smartphone/Tablet	23.6	25.5	29.4	36.8	37.0	54.5	44.0	70.8	54.0	76.1	81.9
Satellite	7.6	16.6	28.2	56.0	55.0	82.0	79.0	102.0	94.0	114.0	127.4
AI server	-	4.2	7.3	20.0	17.0	45.0	38.0	60.0	60.0	82.0	99.2
Modules	-	9.2	9.2	13.0	13.0	16.0	16.0	18.0	18.0	20.0	22.0
Amusement/Smart appliance/Others	14.1	23.7	24.4	27.8	29.0	28.2	28.0	30.1	29.0	32.1	34.4
Electronic equipment	30.3	36.2	42.6	66.0	66.0	79.2	76.0	89.2	86.0	99.2	109.2
IC Package	0.7	1.0	2.0	3.0	3.0	4.0	8.0	5.0	15.0	6.0	7.0
OP	11.66	19.083	24.572	38.6	38.0	48.7	51.0	62.5	65.0	73.4	83.8
OPM	6.5%	9.2%	10.2%	11.9%	11.9%	11.8%	13.1%	13.0%	14.1%	13.6%	14.1%
ROE	12.9%	14.5%	16.2%	18.9%	17.1%	20.5%	20.1%	22.3%	21.1%	21.9%	21.2%
EPS	428.7	569.5	758.6	1,063.6	1,039.7	1,336.4	1,362.0	1,753.2	1,749.0	2,084.4	2,427.3
FCF	1.8	-2.7	-27.9	-42.7	-	-19.6	-	4.4	-	15.3	23.7

Note: Ce=Company guidance, MSe=Morgan Stanley Research estimates

Source: Company data, Morgan Stanley Research

HDD Market by Company (Quarterly, Fiscal Year)

(mn units)	CY23				CY24				CY25				CY26				CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	CY26e	CY27e	CY28e
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q																	
HDD Shipment (by Size)																															
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.1	31.7	28.7	30.4	31.6	33.0	33.0	33.0	564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2	124.9		
Enterprise	1.4	1.6	1.4	1.4	1.5	1.0	1.0	0.9	0.8	0.8	0.7	0.8	0.8	0.6	32.4	32.1	24.8	22.6	21.5	17.4	12.9	13.4	8.4	5.8	4.4	3.2	2.7	1.9	1.1		
NL	11.4	9.2	9.0	9.9	12.3	13.8	15.9	16.5	14.8	16.5	17.3	19.2	18.7	19.1	36.9	37.2	41.0	42.4	51.3	52.7	59.1	73.5	61.5	39.4	58.6	67.6	76.4	78.6	81.8		
3.5"ATA	11.2	12.1	11.2	10.7	9.3	9.4	9.0	7.9	7.6	7.9	8.0	7.6	8.3	8.7	202.6	161.0	142.9	135.4	120.5	103.9	87.0	89.9	55.0	45.2	35.6	30.9	31.5	27.5	26.0		
2.5"	9.5	8.4	7.1	6.9	6.6	6.1	6.2	6.4	5.7	5.3	5.6	5.5	5.3	4.7	291.7	238.6	215.1	202.7	182.0	143.6	101.0	82.5	46.3	32.0	25.3	22.0	19.8	17.3	16.0		
1.8"	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
YoY % Change																															
Total	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%	2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-33.6%	-29%	1%	0%	5%	-4%	0%		
Enterprise	-48%	-34%	-15%	-16%	6%	-36%	-30%	-33%	-45%	-21%	-23%	-16%	-7%	-26%	2%	-1%	-23%	-8%	-5%	-19%	-26%	3%	-37%	-32%	-23%	-29%	-15%	-30%	-41%		
NL	-39%	-49%	-37%	-7%	8%	51%	78%	67%	19%	19%	9%	16%	28%	16%	4%	1%	10%	3%	21%	3%	12%	24%	-16.3%	-36%	49%	15.5%	12.9%	2.9%	4.1%		
3.5"ATA	-33%	-4%	-14%	-16%	-16%	-22%	-19%	-26%	-19%	-17%	-12%	-4%	9%	11%	0%	-21%	-11%	-5%	-11%	-14%	-16%	3%	-39%	-18%	-21%	-13%	2%	-13%	-5%		
2.5"	-36%	-20%	-26%	-39%	-30%	-28%	-13%	-7%	-14%	-13%	-10%	-14%	-7%	-11%	4%	-18%	-10%	-6%	-10%	-21%	-30%	-18%	-44%	-31%	-21%	-13%	-10%	-13%	-8%		
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-6%	-98%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Sequential % Change																															
Total	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Enterprise	-16%	13%	-16%	4%	7%	-32%	-7%	-1%	-13%	-1%	-10%	8%	-4%	-21%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
NL	8%	-20%	-2%	11%	24%	12%	15%	12%	-12%	13%	5%	16%	-3%	2%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
3.5"ATA	-13%	9%	-7%	-4%	-13%	1%	-4%	-16%	-4%	-4%	2%	5%	10%	5%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2.5"	-16%	-11%	-16%	-3%	-4%	-8%	2%	3%	-11%	-8%	6%	-1%	-3%	-12%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
1.8"	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Enterprise																															
NL	4%	5%	5%	5%	5%	3%	3%	3%	3%	3%	2%	2%	2%	2%	6%	7%	6%	6%	6%	5%	5%	5%	5%	5%	4%	3%	2%	1%			
3.5"ATA	34%	29%	31%	34%	41%	46%	50%	52%	51%	54%	55%	58%	57%	58%	7%	8%	10%	11%	14%	17%	23%	28%	36%	32%	47%	55%	59%	63%	65%		
2.5"	33%	39%	37%	37%	31%	28%	25%	26%	26%	26%	25%	23%	22%	26%	36%	34%	34%	34%	32%	33%	33%	35%	32%	37%	29%	25%	24%	22%	21%		
1.8"	28%	27%	25%	24%	22%	20%	19%	20%	20%	17%	18%	17%	16%	14%	52%	51%	51%	50%	48%	45%	39%	32%	27%	26%	20%	18%	15%	14%	13%		
	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
HDD Shipment (by Company)																															
Total	33.5	31.3	28.6	28.9	29.7	30.3	32.2	31.7	28.7	30.4	31.6	33.0	33.0	33.0	564.1	468.8	423.9	403.1	375.4	317.7	260.0	259.3	172.2	122.3	123.9	123.8	130.4	125.2	124.9		
Seagate	14.5	14.3	12.0	11.6	11.4	12.3	13.0	13.0	11.4	12.5	12.5	13.1	13.1	13.3	224.1	188.6	155.2	148.2	150.0	129.1	126.7	110.3	74.1	52.4	49.7	49.4	-	-	-	-	
Western Digital	12.6	11.8	10.4	10.8	11.7	12.1	13.2	13.5	12.1	12.9	13.9	14.3	14.4	14.6	249.3	215.6	191.4	161.8	139.7	114.2	110.8	94.3	63.9	45.6	50.5	53.2	-	-	-	-	
Toshiba	6.4	5.2	6.2	6.5	6.7	5.9	6.0	5.2	5.2	5.0	5.2	5.7	5.6	5.2	90.7	75.8	93.5	93.0	85.7	74.4	70.0	54.7	34.3	24.3	23.8	21.1	-	-	-	-	
Market Share																															
Seagate	43%	46%	42%	40%	38%	41%	40%	41%	40%	41%	39%	39%	40%	40%	40%	40%	37%	37%	40%	41%	49%	43%	43%	43%	40%	40%	-	-	-	-	
Western Digital	38%	38%	36%	37%	39%	40%	41%	43%	42%	42%	44%	43%	44%	44%	44%	46%	45%	40%	37%	36%	43%	36%	37%	37%	41%	43%	-	-	-	-	
Toshiba	19%	17%	22%	22%	23%	20%	19%	17%	18%	16%	17%	17%	17%	16%	16%	16%	22%	23%	23%	23%	27%	21%	20%	20%	19%	17%	-	-	-	-	
YoY % Change																															
Seagate	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%	2%	-17%	-10%	-5%	-7%	-15%	-18%	0%	-34.6%	-29%	1%	0%	5%	-4%	0%		
Western Digital	-37%	-28%	-24%	-25%	-21%	-14%	8%	12%	0%	2%	-4%	0%	15%	6%	1%	-16%	-18%	-4%	1%	-14%	-2%	-13%	-33%	-29%	-5%	0%	-	-	-	-	
Toshiba	-36%	-28%	-29%	-16%	-7%	3%	27%	25%	3%	7%	5%	6%	19%	13%	1%	-14%	-11%	-15%	-14%	-18%	-3%	-15%	-32%	-29%	11%	5%	-	-	-	-	
Sequential % Change																															
Seagate	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Western Digital	-6%	-1%	-16%	-3%	8%	5%	0%	-12%	10%	0%	5%	0%	0%	1%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Toshiba	-2%	-6%	-12%	4%	8%	3%	9%	2%	-10%	7%	8%	3%	0%	1%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	-21%	-19%	20%	5%	3%	-11%	0%	-12%	-1%	-4%	5%	8%	-2%	-6%	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
HDD Shipment Capacity (EB, by Company)																															
Total	236.0	189.8	196.1	214.3	264.6	289.4	340.3	362.3	346.6	391.8	432.5	457.0	477.2	504.3	534.9	535.8	622.6	720.7	855.3	902.6	1,019.1	1,344.8	1,163.5	836.2	1,256.6	1,627.9	2,043.1	2,371.3	2,883.9		
Seagate	118.7	91.2	88.6	95.1	99.1	114.2	137.5	150.8	143.6	162.5	181.7	190.1	199.0	218.8	221.3	222.7	252.2	287.2	386.5	367.0	480.6	614.2	539.5	591.6	677.9	-	-	-	-	-	
Western Digital	96.5	78.8	79.1	90.3	127.3	142.6	163.8	172.5	187.0	204.1	215.0	222.0	231.0		240.1	250.1	286.1	342.7	390.0	422.7	422.3	544.0	476.3	544.7	606.2	768.1	-	-	-	-	
Toshiba	20.8	19.8	27.4	28.9	36.2	32.6	39.2	39.0	41.0	42.3	46.7	51.9	56.1	55.3	73.2	63.0	84.3	90.8	98.8	112.9	116.2	187.5	147.7	96.9	148.0	181.9	-	-	-	-	
YoY % Change																															
Seagate	-28%	-41%	-31%	-6%	12%	52%	74%	69%	31%	35%	27%	26%	38%	29%	16%	0%	16%	16%	19%	6%	13%	32%	-13%	-28%	50%	30%	26%	16%	22%		
Western Digital	-23%	-41%	-24%	-15%	-17%	25%	53%	59%	45%	42%	32%	26%	39%	34%	14%	0%	13%	14%	28%	0%	31%	28%	-12%	-27%	27%	35%	-	-	-	-	
Toshiba	-27%	-39%	-39%	8%	32%	81%	107%	91%	27%	31%	25%	25%	37%	24%	19%	4%	14%	20%	14%	8%	0%	29%	-12%	-28%	76%	27%	-	-	-	-	
	-51%	-45%	-26%	-10%	84%	65%	43%	35%	7%	30%	19%	33%	37%	31%	14%	-14%	34%	8%	9%	14%	3%	61%	-21%	-34%	54%	22%	-	-	-	-	
Sequential % Change																															
Seagate	3%	-20%	3%	9%	23%	9%	18%	6%	-4%	13%</																					

HDD Head Market

	CY23		CY24		CY25		CY26																				
(mn units)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	CY14	CY15	CY16	CY17	CY18	CY19	CY20	CY21	CY22	CY23	CY24	CY25	
HDD Head Shipment																											
Total	262.2	202.9	217.1	237.7	262.5	297.1	327.3	332.4	330.0	348.7	377.6	397.8	401.9	414.8	1,931.6	1,756.1	1,702.8	1,714.4	1,715.4	1,496.2	1,517.9	1,747.9	1,349.0	919.9	1,219.3	1,454.1	
TDK	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	60.6	471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8	
Seagate	118.8	92.1	86.9	101.7	103.3	116.1	132.8	140.7	128.3	140.8	155.1	165.2	165.4	166.9	634.0	616.2	617.1	636.6	683.5	594.1	652.2	755.2	579.1	399.5	492.9	589.4	
Western Digital	103.8	84.4	95.2	95.7	115.9	133.6	145.2	148.3	153.8	162.7	172.8	181.6	183.7	187.3	826.0	763.9	710.7	715.9	701.7	602.5	611.3	654.5	539.1	379.1	543.0	670.9	
Market Share																											
TDK	15%	13%	16%	17%	16%	16%	15%	13%	15%	13%	13%	13%	13%	15%	24%	21%	22%	21%	19%	20%	17%	19%	17%	15%	15%	13%	
Seagate	45%	45%	40%	43%	39%	39%	41%	42%	39%	40%	41%	42%	41%	40%	33%	35%	36%	37%	40%	40%	43%	43%	43%	43%	40%	41%	
Western Digital	40%	42%	44%	40%	44%	45%	44%	45%	47%	47%	46%	46%	46%	45%	43%	43%	42%	42%	41%	40%	40%	37%	40%	41%	45%	46%	
YoY % Change																											
TDK	-38%	-47%	-33%	9%	0%	46%	51%	40%	26%	17%	15%	20%	22%	19%	0%	-9%	-3%	1%	0%	-13%	1%	15%	-23%	-32%	33%	19%	
Seagate	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%	34%	-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%	
Western Digital	-40%	-47%	-32%	26%	-13%	26%	53%	38%	24%	21%	17%	17%	29%	19%	1%	-3%	0%	3%	7%	-13%	10%	16%	-23%	-31%	23%	20%	
Western Digital	-33%	-44%	-31%	2%	12%	58%	53%	55%	33%	22%	19%	22%	19%	15%	2%	-8%	-7%	1%	-2%	-14%	1%	7%	-18%	-30%	43%	24%	
HDD Head/HDD	7.84	6.49	7.59	8.22	8.83	9.80	10.20	10.47	11.50	11.47	11.95	12.04	12.18	12.56	3.42	3.75	4.02	4.25	4.57	4.71	5.84	6.74	7.83	7.52	9.84	11.75	
TDK's Head Shipment by Customer																											
Total	39.6	26.4	35.0	40.3	43.3	47.4	49.3	43.4	47.9	45.2	49.7	51.0	52.8	60.6	471.6	376.0	375.0	361.9	330.2	299.6	254.4	338.2	230.8	141.3	183.4	193.8	
Seagate	4.3	3.7	1.0	0.3	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.2	6.4	17.4	14.0	25.7	24.0	41.3	51.7	43.1	45.5	28.1	9.3	0.3	0.0	
Maxtor	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
HGST	2.9	2.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.7	4.4	13.1	16.0	11.9	21.5	22.6	1.4	0.0	4.8	7.6	5.1	0.0	0.0	
Western Digital	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	43.7	17.5	16.2	12.2	5.7	1.4	0.0	0.0	0.0	0.0	0.0		
Samsung	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	130.2	86.0	21.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Toshiba	32.4	20.7	33.8	40.0	43.2	47.3	49.2	43.4	47.9	45.2	49.7	51.0	51.9	49.8	267.2	242.5	299.8	304.2	260.6	245.1	211.3	287.9	195.1	126.9	183.1	193.8	
Fujitsu	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
Others	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	
TDK's Shipment Share by Customer																											
Seagate	4%	4%	1%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	4%	3%	2%	4%	4%	6%	8%	6%	6%	5%	2%	0%	0%	
Maxtor	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
HGST	3%	3%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	3%	4%	5%	3%	6%	6%	0%	0%	1%	2%	2%	0%	0%	
Western Digital	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	8%	4%	4%	3%	2%	1%	0%	0%	0%	0%	0%	0%	
Samsung	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Toshiba	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	
Fujitsu	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Others	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
YoY % Change	-45%	-53%	-41%	-5%	9%	80%	41%	8%	11%	-5%	1%	18%	10%	34%	-3%	-20%	0%	-3%	-9%	-9%	-15%	33%	-32%	-39%	30%	6%	
Sequential % Change	-7%	-33%	33%	15%	7%	9%	4%	-12%	10%	-6%	10%	3%	4%	15%	-	-	-	-	-	-	-	-	-	-	-	-	

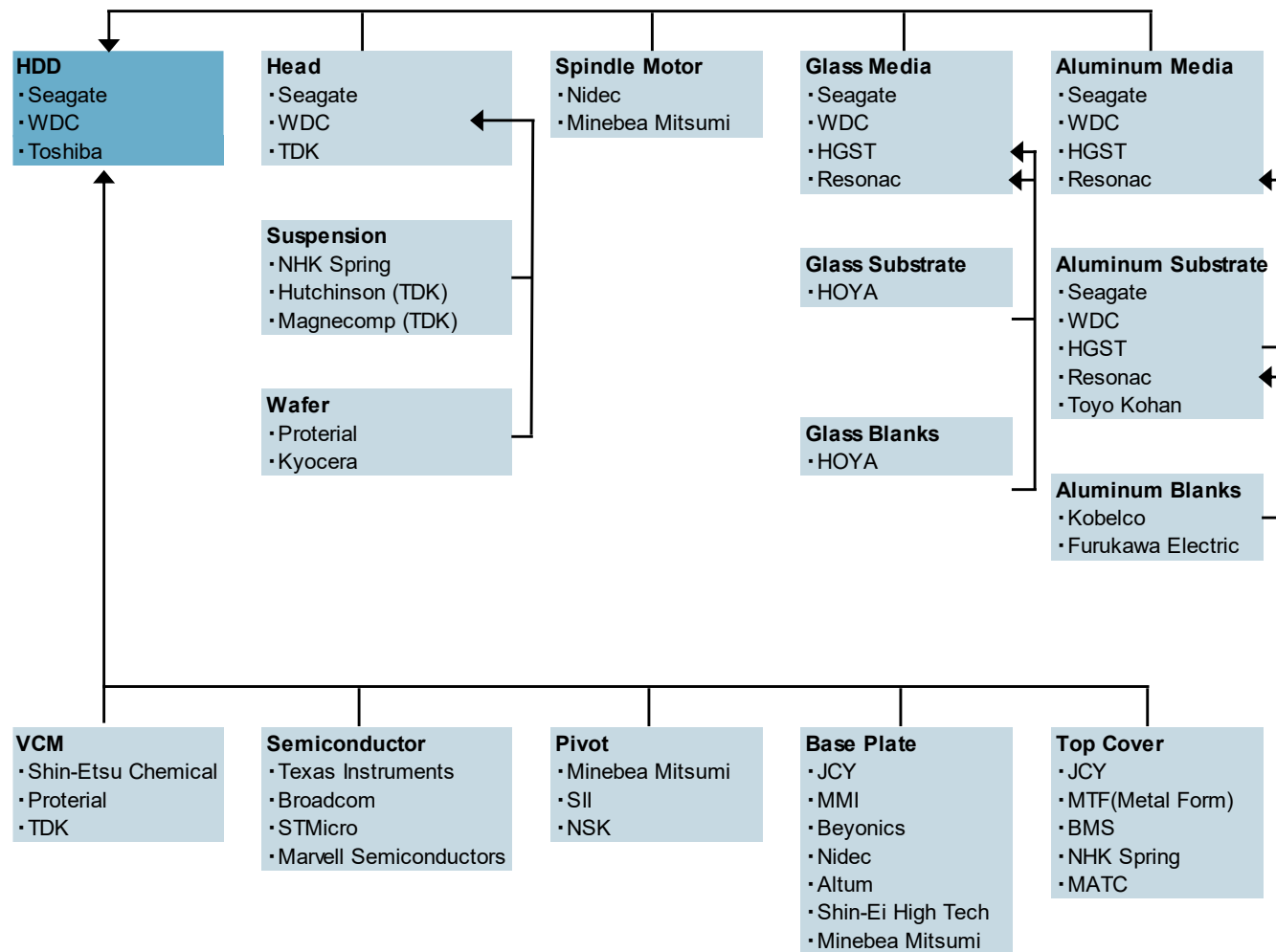
Source: TSR, Morgan Stanley Research

HDD Motor Market

	CY23		CY24		CY25		CY26																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											</
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Source: TSR, Morgan Stanley Research

Supply Chain of HDDs



Source: TSR, Morgan Stanley Research

Ibiden: Key Financial Data and Risk Reward

Income Statement					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	369.4	416.2	567.3	707.0	867.9
COGS	256.1	284.5	327.8	383.2	451.9
Gross Profit	113.3	131.7	239.5	323.8	416.0
SG&A	65.7	69.6	105.7	125.7	148.4
Operating Profit	47.6	62.0	133.8	197.1	267.6
OP Incl. depreciation of inactive noncurrent assets	45.9	58.2	132.9	197.1	267.6
Non-operating Income/Loss	0.3	-1.2	1.5	2.3	2.3
Recurring Profit	47.9	60.8	135.3	199.4	269.9
Extraordinary Income/Loss	3.6	30.3	-2.4	-2.4	-2.4
Pretax Profit	51.5	91.1	132.8	197.0	267.4
Income Tax	17.4	26.9	40.3	59.1	80.2
(Effective Tax Rate)	34%	30%	30%	30%	30%
Minority Interest	0.3	0.4	0.5	0.2	0.2
Net Profit	33.7	63.7	92.0	137.7	187.0
Capex					
Depreciation and Amortization	157.3	64.3	157.6	158.0	208.0
R&D Expenses	54.2	66.2	81.4	95.5	110.5
EBITDA	27.5	29.9	34.8	46.0	55.4
EBITDA	101.8	128.3	215.2	292.6	378.1
ROE	6.8%	12.2%	15.6%	19.8%	22.1%
Average Share Outstanding (Diluted)					
EPS (Diluted, Yen)	279.4	279.3	294.9	294.9	294.9
BPS (Yen)	1,780.0	2,281.1	3,120.0	4,668.8	6,341.1
DPS (Yen)	20.0	30.0	35.0	40.0	45.0
USD/JPY: Average Exchange Rate					
	152.0	149.0	158.0	158.0	158.0
EUR/JPY: Average Exchange Rate					
	162.0	172.0	180.0	180.0	180.0
YoY % Change					
Net Sales	0%	13%	36%	25%	23%
Operating Profit	0%	30%	116%	47%	36%
Net Profit	7%	89%	44%	50%	36%
As % of Sales					
Gross Profit	30.7%	31.6%	42.2%	45.8%	47.9%
Operating Profit	12.9%	14.9%	23.6%	27.9%	30.8%
Net Profit	9.1%	15.3%	16.2%	19.5%	21.5%
Capex	42.6%	15.4%	27.8%	22.3%	24.0%
Depreciation and Amortization	14.7%	15.9%	14.3%	13.5%	12.7%
Balance Sheet					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	1,081.7	960.4	1,052.9	1,189.1	1,375.5
Total Current Assets	549.6	465.5	481.7	555.5	644.4
Cash and Cash Equivalents	390.7	295.7	290.3	328.9	376.7
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	68.2	82.1	100.2	124.8	153.2
Inventory	67.2	71.0	74.5	85.0	97.7
Other Current Assets	22.5	16.8	16.8	16.8	16.8
Total Long Term Assets	532.1	494.9	571.1	633.6	731.1
Net PP&E	460.1	437.7	514.0	576.5	674.0
Investments in Securities	58.8	32.3	32.3	32.3	32.3
Other Assets	13.3	24.9	24.9	24.9	24.9
Total Current Liabilities	327.7	221.6	231.2	240.2	251.5
ST Debt & Current Portion of LT Debt	90.0	15.0	15.0	15.0	15.0
Accounts Payable	38.8	44.1	53.7	62.7	74.0
Other Current Liabilities	108.9	81.6	81.6	81.6	81.6
Total Long Term Liabilities	256.7	181.4	181.4	181.4	181.4
Long Term Debt	253.0	177.5	177.5	177.5	177.5
Other Long Term Liabilities	3.7	3.9	3.9	3.9	3.9
Net Assets	497.3	557.4	640.4	767.6	942.7
Minority Interests	6.8	7.3	7.3	7.3	7.3
Stock Acquisition Rights	0.0	0.0	0.0	0.0	0.0
Net Debt	-106.5	-135.5	-130.1	-168.7	-216.5

Cash Flow Statement					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	118.9	106.4	161.3	207.1	267.6
Net Income	33.7	63.7	92.0	137.7	187.0
Depreciation and Amortization	54.2	66.2	81.4	95.5	110.5
Changes in Working Capital	-9.8	-7.1	-12.1	-26.1	-29.9
Other-net	40.7	-16.4	0.0	0.0	0.0
Net Cash from Investing Activities	-164.2	-82.4	-157.6	-158.0	-208.0
Purchase of PP&E	-197.5	-106.1	-157.6	-158.0	-208.0
Sale/Purchase of Investments	34.6	57.1	0.0	0.0	0.0
Other-net	-1.3	-3.4	0.0	0.0	0.0
Free Cash Flow	-45.3	54.0	3.7	49.1	59.6
Net Cash from Financing Activities	-7.1	-157.5	-9.1	-18.5	-11.9
Change in Interest-bearing Debt	0.0	-150.0	0.0	0.0	0.0
Change in Treasury Stock	-0.5	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-5.6	-7.0	-9.1	-10.5	-11.9
Other-net	-1.0	-0.6	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.5	5.8	0.0	0.0	0.0
Net Change in Cash	-52.9	-97.7	-5.4	38.6	47.7
Cash at End of Period	390.7	292.9	287.5	326.2	373.9
Segment Information					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	369.4	416.2	567.3	707.0	867.9
Electronics	197.2	243.3	390.5	531.5	684.1
FC Package	186.9	219.2	381.5	522.5	675.1
Print Circuit Board	10.3	10.0	9.0	9.0	9.0
FC CSP	0.0	0.0	0.0	0.0	0.0
Ceramics	84.1	82.6	86.0	81.5	87.0
DPP	34.6	32.5	36.4	30.0	30.0
Sealing Materials	30.4	29.3	28.9	30.0	30.0
Graphite Specialty Products	16.2	13.2	16.8	16.5	16.7
Others	3.0	5.7	3.9	5.0	10.3
Others	88.1	90.3	90.8	94.0	96.8
Total Operating Profit	47.6	62.0	133.8	197.1	267.6
Electronics	26.8	45.2	115.0	177.9	246.8
Ceramics	12.2	7.6	8.7	8.5	9.5
Others	8.6	9.1	10.1	10.7	11.2
Operating Profit Margin	12.9%	14.9%	23.6%	27.9%	30.8%
Electronics	13.6%	18.6%	29.4%	33.5%	36.1%
Ceramics	14.5%	9.3%	10.1%	10.4%	10.9%
Others	9.7%	10.1%	11.1%	11.4%	11.6%

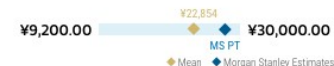
Risk Reward – Ibiden (4062.T)

Probability of upside to F3/31 guidance has increased.

PRICE TARGET ¥27,500

Derived from the base case. DCF assumptions: 2.8% risk-free rate, 1.52 beta, 3.2% risk premium, for 7.7% WACC. Terminal growth of 0% from F3/36.

Consensus Price Target Distribution



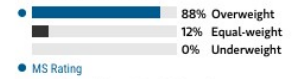
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect sales of ABF packages for AI servers to increase and earnings growth to continue.
- We see Ibiden sustaining a competitive advantage in ABF package technology and keeping dominant market share in GPU ABF packages for AI servers.
- We expect earnings growth to continue due to customer and product mix improvement, even though costs increase due mainly to new lines being activated.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE ¥36,000

Derived from DCF

Earnings ahead of the base case on stronger sales of value-added ABF packages for AI servers: OP of ¥141.9bn in F3/27, ¥237.3bn in F3/28 and ¥500.7bn in F3/31. Same FX and DCF assumptions as the base case. Fair value equates to 108.7x F3/27 diluted EPS of ¥331.2, 64.3x F3/28 diluted EPS of ¥560.3, and 30.4x F3/31 diluted EPS of ¥1,185.5.

BASE CASE ¥27,500

Derived from DCF

Higher sales of advanced FC packages should drive FY3/31 earnings above the company's guidance: OP ¥133.8bn in F3/27, ¥197.1bn in F3/28, ¥392.5bn in F3/31. FX assumptions ¥158/\$, ¥180/€. Fair value is based on DCF as explained in the valuation methodology section. Base case fair value equates to 88.1x our F3/27e diluted EPS of ¥312.0, 58.9x our F3/28e diluted EPS of ¥466.8 and 29.5x our F3/31e diluted EPS of ¥930.8.

BEAR CASE ¥10,000

F3/27 BPS ¥2,037 x P/B 4.9x

Limited earnings improvement from F3/27: OP of ¥77.0bn in F3/27, ¥93.7bn in F3/28 and ¥163.7bn in F3/31. Same FX assumptions as the base case. ROE declines from 12.2% in F3/26 to 9.5% in F3/27, but starts to improve from F3/28 and 11.2% in F3/28, higher than 7.7% cost of capital, but below the market consensus.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Niterra: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	653.0	731.2	813.2	854.8	919.7
COGS	394.9	450.3	509.7	532.6	568.8
Gross profit	258.1	280.9	303.6	322.2	350.9
SG&A	128.4	142.7	145.4	147.9	150.1
Operating Profit	129.7	138.2	158.2	174.4	200.8
Non-operating Income/Loss	3.7	27.3	7.0	7.0	7.0
Recurring Profit	133.3	165.5	165.2	181.4	207.8
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pretax Profit	133.3	165.5	165.2	181.4	207.8
Income Tax	40.5	49.0	46.2	50.8	58.2
(Effective Tax Rate)	30%	30%	28%	28%	28%
Minority Interest	0.2	3.6	1.0	1.0	1.0
Net Profit	92.6	112.9	117.9	129.6	148.6
Capex	37.5	49.0	47.4	45.0	45.0
Depreciation and Amortization	39.7	45.0	42.1	43.0	43.0
R&D	0.0	0.0	30.0	31.0	31.0
EBITDA	169.4	183.2	200.3	217.4	243.8
ROE	14.1%	15.7%	14.9%	15.4%	16.3%
Average Share Outstanding (Diluted)	198.6	197.9	193.4	188.4	183.4
EPS (Diluted, Yen)	466.3	570.4	609.6	687.7	810.0
BPS (Yen)	3,399.4	3,910.2	4,203.2	4,625.4	5,159.9
DPS (Yen)	178.0	205.0	210.0	220.0	250.0
USD/JPY: Average Exchange Rate	153.0	151.0	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	164.0	175.0	180.0	180.0	180.0
YoY % Change					
Sales	6%	12%	11%	5%	8%
Operating Profit	21%	7%	14%	10%	15%
Net Profit	12%	22%	4%	10%	15%
As % of Sales					
Gross profit	39.5%	38.4%	37.3%	37.7%	38.2%
Operating Profit	19.9%	18.9%	19.5%	20.4%	21.8%
Net Profit	14.2%	15.4%	14.5%	15.2%	16.2%
Capex	5.7%	6.7%	5.8%	5.3%	4.9%
Depreciation and Amortization	6.1%	6.2%	5.2%	5.0%	4.7%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	991.0	1,221.1	1,245.9	1,278.0	1,328.6
Total Current Assets	588.5	658.2	677.8	707.9	756.4
Cash and Cash Equivalents	208.2	187.7	173.2	197.7	223.2
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	147.6	172.2	191.5	201.3	216.5
Inventory	183.9	217.7	232.5	228.3	236.1
Other Current Assets	48.8	80.6	80.6	80.6	80.6
Total Long Term Assets	402.4	562.9	568.2	570.2	572.2
Net PP&E	220.0	265.7	271.0	273.0	275.0
Investments in Securities	93.1	81.6	81.6	81.6	81.6
Other Assets	89.4	215.7	215.7	215.7	215.7
Total Current Liabilities	153.6	195.8	204.9	208.5	214.1
ST Debt & Current Portion of LT Debt	16.7	31.4	31.4	31.4	31.4
Accounts Payable	57.7	69.7	78.9	82.5	88.1
Other Current Liabilities	79.3	94.6	94.6	94.6	94.6
Total Long Term Liabilities	162.6	254.1	224.1	194.1	164.1
Long Term Debt	140.0	211.0	181.0	151.0	121.0
Other Long Term Liabilities	22.6	43.1	43.1	43.1	43.1
Net Assets	674.7	771.3	816.9	875.4	950.4
Minority Interests	0.1	3.8	3.8	3.8	3.8
Net Debt	-144.6	-26.8	-42.3	-96.7	-152.3

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	132.9	109.4	135.2	170.5	174.2
Net Income	92.6	112.9	117.9	129.6	148.6
Depreciation and Amortization	39.7	45.0	42.1	43.0	43.0
Changes in Working Capital	-6.2	-27.3	-24.8	-2.1	-17.4
Other-net	6.8	-21.2	0.0	0.0	0.0
Net Cash from Investing Activities	-34.2	-165.5	-47.4	-45.0	-45.0
Purchase of PP&E	-27.8	-32.3	-47.4	-45.0	-45.0
Sale/Purchase of Investments	-3.4	7.7	0.0	0.0	0.0
Other-net	-3.0	-140.9	0.0	0.0	0.0
Free Cash Flow	98.7	-56.1	87.8	125.5	129.2
Net Cash from Financing Activities	-71.0	28.6	-102.3	-101.0	-103.7
Change in Interest-bearing Debt	-25.9	86.4	-30.0	-30.0	-30.0
Change in Treasury Stock	-10.0	-16.8	-30.0	-30.0	-30.0
Payment of Cash Dividends	-34.4	-36.4	-42.3	-41.0	-43.7
Other-net	-0.7	-4.6	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.2	7.1	0.0	0.0	0.0
Net Change in Cash	27.5	-20.4	-14.5	24.4	25.5
Cash at End of Period	208.2	187.7	173.2	197.7	223.2
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net sales	653.0	731.2	813.2	854.8	919.7
Automotive Components	542.8	587.5	648.6	685.2	735.6
Spark & Glow Plugs	362.6	390.7	426.2	458.4	501.9
Automotive Sensors	180.2	196.8	222.4	226.9	233.7
Ceramics	103.1	130.7	154.6	163.6	178.1
New Business	0.0	0.0	0.0	0.0	0.0
Others/Elimination	7.1	13.0	10.0	6.0	6.0
Total Operating Profit	129.7	138.2	158.2	174.4	200.8
Automotive Components	134.2	135.3	149.2	161.5	180.0
Ceramics	-6.3	-4.6	9.0	12.9	20.4
New Business	0.0	0.0	0.0	0.0	0.0
Others/Elimination	1.8	7.4	0.0	0.0	0.4
Operating Profit Margin	19.9%	18.9%	19.5%	20.4%	21.8%
Automotive Components	24.7%	23.0%	23.0%	23.6%	24.5%
Ceramics	-6.1%	-3.5%	5.8%	7.9%	11.5%

Risk Reward – Niterra (5334.T)

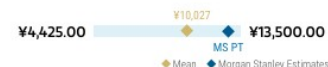
Expect earnings growth in auto components with volume & unit price going up

PRICE TARGET ¥12,600

Calculated from our DCF model derived from our base case. Our DCF model assumes a 2.6% risk-free rate, 111 equity beta, and 3.2% risk premium, yielding a 5.7% WACC. We assume a 0% terminal growth rate from F3/36.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- We expect the firm to maintain competitiveness in automotive plugs and exhaust gas sensors, and market share to keep gradually expanding.
- Although demand for plugs is unlikely to grow substantially (Niterra boasts the top global share at 47%), we expect Niterra's market share to increase gradually and the firm to book large profits on aftermarket products.
- In automotive exhaust gas oxygen sensors, we expect Niterra to gradually increase its 47% global market share, and earnings to keep growing alongside the tightening of environmental regulations.
- In non-auto business, we believe earnings from electrostatic chucks for SPE will continue growing, having bottomed in F3/24.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Contrarian: Positive

Pricing Power: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥14,000

Derived from DCF

Automotive plug & exhaust gas sensor sales exceed the base case by 4% and earnings continue to grow further: This scenario assumes OP of ¥183.7bn in F3/27, ¥202.4bn in F3/28, and ¥293.4bn in F3/31, and uses the same FX & DCF assumptions as the base case. Bull case fair value equates to P/E of 19.9x on F3/27e diluted EPS of ¥704.4, 17.6x on F3/28e diluted EPS of ¥794.9 and 11.6x on F3/31e diluted EPS of ¥1,206.6.

BASE CASE

¥12,600

Derived from DCF

Expect profit to continue to reach new heights, driven by increased sales of automobile plugs and SPE electrostatic chucks: We forecast OP of ¥158.2bn in F3/27, ¥174.4bn in F3/28, and ¥254.4bn in F3/31, assuming FX rates of ¥155/US\$. Our PT is calculated from our DCF model, the conditions for which are outlined in our valuation methodology. Our base case PT equates to P/E of 20.7x on F3/27e diluted EPS of ¥609.6, 18.3x on F3/28e diluted EPS of ¥687.7 and 12.0x on F3/31e diluted EPS of ¥1,049.

BEAR CASE

¥3,700

F3/27e diluted BPS ¥4,036 x P/B 0.9

Lack of growth in autos-related volumes, limited improvement in SPE-related earnings: This scenario assumes OP of ¥113.2bn in F3/27, ¥101.5bn in F3/28, and ¥73.7bn in F3/31, and uses the same FX assumptions as the base case. In this case, the market consensus moves lower and the share price drops to P/B 0.9x.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Minebea Mitsumi: Key Financial Data and Risk Reward

Income Statement (¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,522.7	1,664.4	1,711.2	1,766.0	1,839.2
Cost of Goods Sold	1,252.3	1,371.4	1,393.9	1,429.4	1,482.4
Gross Profit	270.4	293.0	317.3	336.6	356.8
SG&A Expenses	175.9	189.0	195.8	198.9	202.5
Operating Profit	94.5	104.0	121.5	137.8	154.3
Interest/Dividend Income	6.9	40.9	4.0	1.8	1.8
Interest Expense	18.8	11.1	12.0	2.5	2.5
Other Non-Operating Income/Loss	0.0	0.0	0.0	0.0	0.0
Recurring Profit	82.6	133.8	113.5	137.1	153.6
Extraordinary Income/Loss	0.0	0.0	0.0	0.0	0.0
Pretax Profit	82.6	133.8	113.5	137.1	153.6
Income Tax	22.8	34.5	29.5	35.6	39.9
(Effective Tax Rate)	28%	26%	26%	26%	26%
Minority Interest	0.4	0.2	0.0	0.0	0.0
Net Profit	59.5	99.0	84.0	101.4	113.7

Capex	94.8	96.5	105.0	106.0	110.4
Depreciation and Amortization	66.2	70.4	75.0	77.4	80.6
R&D	45.5	49.8	51.0	53.0	55.2
EBITDA	160.7	174.4	196.5	215.2	234.9
ROE	8.2%	12.1%	9.0%	10.1%	10.2%

Average Share Outstanding (mil, Diluted)	406.3	401.6	401.6	401.6	401.6
EPS (Diluted, Yen)	147.7	246.6	209.1	252.6	283.1
BPS (Yen)	1,630.0	2,238.6	2,394.6	2,591.2	2,815.3
DPS (Yen)	45.0	50.0	53.0	56.0	59.0

USD/JPY: Average Exchange Rate	152.6	150.0	155.0	155.0	155.0
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YoY % Change					
Net Sales	0%	0%	0%	0%	0%
Operating Profit	28%	10%	17%	13%	12%
Net Profit	10%	67%	-15%	21%	12%

As % of Sales					
Gross Profit	17.8%	17.6%	18.5%	19.1%	19.4%
Operating Profit	6.2%	6.2%	7.1%	7.8%	8.4%
Net Profit	3.9%	6.0%	4.9%	5.7%	6.2%

Balance Sheet (¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	1,584.8	1,614.8	1,674.9	1,962.3	2,064.8
Total Current Assets	904.1	1,028.7	1,058.8	1,117.7	1,190.4
Cash and Cash Equivalents	214.3	227.5	227.2	263.6	305.0
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	293.3	356.5	380.5	392.8	409.0
Inventory	350.9	391.3	397.7	407.9	423.0
Other Current Assets	170.3	220.9	220.9	220.9	220.9
Total Long Term Assets	680.7	786.1	816.1	844.7	874.4
Net PP&E	529.0	585.3	615.3	643.9	673.6
Investments in Securities	27.1	33.3	33.3	33.3	33.3
Other assets	0.0	0.0	0.0	0.0	0.0
Total Current Liabilities	477.2	602.2	599.6	608.0	620.5
ST Debt & Current Portion of LT Debt	175.1	264.1	264.1	264.1	264.1
Accounts Payable	213.6	235.6	223.7	229.4	237.9
Other Current Liabilities	88.5	102.5	111.8	114.5	118.4
Total Long Term Liabilities	353.0	301.6	301.6	301.6	301.6
Long Term Debt	288.5	218.4	218.4	218.4	218.4
Other Long Term Liabilities	64.4	83.2	83.2	83.2	83.2
Accrued pension cost	0.0	0.0	0.0	0.0	0.0
Minority Interests	11.2	12.1	12.1	12.1	12.1
Total Equity	754.6	911.0	973.7	1,052.7	1,142.7
Net Debt	249.3	255.0	255.3	218.8	177.5

Cash Flow Statement (¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	133.7	94.9	116.7	162.2	171.4
Net Income After Tax	59.5	99.0	84.0	101.4	113.7
Depreciation and Amortization	66.2	70.4	75.0	77.4	80.6
Changes in Working Capital:	9.7	-53.9	-42.3	-16.6	-22.9
Other-net	-1.7	-20.7	0.0	0.0	0.0
Net Cash from Investing Activities	-125.8	-82.8	-95.7	-103.2	-106.4
Purchase of PP&E	-77.4	-77.1	-105.0	-106.0	-110.4
Sale/Purchase of Investments	-0.4	-0.2	0.0	0.0	0.0
Other-net	-48.0	-5.5	9.3	2.7	4.0
Free cash flow	7.9	12.1	20.9	59.0	65.0
Net Cash from Financing Activities	64.0	-16.1	-21.3	-22.5	-23.7
Change in Interest-bearing Debt	96.4	13.0	0.0	0.0	0.0
Change in Treasury Stock	-8.7	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-16.1	-20.1	-21.3	-22.5	-23.7
Other-net	-7.6	-9.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-4.3	17.3	0.0	0.0	0.0
Cash at End of Period	214.3	227.5	227.2	263.6	305.0

Segment Information (¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales by Segment					
Total Sales	1,522.7	1,664.4	1,711.2	1,766.0	1,839.2
Precision Technologies	255.7	281.2	323.2	336.0	356.2
Ball bearings	171.8	186.4	207.9	215.0	229.4
Rod-ends/Fasteners	59.1	66.0	83.0	88.0	93.3
Pivot assemblies	24.8	28.7	32.3	32.9	33.6
Motor, Lighting & Sensing	407.7	456.5	490.0	509.9	530.8
Motors	322.4	340.3	375.0	383.8	413.4
Electronic devices	41.1	68.0	76.0	76.0	76.0
Measuring components	36.9	38.5	39.0	40.2	41.4
Other	7.4	9.8	0.0	0.0	0.0
Semiconductors & Electronics	527.6	590.3	541.0	549.6	563.1
Access Solutions	328.1	332.2	352.0	369.6	388.1
Others	3.5	4.2	5.0	1.0	1.0

Operating Profit by Segment (¥ bil)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Operating Profit	94.5	104.0	121.5	137.8	154.3
Machined Components Business	55.7	62.2	71.0	76.4	83.5
Electronic Devices & Components Business	23.0	26.9	31.5	35.5	39.7
Semiconductors & Electronics	22.0	26.7	28.5	34.1	37.2
Access Solutions	15.9	17.1	19.0	20.3	21.3
Others	-1.2	-2.7	-2.0	-1.0	-1.0
Adjustments	-20.9	-26.3	-26.5	-26.5	-26.5

Operating Profit Margin	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Machined Components Business	6.2%	6.2%	7.1%	7.8%	8.4%
Electronic Devices & Components Business	21.8%	22.1%	22.0%	22.4%	23.5%
Semiconductors & Electronics	5.9%	5.9%	6.4%	7.0%	7.5%
Access Solutions	4.2%	4.5%	5.3%	6.2%	6.6%
Access Solutions	4.9%	5.1%	5.4%	5.5%	5.5%
Others	-33.8%	-63.2%	-40.0%	-100.0%	-100.0%

Risk Reward – Minebea Mitsumi (6479.T)

Earnings should recover gradually, but this is mostly priced into the stock

PRICE TARGET ¥5,300

Derived from the base case. DCF assumptions: 2.6% risk-free rate, 1.27 equity beta, 3.2% risk premium, yielding 5.8% WACC; zero growth from F3/36.



Source: Refinitiv, Morgan Stanley Research

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥6,000	BASE CASE	¥5,300	BEAR CASE	¥2,400
Derived from DCF		Derived from DCF		F3/27e BPS ¥2,346 x P/B 1.0	
Semiconductor & Electronics and Precision Technology sales higher than in our base case: OP ¥14.14bn in F3/27, ¥155.9bn in F3/28, and ¥219.8bn in F3/31. Same FX and DCF model assumptions as the base case. Fair value for the shares equates to 24.4x F3/27e diluted EPS of ¥245.8, 21.0x F3/28e diluted EPS of ¥286.1, and 14.9x F3/28e diluted EPS of ¥403.8.		Earnings continue to grow, driven by rising sales of core products, as the market expects: OP ¥121.5bn in F3/27, ¥137.8bn in F3/28, and ¥186.3bn in F3/31. FX assumptions of ¥155/\$, ¥180/€, DCF model methodology section. Base case fair value equates to 25.3x F3/27e diluted EPS of ¥209.1, 21.0x F3/28e diluted EPS of ¥252.6, and 15.5x F3/31e diluted EPS of ¥342.0.		Earnings especially in Semiconductor & Electronics and Motor, Lighting & Sensing weaker than base case: OP ¥95.3bn in F3/27, ¥100.4bn in F3/28, and ¥116.9bn in F3/31. 6.8% ROE in F3/27 is higher than cost of capital of 6.7%, but earnings are lower than market consensus and the share price declines to 1.0 x P/B.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Mabuchi Motor: Key Financial Data and Risk Reward

Income Statement					
(¥ bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	218.8	233.7	249.0
COGS	143.3	140.6	154.1	166.0	178.3
Gross Profit	52.9	59.8	64.7	67.7	70.8
SG&A Expenses	31.2	34.3	37.5	37.8	38.4
Operating Profit	21.6	25.5	27.2	29.9	32.4
Non-operating Income/Loss	10.8	9.6	5.8	5.8	4.4
Recurring Profit	32.4	35.1	33.1	35.7	36.8
Extraordinary Income/Loss	-8.7	-0.1	-0.2	-0.2	-0.2
Pretax Profit	23.7	35.0	32.8	35.5	36.6
Income Tax	10.9	8.7	9.3	10.3	10.6
(Effective Tax Rate)	46%	25%	28%	29%	29%
Minority Interest	0.0	0.0	0.0	0.0	0.0
Net Profit	12.8	26.3	23.5	25.2	26.0
Capex	14.1	10.3	12.6	15.0	15.0
Depreciation and Amortization	14.2	13.3	13.9	14.0	14.0
R&D Expenses	7.0	7.8	8.8	9.0	9.0
EBITDA	35.8	38.8	41.1	43.9	46.4
ROE	4.1%	8.0%	7.2%	7.9%	8.4%
Average Share Outstanding (Diluted)	254.1	248.1	234.1	234.1	234.1
EPS (Diluted, Yen)	50.5	105.9	95.8	107.7	110.9
BPS (Yen)	1,257.7	1,347.0	1,381.9	1,344.1	1,303.5
DPS (Yen)	38.0	53.0	56.0	64.0	68.0
USD/JPY: Average Exchange Rate	151.6	149.7	155.5	155.0	155.0
YoY % Change					
Net Sales	10%	2%	9%	7%	7%
Operating Profit	39%	18%	7%	10%	8%
Net Profit	-34%	105%	-10%	7%	3%
As % of Sales					
Gross Profit	26.9%	29.8%	29.6%	29.0%	28.4%
Operating Profit	11.0%	12.7%	12.4%	12.8%	13.0%
Net Profit	6.5%	13.1%	10.8%	10.8%	10.4%
Capex	7.2%	5.1%	5.8%	6.4%	6.0%
Depreciation and Amortization	7.2%	6.6%	6.4%	6.0%	5.6%
Balance Sheet					
(¥ bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Assets	355.0	370.2	399.4	351.3	342.7
Total Current Assets	238.5	247.7	238.2	229.2	219.5
Cash and Cash Equivalents	134.9	143.4	130.0	118.7	107.4
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	37.3	38.7	38.0	40.0	41.9
Inventory	59.8	57.8	62.4	62.7	62.5
Other Current Assets	6.4	7.8	7.8	7.8	7.8
Total Long Term Assets	116.5	122.4	121.1	122.1	123.1
Net PP&E	95.5	95.9	94.5	95.6	96.6
Investments in Securities	12.4	16.4	16.4	16.4	16.4
Other Assets	8.6	10.2	10.2	10.2	10.2
Total Current Liabilities	28.8	28.3	28.3	29.1	29.9
ST Debt and Current Portion of LT Debt	0.0	0.0	0.0	0.0	0.0
Accounts Payable	7.7	10.4	10.3	11.1	11.9
Other Current Liabilities	21.1	17.9	17.9	17.9	17.9
Total Long Term Liabilities	6.5	7.6	7.6	7.6	7.6
Long Term Debt	0.5	1.3	1.3	1.3	1.3
Other Long Term Liabilities	6.0	6.3	6.3	6.3	6.3
Net Assets	319.6	334.2	323.4	314.6	305.1
Minority Interests	0.0	0.0	0.0	0.0	0.0
Net Debt	-146.8	-158.5	-145.0	-133.8	-122.4

Cash Flow Statement					
(¥ bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Cash from Operations	40.1	35.4	33.5	37.8	39.1
Net Income	12.8	26.3	23.5	25.2	26.0
Depreciation and Amortization	14.2	13.5	13.9	14.0	14.0
Changes in Working Capital	2.2	8.1	-4.0	-1.4	-0.9
Other-net	10.9	-12.5	0.0	0.0	0.0
Net Cash from Investing Activities	-15.8	-10.5	-12.6	-15.0	-15.0
Purchase of PP&E	-14.0	-10.2	-12.6	-15.0	-15.0
Other-net	-1.2	-0.3	0.0	0.0	0.0
Free Cash Flow	24.4	24.8	20.9	22.8	24.1
Net Cash from Financing Activities	-16.2	-17.4	-34.3	-34.0	-35.4
Change in Interest-bearing Debt	0.0	0.0	0.0	0.0	0.0
Change in Treasury Stock	-5.7	-6.7	-19.0	-20.0	-20.0
Payment of Cash Dividends	-10.2	-9.7	-15.3	-14.0	-15.4
Other-net	-0.3	-1.0	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	10.1	1.9	0.0	0.0	0.0
Net Change in Cash	18.3	9.4	-13.4	-11.3	-11.4
Cash at End of Period	130.6	139.9	126.5	115.2	103.9
Sales by Application					
(¥ bn)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Net Sales	196.2	200.4	218.8	233.7	249.0
Automotive Products	152.5	154.5	166.3	178.1	190.6
Life & Industrial Products	43.7	45.9	52.5	55.6	58.4
YoY % Change					
Net Sales	10%	2%	9%	7%	7%
Automotive Products	9%	1%	8%	7%	7%
Life & Industrial Products	13%	5%	15%	6%	5%
As % of Sales					
Automotive Products	78%	77%	76%	76%	77%
Life & Industrial Products	22%	23%	24%	24%	23%
Shipment Volume by Application					
(mn units)	F12/24	F12/25	F12/26e	F12/27e	F12/28e
Total Sales Quantity	1,348	1,340	1,336	1,414	1,496
Automotive Products	1,055	1,071	1,073	1,137	1,206
Life & Industrial Products	293	269	263	276	290
Precision & Office Equipment					
Total Sales Quantity	2%	-1%	0%	6%	6%
Automotive Products	2%	2%	0%	6%	6%
Life & Industrial Products	-1%	-8%	-2%	5%	5%
As % of Sales					
Automotive Products	78%	80%	80%	80%	81%
Life & Industrial Products	22%	20%	20%	20%	19%

Risk Reward – Mabuchi Motor (6592.T)

Gradual profit growth on auto output recovery, headcount increases

PRICE TARGET ¥1,750

Our PT is calculated from our DCF model derived from our base case, and assumes a risk-free rate of 2.6%, equity beta of 0.83, and risk premium of 3.2%, yielding WACC of 5.3%. We assume a 0% perpetual growth rate for F12/36 onwards.

Consensus Price Target Distribution



RISK REWARD CHART



EQUAL-WEIGHT THESIS

- ROE improves gradually due to moderate recovery in profit, driven by global auto production recovery and headcount increases.
- Higher automotive motor volumes drive earnings growth over the longer term.

Consensus Rating Distribution



Risk Reward Themes

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥2,300

Derived from DCF

Automotive motor output exceeds our base case by 8% thanks to demand growth and market share gains: This scenario assumes OP of ¥35.0bn in F12/26, ¥37.7bn in F12/27 and ¥47.7bn in F12/30 using the same FX and DCF assumptions as the base case. Our bull case fair value estimate equates to P/E of 18.5x (13.7x excluding net financial assets) on F12/26e diluted EPS of ¥124.3.

BASE CASE

¥1,750

Derived from DCF

Earnings recover gradually due mainly to global auto production recovery: We forecast OP of ¥27.2bn in F12/26, ¥29.9bn in F12/27 and ¥45.2bn in F12/30, assuming an FX rate of ¥155/\$. Our PT is derived from our DCF model, the conditions for which are outlined in our Valuation Methodology. Our PT equates to P/E of 18.3x (11.8x excluding net financial assets) on F12/26e diluted EPS of ¥95.8, 16.3x on F12/27e diluted EPS of ¥107.7 and 11.6x on F12/30e diluted EPS of ¥150.5.

BEAR CASE

¥1,200

End-F12/26e diluted BPS ¥1,379 x P/B 0.9
Profit drops in F12/26 due to rising costs and automobile production decline: This scenario is for OP of ¥23.0bn in F12/26, ¥23.1bn in F12/27 and ¥26.7bn in F12/30, assuming ROE declines to 6.3% in F12/26, remaining above the cost of capital of 5.3%; however, the market consensus has been revised down sharply, and the share price is expected to fall to a P/B of 0.9x, but end-F12/25 net financial assets/share of ¥645 and shareholder returns underpin the share price.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

TDK: Key Financial Data and Risk Reward

Income Statement					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	2,768.0	3,020.8	3,295.6
Cost of Goods Sold	1,516.9	1,721.4	1,879.0	2,075.6	2,281.3
Gross Profit	688.0	783.4	889.0	945.0	1,014.3
SG&A Expenses	494.0	544.0	553.2	575.8	599.6
Others*	30.2	33.0	0.0	0.0	0.0
Operating Profit	224.2	272.4	335.8	369.1	414.7
Non-operating Income/Loss	13.6	4.4	4.0	7.6	8.7
Pretax Profit	237.8	276.8	339.8	376.7	423.4
Income Tax	67.4	77.6	91.7	98.0	105.9
(Effective Tax Rate)	28%	28%	27%	26%	25%
Minority Interest	3.2	3.5	3.0	4.0	4.0
Net Profit	167.2	195.7	245.0	274.8	313.6
Capex	225.3	298.6	370.0	350.0	350.0
Depreciation and Amortization	196.2	204.2	240.0	260.0	280.0
R&D Expenses	253.6	289.7	310.0	320.0	330.0
EBITDA	420.4	476.6	575.8	629.1	694.7
ROE	9.5%	9.7%	10.7%	11.2%	11.8%
Average Share Outstanding (Diluted)	1,900.0	1,900.2	1,896.7	1,898.2	1,898.2
EPS (Diluted, Yen)	88.0	103.0	129.1	144.8	165.2
BPS (Yen)	953.3	1,159.6	1,244.4	1,346.2	1,465.4
DPS (Yen)	30.0	36.0	40.0	44.0	48.0
USD/JPY: Average Exchange Rate	152.7	150.8	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	163.9	174.8	180.0	180.0	180.0
YoY % Change					
Net Sales	5%	14%	11%	9%	9%
Operating Profit	30%	22%	23%	10%	12%
Net Profit	34%	17%	25%	12%	14%
As % of Sales					
Gross Profit	31.2%	31.3%	32.1%	31.3%	30.8%
Operating Profit	10.2%	10.9%	12.1%	12.2%	12.6%
Net Profit	7.6%	7.8%	8.9%	9.1%	9.5%
Capex	10.2%	11.9%	13.4%	11.6%	10.6%
Depreciation and Amortization	8.9%	8.2%	8.7%	8.6%	8.5%
Balance Sheet					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,541.4	4,415.2	4,298.1	4,540.4	4,816.0
Total Current Assets	1,836.9	2,455.2	2,208.1	2,360.5	2,566.0
Cash and Cash Equivalents	697.3	842.8	894.4	996.4	1,154.6
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	583.1	780.6	602.5	632.6	663.2
Inventory	410.0	585.4	464.8	485.0	501.9
Other Current Assets	146.4	246.4	246.4	246.4	246.4
Total Long Term Assets	1,704.6	1,959.9	2,089.9	2,179.9	2,249.9
Net PP&E	1,030.1	1,225.8	1,355.8	1,445.8	1,515.8
Investments in Securities	183.8	226.8	226.8	226.8	226.8
Other Assets	276.6	269.2	269.2	269.2	269.2
Total Current Liabilities	1,098.5	1,574.5	1,281.5	1,324.8	1,370.1
ST Debt and Current Portion of LT Debt	187.1	211.0	211.0	211.0	211.0
Accounts Payable	392.5	706.7	413.7	457.0	502.3
Other Current Liabilities	518.9	656.9	656.9	656.9	656.9
Total Long Term Liabilities	631.7	637.1	637.1	637.1	637.1
Long Term Debt	346.0	332.7	332.7	332.7	332.7
Other Long Term Liabilities	285.7	304.4	304.4	304.4	304.4
Minority Interests	11.2	16.3	19.3	23.3	27.3
Total Equity	1,811.3	2,203.5	2,360.2	2,555.3	2,781.5
Net Debt	-348.0	-525.9	-577.5	-679.6	-837.8

Cash Flow Statement					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	445.8	507.7	493.7	531.7	595.5
Net Income	167.2	195.7	245.0	274.8	313.6
Depreciation and Amortization	196.2	204.2	240.0	260.0	280.0
Changes in Working Capital	2.5	-8.9	5.7	-7.1	-2.1
Other-net	80.0	116.7	3.0	4.0	4.0
Net Cash from Investing Activities	-244.8	-377.8	-370.0	-350.0	-350.0
Purchase of PP&E	-225.3	-298.6	-370.0	-350.0	-350.0
Sale/Purchase of Investments	9.9	-4.3	0.0	0.0	0.0
Other-net	-29.5	-74.9	0.0	0.0	0.0
Free Cash Flow	201.0	129.9	123.7	181.7	245.5
Net Cash from Financing Activities	-143.3	-64.7	-72.1	-79.7	-87.3
Change in Interest-bearing Debt	-81.3	9.9	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-48.5	-60.7	-72.1	-79.7	-87.3
Other-net	-13.5	-13.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-10.4	80.3	0.0	0.0	0.0
Net Change in Cash	47.3	145.5	51.6	102.0	158.2
Cash at End of Period	697.3	842.8	894.4	996.4	1,154.6
Segment Information					
(¥ bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,204.8	2,504.8	2,768.0	3,020.8	3,295.6
Passive Components	559.6	593.2	647.4	674.9	704.0
Capacitors	234.3	257.5	280.9	294.9	309.6
Inductive Devices	204.3	216.2	244.9	254.7	264.9
Other Passive Components	121.1	119.5	121.6	125.3	129.4
Sensor Application Products	189.5	224.6	233.5	249.4	266.5
Magnetic Application Products	223.6	262.9	370.8	404.1	436.0
Energy Application Products	1,176.5	1,370.3	1,466.2	1,642.1	1,839.2
Other	55.6	53.8	50.0	50.0	50.0
Total Operating Profit	224.2	272.4	335.8	369.1	414.7
Passive Components	34.1	41.8	54.0	66.3	75.0
Sensor Application Products	5.0	20.7	22.2	30.2	35.8
Magnetic Application Products	3.4	27.0	61.5	68.2	74.4
Energy Application Products	234.4	246.7	256.1	262.3	287.6
Other	-4.4	-10.2	-6.0	-6.0	-6.0
Corporate and Eliminations	-48.3	-53.6	-52.0	-52.0	-52.0
Profit Margin	10.2%	10.9%	12.1%	12.2%	12.6%
Passive Components	6.1%	7.1%	8.3%	9.8%	10.6%
Sensor Application Products	2.6%	9.2%	9.5%	12.1%	13.4%
Magnetic Application Products	1.5%	10.3%	16.6%	16.9%	17.1%
Energy Application Products	19.9%	18.0%	17.5%	16.0%	15.9%
Other	-8.0%	-19.0%	-12.0%	-12.0%	-12.0%

Risk Reward – TDK (6762.T)

We expect LiB profit growth and continued improvement beyond LiB

PRICE TARGET ¥4,900

Derived from the base case. DCF assumptions: risk-free rate 2.6%, equity beta 1.31, risk premium 3.2%, yielding WACC 6.5%. 0% perpetual growth rate from F3/36.

Consensus Price Target Distribution



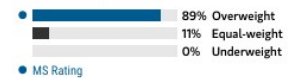
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect Passive Components and Sensor Application earnings to continue to improve after the bottom in F3/25.
- We envision only modest LiB profit growth in F3/27 due to higher cobalt prices, etc., but sustained LiB profit growth in the mid-long term as sales expand in high value-added silicon cathode products and DC BBU batteries.
- HDD head and suspension earnings continue to expand due to increased demand for NL HDD for DC.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥6,000	BASE CASE	¥4,900	BEAR CASE	¥1,500
Derived from DCF					
Earnings from rechargeable batteries, passive components, sensors, and HDD heads ahead of base case. OP ¥373.2bn in F3/27, ¥436.8bn in F3/28, ¥730.1bn in F3/31. Same currency and DCF model assumptions as base case. Fair value equates to P/E 41.9x on F3/27e diluted EPS ¥143.2, P/E 35.1x on F3/28e diluted EPS ¥170.8, P/E 20.6x on F3/31e diluted EPS ¥290.8.					
Derived from DCF					
Modest rechargeable battery profit gains in F3/27, but mid-long term profit growth not only for rechargeable batteries, but also HDDs, sensors & passive components. OP ¥335.8bn in F3/27, ¥369.1bn in F3/28, ¥518.6bn in F3/31. Assumes ¥155/\$; DCF derived from assumptions listed in valuation methodology. PT equates to P/E 38.0x F3/27e diluted EPS ¥129.1, 33.8x F3/28e diluted EPS ¥144.8, 23.6x F3/31e diluted EPS ¥207.6.					
End-F3/27e diluted BPS ¥1,202 x P/B 1.2x					
Limited profit growth due to rechargeable battery slowdown. OP ¥230.1bn in F3/27, ¥251.7bn in F3/28, ¥373.9bn in F3/31. ROE 7.5% in F3/27 and 7.9% in F3/28, above the cost of capital of 6.8%. P/B sinks to 1.2x due to earnings missing market consensus.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Alps Alpine: Key Financial Data and Risk Reward

Income Statement (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,064.5	1,157.6	1,222.2
Cost of Goods Sold	815.2	837.6	859.9	923.3	967.7
Gross Profit	175.2	181.9	204.6	234.2	254.5
SG&A Expenses	141.1	139.8	155.5	165.5	174.9
Operating Profit	34.1	42.0	49.1	68.7	79.7
Non-Operating Income/Loss	-3.6	7.1	-2.8	2.2	2.2
Pretax profit	57.9	43.9	44.3	68.9	79.9
Income Tax	19.8	16.7	12.8	20.0	23.2
(Effective Tax Rate)	34%	38%	29%	29%	29%
Minority Interest	0.2	0.4	0.4	0.0	0.0
Net Profit	37.8	26.9	31.1	48.9	56.7

Capex	51.8	61.6	75.2	60.0	60.0
Depreciation and Amortization	35.1	34.0	44.8	42.0	42.0
R&D	24.3	21.9	28.0	33.0	35.0
EBITDA	69.2	76.0	93.9	110.7	121.7
ROE	9.4%	6.2%	6.8%	10.1%	10.9%

Average Share Outstanding (Diluted)	205.6	199.4	195.1	195.1	195.1
EPS (Diluted, Yen)	184.0	134.8	159.2	250.7	290.7
BPS (Yen)	2,013.1	2,293.7	2,388.8	2,569.6	2,785.3
DPS (Yen)	60.0	62.0	64.0	70.0	75.0

USD/JPY: Average Exchange Rate	151.3	150.8	155.0	155.0	155.0
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YoY % Change					
Net Sales	3%	3%	4%	9%	6%
Operating Profit	73%	23%	17%	40%	16%
Net Profit	-	-29%	16%	58%	16%

As % of Sales					
Gross Profit	17.7%	17.8%	19.2%	20.2%	20.8%
Operating Profit	3.4%	4.1%	4.6%	5.9%	6.5%
Net Profit	3.8%	2.8%	2.9%	4.2%	4.6%
Capex	5.2%	6.0%	7.1%	5.2%	4.9%
Depreciation and Amortization	3.5%	3.3%	4.2%	3.6%	3.4%

Balance Sheet (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	740.7	783.2	802.8	845.0	891.9

Total Current Assets	494.9	500.7	489.9	514.1	543.1
Cash and Cash Equivalents	147.9	153.6	150.0	148.4	159.4
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	166.8	169.4	175.5	190.9	201.5
Inventory	138.6	138.3	142.2	152.6	160.0
Other Current Assets	41.5	39.4	22.2	22.2	22.2
Total Long Term Assets	245.8	282.4	312.8	330.8	348.8
Net PP&E	136.7	158.0	188.4	206.4	224.4
Investments in Securities	60.0	67.7	67.7	67.7	67.7
Other assets	49.1	56.8	56.8	56.8	56.8

Total Current Liabilities	226.9	240.7	241.8	248.7	253.6
ST Debt & Current Portion of LT Debt	44.3	37.1	37.1	37.1	37.1
Accounts Payable	90.4	93.2	94.2	101.2	106.0
Other Current Liabilities	92.2	110.5	110.5	110.5	110.5
Total Long Term Liabilities	98.3	93.0	93.0	93.0	93.0
Long Term Debt	56.5	55.2	55.2	55.2	55.2
Other Long Term Liabilities	41.8	37.8	37.8	37.8	37.8
Minority Interests	1.5	1.9	1.9	1.9	1.9
Total Equity	415.5	449.4	468.0	503.2	545.3

Net Debt	-107.2	-129.0	-125.4	-123.8	16.8
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Cash Flow Statement (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	65.8	95.9	66.9	72.0	85.6
Net Income	37.8	26.9	31.1	48.9	56.7
Depreciation and Amortization	35.1	34.0	44.8	42.0	42.0
Changes in Working Capital:	9.0	16.0	-9.0	-18.9	-13.1
Other-net	-16.2	19.1	0.0	0.0	0.0
Net Cash from Investing Activities	-1.7	-58.4	-58.0	-60.0	-60.0
Purchase of PP&E	-43.0	-43.6	-75.2	-60.0	-60.0
Sale/Purchase of Investments	0.0	0.0	0.0	0.0	0.0
Other-net	41.4	-14.8	17.2	0.0	0.0

Free cash flow	64.1	37.5	8.9	12.0	25.6
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Net Cash from Financing Activities	-37.3	-41.1	-12.5	-13.7	-14.6
Change in Interest-bearing Debt	-28.5	-9.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-8.2	-12.2	-12.5	-13.7	-14.6
Other-net	-0.5	-19.4	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-1.7	9.5	0.0	0.0	0.0
Net Change in Cash	25.2	5.9	-3.6	-1.6	11.0

Segment Information (Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	990.4	1,019.5	1,064.5	1,157.6	1,222.2
Components					
Sensor & Communication	84.2	85.3	87.0	91.4	95.9
Mobility	537.2	555.1	544.0	578.2	616.9
Logistics/Others	21.0	20.8	20.0	20.0	20.0

Total Operating Profit	34.1	42.0	49.1	68.7	79.7
Components					
Sensor & Communication	30.4	30.2	37.1	51.7	60.2
Mobility	-3.4	-3.5	-1.2	2.0	2.5
Logistics/Others	1.5	1.3	1.0	1.0	1.0

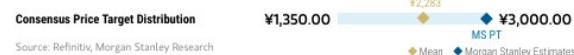
Operating Profit Margin	3.4%	4.1%	4.6%	5.9%	6.5%
Components					
Sensor & Communication	8.7%	8.4%	9.0%	11.0%	12.3%
Mobility	-4.0%	-4.1%	-1.4%	2.2%	2.6%
Logistics/Others	7.0%	6.0%	5.0%	5.0%	5.0%

Risk Reward – ALPS ALPINE (6770.T)

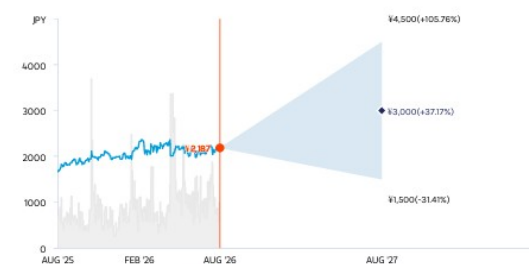
Expect earnings to expand in line with company mid-term plan, above consensus.

PRICE TARGET ¥3,000

Derived from the base case. DCF model assumptions: risk-free rate 2.6%, equity beta 0.91, risk premium 3.2%, yielding WACC 5.0%. 0% perpetual growth rate from F3/36.



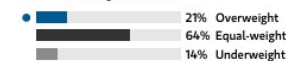
RISK REWARD CHART



OVERWEIGHT THESIS

- We expect a significant expansion in profits from smartphone camera actuators and components starting in F3/27.
- We expect only limited improvement in earnings for the Module & System and Mobility segments.
- We expect continued earnings expansion in line with the mid-term plan's final-year (F3/28), which significantly exceeds market consensus.

Consensus Rating Distribution



Risk Reward Themes

Contrarian: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE ¥4,500 BASE CASE ¥3,000 BEAR CASE ¥1,500

Derived from DCF

Earnings improve at a pace exceeding the mid-term management plan. OP ¥67.6bn in F3/27, ¥92.2bn in F3/28, ¥127.3bn in F3/31. Same currency and DCF model assumptions as base case. Fair value equates to P/E 20.9x F3/27e diluted EPS ¥214.9, 14.0x F3/28e diluted EPS ¥321.9, 9.7x F3/31e diluted EPS ¥463.9.

Derived from DCF

We expect continued strong profit growth in line with the mid-term plan, driven by camera actuators. We forecast OP ¥49.1bn in F3/27, ¥68.7bn in F3/28, ¥103.8bn in F3/31. Base case assumes ¥155/\$ PT equates to P/E 18.9x F3/27e diluted EPS ¥159.2, 12.0x F3/28e diluted EPS ¥250.7 and 7.9x F3/31e EPS ¥378.4.

F3/27e diluted BPS ¥2,204 x P/B 0.7x

Sluggish output of high-end smartphones and declining market share cause smartphone camera actuator sales to fall 10% short of our base case. OP ¥31.8bn in F3/27, ¥34.7bn in F3/28, ¥45.2bn in F3/31. ROE 4.2% in F3/27, lower than the 5.5% cost of capital we assume, and the share drops to 0.7x F3/27e diluted BPS from earnings missing market consensus.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nihon Dempa Kogyo: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	53.1	54.6	61.9	66.6	72.4	Cash flow from operating activities	6.1	4.2	8.2	7.8	9.0
Costs of Sales	37.0	38.9	43.0	46.2	49.8	EBIT	3.0	2.6	4.1	4.8	6.0
Gross Profit	16.1	15.7	18.9	20.4	22.6	Depreciation	3.4	3.9	4.4	4.9	5.4
SG&A (incl. other income/losses)	11.4	12.4	14.3	15.1	16.0	Changes in Receivables	0.2	-0.5	-0.6	-1.1	-1.3
Operating Income	4.6	3.4	4.6	5.3	6.6	Change in inventory	0.6	-1.5	-0.3	-0.6	-0.8
Financial income/losses	-1.7	-0.8	-0.5	-0.5	-0.6	Changes in Payables	0.0	1.2	1.4	0.9	1.1
Profit before income taxes	3.0	2.6	4.1	4.8	6.0	Other	-1.1	-1.5	-0.8	-1.1	-1.4
Income taxes	1.2	0.5	1.3	1.5	1.8	Cash flow from investing activities	-4.5	-7.4	-10.0	-9.0	-8.8
(Effective tax rate)	39%	19%	32%	31%	30%	Purchase of PP&E	-3.7	-5.7	-9.5	-8.6	-9.3
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.1	0.0	0.0	0.0	0.0
Net Profit	1.8	2.1	2.8	3.3	4.2	Other	-0.9	-1.7	-0.5	-0.4	-0.5
Capex	4.6	6.5	10.0	9.0	9.8	Cash flow from financing activities	1.9	-2.4	1.7	0.6	1.6
Depreciation & amortization	3.4	3.9	4.4	4.9	5.4	Net Change in Borrowings	3.3	-0.7	2.4	1.5	2.5
R&D	1.8	2.1	2.9	3.7	4.0	Purchase of treasury shares	0.0	-0.2	0.0	0.0	0.0
EBITDA	8.0	7.3	9.0	10.2	12.0	Cash dividend paid	-0.7	-0.7	-0.7	-0.9	-0.9
ROE	6.3%	6.8%	8.5%	9.4%	11.0%	Other	-0.7	-0.9	0.0	0.0	0.0
No of average shares (mn)	23.0	23.0	22.9	22.9	22.9	Effect of exchange rate change	0.0	0.5	0.0	0.0	0.0
EPS (JPY)	77.8	89.7	122.1	143.9	163.2	Changes in cash and cash equivalents	3.6	-5.1	-0.1	-0.6	0.8
BPS (JPY)	1,265.0	1,392.1	1,484.2	1,588.1	1,731.3	Cash and cash equivalents at end of the period	15.9	10.8	10.7	10.1	10.9
DPS (JPY)	30.0	30.0	35.0	40.0	45.0	FCF	1.7	-3.2	-1.8	-1.2	-0.8
USD/JPY (Average)	152.5	151.0	155.0	155.0	155.0	Sales by Application					
Balance Sheet						(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	71.5	76.3	82.2	87.1	94.0	Automotive	29.2	30.1	32.1	33.1	34.0
Current assets	43.5	44.0	44.7	45.8	48.7	Mobile communications / IoT	13.5	13.0	13.5	13.8	14.1
Cash and cash equivalents	15.9	10.8	10.7	10.1	10.9	Industrial equipment	3.7	4.4	8.4	11.3	15.3
Accounts receivable	12.7	13.8	14.4	15.5	16.8	Special products	1.4	1.7	2.3	2.7	3.1
Inventories	10.5	12.4	12.7	13.3	14.1	Optical products	1.4	1.2	1.5	1.6	1.8
Other financial assets	4.4	6.9	6.9	6.9	6.9	Others	3.9	4.3	4.1	4.1	4.1
Fixed assets	28.0	32.3	37.5	41.2	45.3	(YoY)					
Tangible fixed assets	19.7	23.2	28.1	31.8	35.7	Automotive	4%	3%	7%	3%	3%
Intangible fixed assets	2.3	3.2	3.4	3.5	3.7	Mobile communications / IoT	22%	-4%	4%	2%	2%
Other non-current assets	6.0	5.9	5.9	5.9	5.9	Industrial equipment	28%	19%	91%	35%	35%
Current liabilities	12.2	14.0	15.4	16.4	17.6	Special products	-	21%	36%	15%	15%
Short term borrowings	0.8	1.3	1.3	1.4	1.4	Optical products	-	-18%	28%	12%	12%
Accounts payable	9.5	10.3	11.7	12.6	13.7	Others	-	11%	-4%	0%	0%
Other current liabilities	1.9	2.3	2.4	2.4	2.5						
Fixed liabilities	30.2	30.4	32.8	34.3	36.7						
Long term borrowings	25.8	24.8	27.2	28.7	31.1						
Other fixed liabilities	4.4	5.6	5.6	5.6	5.6						
Net assets	29.2	31.9	34.0	36.4	39.7						
Total liabilities and equity	71.5	76.3	82.2	87.1	94.0						

Risk Reward – Nihon Dempa Kogyo (6779.T)

Ensuring profitability of DC, server products is key

PRICE TARGET ¥4,900

Our price target is ¥4,900, based on a DCF valuation (risk-free rate 2.6%; beta 1.2; risk premium 3.2%; WACC 4.4%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥350; our price target equates to P/E 40.1x our F3/27e EPS of ¥122.1, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



BULL CASE	¥9,300	BASE CASE	¥4,900	BEAR CASE	¥1,000
Accelerated sales growth for industrials/defence		AI/DC oscillator demand grows, mainly in optics		Weak demand for automotive, N America smartphones	
OP and EPS forecasts of ¥4.6bn/¥122.1 in F3/27, ¥7.0bn/¥196.3 in F3/28, ¥9.3bn/¥261.7 in F3/29, ¥12.3bn/¥353.3 in F3/30, and ¥16.6bn/¥479.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥600. Our fair value equates to 76.2x our F3/27e EPS of ¥122.1 and 6.3x BPS of ¥1,484.2.		OP and EPS forecasts of ¥4.6bn/¥122.1 in F3/27, ¥5.3bn/¥143.9 in F3/28, ¥6.6bn/¥183.2 in F3/29, ¥8.1bn/¥226.8 in F3/30, and ¥10.3bn/¥292.2 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥350. Our PT equates to 40.1x our F3/27e EPS of ¥122.1 and 3.3x BPS of ¥1,484.2.		OP and EPS forecasts of ¥4.6bn/¥122.1 in F3/27, ¥4.5bn/¥122.1 in F3/28, ¥4.5bn/¥122.1 in F3/29, ¥4.6bn/¥122.1 in F3/30, and ¥4.7bn/¥122.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥120. Our fair value equates to 8.2x our F3/27e EPS of ¥122.1 and 0.7x BPS of ¥1,484.2.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Meiko Electronics: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	206.8	240.6	323.8	412.9	482.1	Cash flow from operating activities	21.7	27.5	27.3	50.4	74.4
Costs of Sales	167.0	190.1	250.8	321.0	369.6	EBIT	18.5	27.4	35.6	44.7	58.6
Gross Profit	39.8	50.4	73.0	91.9	112.5	Depreciation	12.5	13.7	17.5	34.8	42.0
SG&A	20.7	25.9	34.4	43.2	50.0	Changes in Receivables	-3.3	-8.3	-14.5	-14.2	-9.1
Operating Income	19.1	24.6	38.6	48.7	62.5	Change in inventory	2.9	6.5	13.5	14.5	11.2
Non-operating income/expenses	-0.3	1.9	-3.0	-4.0	-3.9	Changes in Payables	-3.4	-3.3	-8.1	-10.2	-13.4
Recurring Profit	18.8	26.5	35.6	44.7	58.6	Other	-5.5	-8.5	-16.8	-19.1	-14.9
Extraordinary income	-0.2	0.9	0.0	0.0	0.0	Cash flow from investing activities	-24.3	-55.5	-70.0	-70.0	-70.0
Profit before income taxes	18.5	27.4	35.6	44.7	58.6	Purchase of PP&E	-24.4	-49.8	-69.8	-69.8	-69.8
Income taxes	3.4	7.5	8.1	10.2	13.4	Sales of PP&E	0.2	0.0	0.0	0.0	0.0
(Effective tax rate)	19%	27%	23%	23%	23%	Other	-0.1	-5.7	-0.2	-0.2	-0.2
Profit attributable to non-controlling interests	0.2	0.2	0.2	0.2	0.2	Cash flow from financing activities	4.1	30.1	54.4	32.2	-11.8
Net Profit	14.9	19.8	27.3	34.3	45.0	Net Change in Borrowings	7.4	33.1	58.3	36.8	-5.9
Capex	29.8	49.7	70.0	70.0	70.0	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	12.5	13.7	17.5	34.8	42.0	Cash dividend paid	-2.4	-2.7	-3.9	-4.6	-5.9
R&D	5.3	6.3	6.0	9.7	10.3	Other	-0.8	-0.3	0.0	0.0	0.0
EBITDA	31.6	38.3	56.1	83.5	104.5	Effect of exchange rate changes	0.1	1.4	-7.5	0.0	0.0
ROE	14.5%	16.2%	18.9%	20.5%	22.3%	Changes in cash and cash equivalents	1.6	3.6	4.2	12.6	-7.3
						Cash and cash equivalents at end of the period	22.9	26.5	30.7	43.3	36.0
No of average shares (mn)	26.2	26.1	25.7	25.7	25.7	FCF	-27.9	-42.7	-19.6	4.4	
EPS (JPY)	569.5	758.6	1,063.6	1,336.4	1,753.2						
BPS (JPY)	4,218.2	5,309.2	5,932.4	7,088.8	8,612.0	Sales by Application					
DPS (JPY)	88.0	115.0	160.0	200.0	260.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	151.1	155.0	155.0	155.0	Automotive	90.4	97.5	101.3	104.0	107.0
						Smartphone/Tablet	25.5	29.4	36.8	54.5	70.8
						Satellite	16.6	28.2	56.0	82.0	102.0
						AI server	4.2	7.3	20.0	45.0	60.0
						Moldes	9.2	9.2	13.0	16.0	18.0
						Amusement/Smart appliance/Others	23.7	24.4	27.8	28.2	30.1
						Electronic equipment	36.2	42.6	66.0	79.2	89.2
						Semi Package	1.0	2.0	3.0	4.0	5.0
						(YoY)					
						Automotive	-5%	8%	4%	3%	3%
						Smartphone/Tablet	8%	15%	25%	48%	30%
						Satellite	118%	70%	99%	46%	24%
						AI server	-	74%	174%	125%	33%
						Moldes	-	0%	41%	23%	13%
						Amusement/Smart appliance/Others	69%	3%	14%	1%	7%
						Electronic equipment	19%	18%	55%	20%	13%
						Semi Package	43%	100%	50%	33%	25%

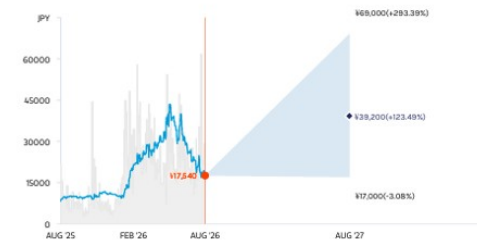
Risk Reward – Meiko Electronics (6787.T)

Rapid Earnings Growth Driven by Aggressive Investment Largely Priced in

PRICE TARGET ¥39,200

Our price target is ¥39,200, based on a DCF valuation (risk-free rate 2.6%; beta 1.5; risk premium 3.2%; WACC 4.6%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥2,800; our price target equates to 36.9x our F3/27e EPS of ¥1,063.6, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥69,000	BASE CASE	¥39,200	BEAR CASE	¥17,000
Faster demand growth in buildup/semi PKG substrate					
OP and EPS forecasts of ¥42.4bn/¥1,176.6 in F3/27, ¥61.9bn/¥1,726.0 in F3/28, ¥87.5bn/¥2,497.4 in F3/29, ¥111.3bn/¥3,214.3 in F3/30, and ¥132.2bn/¥3,868.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥4,500. Our fair value equates to 58.6x our F3/27e EPS of ¥1,176.6 and 114x BPS of ¥6,045.4.					
Continued growth in high layer/density buildup PCB					
OP and EPS forecasts of ¥38.6bn/¥1,063.6 in F3/27, ¥48.7bn/¥1,336.4 in F3/28, ¥62.5bn/¥1,753.2 in F3/29, ¥73.4bn/¥2,084.4 in F3/30, and ¥83.8bn/¥2,427.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥2,800. Our PT equates to 36.9x our F3/27e EPS of ¥1,063.6 and 6.6x BPS of ¥5,932.4.					
AI server/satellite demand weakens					
OP and EPS forecasts of ¥36.4bn/¥997.4 in F3/27, ¥34.5bn/¥919.5 in F3/28, ¥38.6bn/¥1,048.1 in F3/29, ¥43.0bn/¥1,184.4 in F3/30, and ¥47.6bn/¥1,351.9 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥1,550. Our fair value equates to 17.0x our F3/27e EPS of ¥997.4 and 2.9x BPS of ¥5,866.2.					

EQUAL-WEIGHT THESIS

Our scenario has the firm benefitting from production capacity enhancements undertaken to meet growing demand. We evaluate positively its proactive stance on expanding production capacity for build-up substrates, including those for satellite communications and high-end smartphones. In our view, Meiko's expertise in producing build-up substrates in Vietnam will enable it to benefit from growing demand for high layer/density buildup PCBs, as well as from efforts to move supply chains out of Greater China.

Risk Reward Themes

New Data Era:
Secular Growth: Positive
Positive

View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Hirose Electric: Key Financial Data and Risk Reward

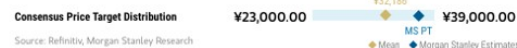
Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	189.4	211.3	234.6	254.6	275.4	Cash flow from operating activities	55.7	45.9	47.9	57.5	63.6
Costs of Sales	104.0	122.3	139.2	149.5	160.3	EBIT	46.2	46.6	52.1	58.2	64.7
Gross Profit	85.4	89.0	95.4	105.1	115.1	Depreciation	17.9	19.5	21.3	23.1	25.0
SG&A (incl. other income/losses)	42.7	46.0	48.1	51.7	55.2	Changes in Receivables	-1.9	-7.2	-5.5	-4.7	-4.9
Operating Income	42.7	43.0	47.3	53.4	59.9	Change in inventory	-1.4	-3.5	-3.3	-2.9	-3.0
Financial income/losses	3.5	3.6	4.8	4.8	4.8	Changes in Payables	2.0	3.5	1.8	1.6	1.6
Profit before income taxes	46.2	46.6	52.1	58.2	64.7	Other	-7.2	-13.0	-18.5	-17.8	-19.9
Income taxes	13.2	13.5	15.9	17.8	19.8	Cash flow from investing activities	-42.9	-9.2	-25.8	-28.0	-30.3
(Effective tax rate)	29%	29%	31%	31%	31%	Purchase of PP&E	-26.7	-20.0	-25.8	-28.0	-30.3
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
Net Profit	33.0	33.1	36.2	40.4	44.9	Other	-16.3	10.8	0.0	0.0	0.0
Capex	26.7	20.0	25.8	28.0	30.3	Cash flow from financing activities	-16.7	-37.9	-39.0	-31.6	-32.2
Depreciation & amortization	17.9	19.5	21.3	23.1	25.0	Net Change in Borrowings	0.0	0.0	0.0	0.0	0.0
R&D	11.4	13.5	13.5	14.0	15.2	Purchase of treasury shares	0.0	-20.4	-22.5	-15.0	-15.0
EBITDA	60.6	62.5	68.6	76.5	84.9	Cash dividend paid	-15.7	-16.5	-16.5	-16.6	-17.2
ROE	9.0%	8.9%	9.6%	10.6%	11.5%	Other	-0.9	-1.1	0.0	0.0	0.0
No of average shares (mn)	33.8	33.4	32.5	31.6	30.9	Effect of exchange rate change	-0.7	3.0	0.0	0.0	0.0
EPS (JPY)	976.3	991.9	1,114.9	1,280.5	1,453.9	Changes in cash and cash equivalents	-4.7	1.7	-16.9	-2.1	1.1
BPS (JPY)	10,940.1	11,554.4	11,831.8	12,369.7	13,061.1	Cash and cash equivalents at end of the period	85.7	87.3	70.4	68.3	69.4
DPS (JPY)	490.0	520.0	520.0	550.0	580.0						
USD/JPY (Average)	152.6	150.8	155.0	155.0	155.0	FCF	12.7	36.6	22.1	29.5	33.3
Balance Sheet (JPY bn)						Sales by Application (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total assets	416.9	431.2	427.6	438.0	452.2	General Industrial	46.9	62.0	73.9	80.6	87.8
Current assets	248.8	233.8	225.7	231.2	240.2	Smartphone	46.4	44.3	43.0	44.2	45.6
Cash and cash equivalents	85.7	87.3	70.4	68.3	69.4	Consumer/Mobile equipment	34.7	34.7	34.1	34.8	35.5
Marketable securities	87.6	57.6	57.6	57.6	57.6	Automotive/Mobility related	49.2	54.3	60.0	67.2	75.3
Accounts receivable	40.4	49.9	55.4	60.1	65.0	Telecommunications infrastructure equipment	5.4	6.7	10.7	12.3	14.1
Inventories	25.1	30.2	33.5	36.4	39.4	OA equipment and Others	6.8	9.3	13.0	15.5	17.1
Other financial assets	97.6	66.4	66.4	66.4	66.4	(YoY)					
Fixed assets	168.1	197.4	201.9	206.8	212.1	General Industrial	-7%	32%	19%	9%	9%
Tangible fixed assets	91.3	96.3	100.8	105.7	111.0	Smartphone	31%	-5%	-3%	3%	3%
Intangible fixed assets	6.6	7.7	7.7	7.7	7.7	Consumer/Mobile equipment	37%	0%	-2%	2%	2%
Other non-current assets	70.2	93.4	93.4	93.4	93.4	Automotive/Mobility related	15%	10%	11%	12%	12%
Current liabilities	29.1	32.8	32.0	33.6	35.1	Telecommunications infrastructure equipment	20%	24%	59%	15%	15%
Short term borrowings	0.0	0.0	0.0	0.0	0.0	OA equipment and Others	0%	36%	40%	20%	10%
Accounts payable	12.9	16.7	18.6	20.2	21.8						
Other current liabilities	16.2	16.1	13.4	13.4	13.3						
Fixed liabilities	17.7	20.3	20.3	20.3	20.3						
Long term borrowings	0.0	0.0	0.0	0.0	0.0						
Other fixed liabilities	17.7	20.3	20.3	20.3	20.3						
Net assets	370.1	378.1	375.3	384.1	396.8						
Total liabilities and equity	416.9	431.2	427.6	438.0	452.2						

Risk Reward – Hirose Electric (6806.T)

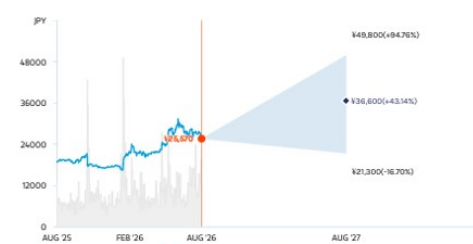
Sales growth to sustain historically high levels while maintaining high margins

PRICE TARGET ¥36,600

Our price target is ¥36,600, based on a DCF valuation (risk-free rate 2.6%; beta 0.9; risk premium 3.2%; WACC 5.1%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥2,100; our price target equates to 32.8x our F3/27E EPS of ¥1,114.9, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.



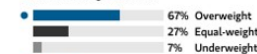
RISK REWARD CHART



OVERWEIGHT THESIS

The equity story is the continued top-line growth across applications through the acquisition of high-margin projects. We like the continued high level of sales growth relative to historical levels, while maintaining a high-profitability structure enabled by its semi-fabrics production model. We believe that if the equity market comes to view profit growth as a mid-term trend, there is further upside to valuations.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive
 Secular Growth: Positive
 Technology Diffusion: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥49,800	BASE CASE	¥36,600	BEAR CASE	¥21,300
Accelerated profit growth through new project wins		Sales growth continues for industrial/auto/AI DC		Lower NA phone share, slower growth of ind. sales	
OP and EPS forecasts of ¥52.7bn/¥1,228.8 in F3/27, ¥62.2bn/¥1,473.8 in F3/28, ¥73.2bn/¥1,751.8 in F3/29, ¥85.1bn/¥2,065.3 in F3/30, and ¥100.9bn/¥2,484.4 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥3,000. Our fair value equates to 40.5x our F3/27E EPS of ¥1,228.8 and 4.2x BPS of ¥11,948.		OP and EPS forecasts of ¥47.3bn/¥1,114.9 in F3/27, ¥53.4bn/¥1,280.5 in F3/28, ¥59.9bn/¥1,453.9 in F3/29, ¥67.0bn/¥1,648.3 in F3/30, and ¥75.4bn/¥1,885.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥2,100. Our PT equates to 32.8x our F3/27E EPS of ¥1,114.9 and 3.1x BPS of ¥11,831.8.		OP and EPS forecasts of ¥38.7bn/¥930.1 in F3/27, ¥39.0bn/¥963.5 in F3/28, ¥39.9bn/¥1,003.8 in F3/29, ¥40.0bn/¥1,029.3 in F3/30, and ¥39.8bn/¥1,049.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥1,050. Our fair value equates to 22.9x our F3/27E EPS of ¥930.1 and 1.8x BPS of ¥11,642.6.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
 Source: Company data, Morgan Stanley Research

Japan Aviation Electronics: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	221.6	227.9	235.1	236.9	245.1	Cash flow from operating activities	36.3	17.0	26.8	27.8	26.7
Costs of Sales	179.6	191.4	197.9	198.8	205.6	EBIT	15.2	10.2	9.6	10.1	10.7
Gross Profit	42.1	36.4	37.2	38.1	39.5	Depreciation	20.9	19.7	22.0	21.7	21.4
SG&A	26.4	27.5	28.3	28.5	29.4	Changes in Receivables	3.8	-0.1	-1.2	-0.3	-1.4
Operating Income	15.6	8.9	8.9	9.6	10.1	Change in inventory	-1.0	-2.6	-0.4	0.4	-0.5
Non-operating income/expenses	-0.8	-0.7	0.7	0.5	0.6	Changes in Payables	1.6	-2.9	0.8	0.2	0.9
Recurring Profit	14.8	8.2	9.6	10.1	10.7	Other	-4.0	-7.2	-4.0	-4.3	-4.5
Extraordinary income	0.3	1.9	0.0	0.0	0.0	Cash flow from investing activities	-19.2	-24.4	-19.5	-19.5	-19.5
Profit before income taxes	15.2	10.2	9.6	10.1	10.7	Purchase of PP&E	-18.0	-23.7	-21.0	-21.0	-21.0
Income taxes	3.6	3.1	2.9	3.1	3.3	Sales of PP&E	0.4	0.0	0.0	0.0	0.0
(Effective tax rate)	24%	30%	30%	31%	31%	Other	-1.5	-0.7	1.5	1.5	1.5
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Cash flow from financing activities	-31.6	0.5	-8.8	-6.0	-5.0
Net Profit	11.6	7.1	6.7	7.0	7.4	Net Change in Borrowings	-27.5	4.8	-5.1	-2.6	-1.7
Capex	18.0	23.7	21.0	21.0	21.0	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	20.9	19.7	22.0	21.7	21.4	Cash dividend paid	-4.0	-4.0	-3.7	-3.4	-3.4
R&D	10.9	11.3	12.1	12.5	12.6	Other	0.0	-0.3	0.0	0.0	0.0
EBITDA	36.5	28.7	30.9	31.3	31.5	Effect of exchange rate changes	-1.0	2.6	-1.2	0.0	0.0
ROE	8.9%	5.1%	4.6%	4.7%	4.9%	Changes in cash and cash equivalents	-15.4	-4.4	-2.7	2.3	2.1
No of average shares (mn)	67.4	67.4	67.4	67.4	67.4	Cash and cash equivalents at end of the period	52.9	48.5	45.8	48.1	50.2
EPS (JPY)	172.0	104.9	99.3	103.8	109.7	FCF	17.1	-7.4	7.2	8.3	7.2
BPS (JPY)	1,983.6	2,141.0	2,168.2	2,222.0	2,281.7	Sales by Product/Application (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
DPS (JPY)	60.0	60.0	50.0	50.0	50.0	Connector	201.8	206.9	212.1	212.9	220.1
USD/JPY (Average)	152.6	150.7	155.0	155.0	155.0	Handhelds	66.9	65.2	61.8	59.1	56.5
Balance Sheet (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Automotive	112.6	115.7	122.5	125.0	127.5
Total assets	215.4	232.1	229.8	231.0	234.3	Industrial & Infrastructure	18.0	20.6	22.3	23.4	24.6
Current assets	125.3	130.7	129.6	131.8	135.8	Aviation & Space (Defence)	0.0	0.0	0.0	0.0	6.0
Cash and cash equivalents	52.9	48.5	45.8	48.1	50.2	Others	4.3	5.4	5.5	5.5	5.5
Marketable securities	0.0	0.0	0.0	0.0	0.0	Aerospace	19.3	20.5	22.5	23.5	24.5
Accounts receivable	37.7	38.9	40.1	40.5	41.9	Automotive	0.2	0.1	0.1	0.1	0.1
Inventories	29.2	32.8	33.2	32.8	33.3	Industrial & Infrastructure	10.7	9.3	10.5	11.1	11.8
Other financial assets	5.6	10.4	10.4	10.4	10.4	Aviation & Space (Defence)	8.4	11.1	11.9	12.3	12.6
Fixed assets	90.2	101.5	100.2	99.2	98.5	Others	0.5	0.5	0.5	0.5	0.5
Tangible fixed assets	70.3	77.6	76.6	75.9	75.5	Connector	-2%	3%	3%	0%	3%
Intangible fixed assets	2.7	3.1	2.8	2.5	2.3	Handhelds	-11%	-2%	-5%	-4%	-4%
Other non-current assets	17.2	20.8	20.8	20.8	20.8	Automotive	-1%	24%	1%	0%	0%
Current liabilities	48.9	52.9	53.9	54.1	55.3	Industrial & Infrastructure	-79%	-56%	2%	0%	0%
Short term borrowings	6.0	11.0	11.2	11.1	11.4	Aviation & Space (Defence)	-22%	-13%	13%	6%	6%
Accounts payable	28.4	25.5	26.3	26.5	27.4	Others	-29%	-2%	1%	0%	0%
Other current liabilities	14.5	16.4	16.5	16.5	16.5	OP by Segment (JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Fixed liabilities	32.8	34.9	29.6	27.1	25.2	Connector	18.0	12.0	11.5	12.0	12.3
Long term borrowings	31.0	30.8	25.5	22.9	21.1	Aerospace	2.6	2.2	2.7	2.9	3.1
Other fixed liabilities	1.8	4.1	4.1	4.1	4.1	Others	0.1	0.1	0.1	0.1	0.1
Net assets	133.7	144.4	146.2	149.8	153.9	OPM					
Total liabilities and equity	215.4	232.1	229.8	231.0	234.3	Connector	9%	6%	5%	6%	6%
						Aerospace	13%	11%	12%	13%	13%
						Others	18%	23%	20%	20%	20%

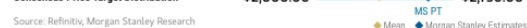
Risk Reward – Japan Aviation Electronics Industry (6807.T)

Key to profitability recovery is regaining North American smartphone share

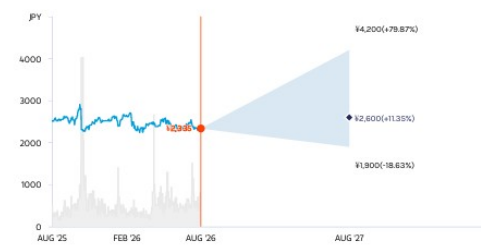
PRICE TARGET ¥2,600

Our price target is ¥2,600, based on a DCF valuation (risk-free rate 2.6%; beta 1.0; risk premium 3.2%; WACC 4.7%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥125; our price target equates to 26.2x our F3/27E EPS of ¥99.3, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

Consensus Price Target Distribution



RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

	¥4,200	BASE CASE	¥2,600	BEAR CASE	¥1,900
Growth accelerates led by auto/phone connectors					
OP and EPS forecasts of ¥10.7bn/¥115.7 in F3/27, ¥12.9bn/¥137.9 in F3/28, ¥15.2bn/¥161.6 in F3/29, ¥17.7bn/¥186.8 in F3/30, and ¥20.0bn/¥210.6 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥235. Our fair value equates to 36.3x our F3/27E EPS of ¥115.7 and 1.9x BPS of ¥2,184.5.					
Sales grow, but midterm margin flat					
OP and EPS forecasts of ¥8.9bn/¥99.3 in F3/27, ¥9.6bn/¥103.8 in F3/28, ¥10.1bn/¥109.7 in F3/29, ¥10.6bn/¥115.7 in F3/30, and ¥11.0bn/¥120.1 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥125. Our PT equates to 26.2x our F3/27E EPS of ¥99.3 and 1.2x BPS of ¥2,168.2.					
Auto growth slows; smartphone decline continues					
OP and EPS forecasts of ¥8.0bn/¥89.0 in F3/27, ¥8.5bn/¥91.9 in F3/28, ¥8.4bn/¥91.9 in F3/29, ¥8.5bn/¥93.4 in F3/30, and ¥8.1bn/¥89.0 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥85. Our fair value equates to 21.4x our F3/27E EPS of ¥89.0 and 0.9x BPS of ¥2,157.8.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

CMK: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	95.5	100.2	103.6	106.2	109.2	Cash flow from operating activities	9.1	7.8	10.4	11.3	12.1
Costs of Sales	79.9	85.3	88.1	89.8	91.8	EBIT	5.3	5.3	2.7	3.4	4.0
Gross Profit	15.6	14.9	15.5	16.4	17.4	Depreciation	5.9	6.5	8.5	8.9	9.3
SG&A	11.8	12.1	12.3	12.5	12.9	Changes in Receivables	-0.2	0.5	-0.3	-0.2	-0.2
Operating Income	3.8	2.8	3.2	3.9	4.5	Change in Inventory	0.7	0.3	0.4	0.3	0.4
Non-operating income/expenses	1.7	1.3	-0.5	-0.5	-0.5	Changes in Payables	-0.8	-1.0	-0.8	-1.0	-1.2
Recurring Profit	5.5	4.1	2.7	3.4	4.0	Other	-2.0	-3.8	-0.1	-0.1	-0.2
Extraordinary income	-0.2	1.2	0.0	0.0	0.0	Cash flow from investing activities	-18.8	-6.9	-8.4	-9.0	-9.3
Profit before income taxes	5.3	5.3	2.7	3.4	4.0	Purchase of PP&E	-18.3	-8.6	-7.9	-8.4	-8.7
Income taxes	1.4	1.1	0.8	1.0	1.2	Sales of PP&E	0.3	0.0	0.0	0.0	0.0
(Effective tax rate)	27%	20%	30%	29%	30%	Other	-0.8	1.8	-0.5	-0.6	-0.6
Profit attributable to non-controlling interests	0.1	0.2	0.2	0.2	0.2	Cash flow from financing activities	4.7	-4.1	1.7	0.2	-4.2
Net Profit	3.8	4.0	1.7	2.2	2.6	Net Change in Borrowings	8.2	-1.8	3.3	2.4	-1.8
Capex	18.9	9.2	8.4	9.0	9.3	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	5.9	6.5	8.5	8.9	9.3	Cash dividend paid	-1.4	-1.4	-1.4	-2.0	-2.1
R&D	1.3	0.6	0.5	0.5	0.5	Other	-2.2	-0.9	-0.2	-0.2	-0.2
EBITDA	9.7	9.3	11.7	12.8	13.8	Effect of exchange rate changes	1.8	0.4	-1.7	0.0	0.0
ROE	5.1%	5.0%	2.1%	2.7%	3.2%	Changes in cash and cash equivalents	-3.1	-2.8	2.0	2.5	-1.4
No of average shares (mn)	71.2	71.3	71.3	71.3	71.3	Cash and cash equivalents at end of the period	22.2	19.4	21.4	23.9	22.5
EPS (JPY)	53.2	56.5	23.8	30.9	36.5	FCF	-9.7	0.9	2.0	2.3	2.8
BPS (JPY)	1,110.3	1,171.1	1,151.0	1,153.9	1,160.3	Sales by Product					
DPS (JPY)	20.0	20.0	28.0	30.0	32.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	151.8	149.8	155.0	155.0	155.0	Double-sided/Multi-layer	61.1	63.4	62.7	62.4	62.2
Balance Sheet						Double-sided	10.0	9.5	8.3	7.6	7.0
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Multi-layer	51.0	53.9	54.4	54.8	55.2
Total assets	148.5	151.5	154.2	157.5	156.8	HDI	30.2	33.0	37.2	40.1	43.3
Current assets	62.4	61.7	64.5	67.8	67.2	Others	4.2	3.8	3.7	3.7	3.7
Cash and cash equivalents	22.2	19.5	21.4	23.9	22.5	(YoY)					
Accounts receivable	17.8	19.9	20.6	21.1	21.7	Double-sided/Multi-layer	2%	4%	-1%	0%	0%
Inventories	18.6	18.4	18.8	19.0	19.2	Double-sided	0%	-5%	-13%	-8%	-8%
Other financial assets	3.8	3.8	3.8	3.8	3.8	Multi-layer	2%	6%	1%	1%	1%
Fixed assets	86.1	89.8	89.7	89.7	89.6	HDI	16%	9%	13%	8%	8%
Tangible fixed assets	76.5	80.3	79.7	79.4	78.9	Others	-8%	-10%	-2%	0%	0%
Intangible fixed assets	1.0	1.5	1.8	2.2	2.6						
Other non-current assets	8.6	8.1	8.1	8.1	8.1						
Current liabilities	33.7	38.0	37.5	37.3	37.1						
Short term borrowings	12.8	10.3	9.2	8.5	7.8						
Accounts payable	14.0	12.5	12.9	13.3	13.6						
Other current liabilities	6.9	15.3	15.4	15.5	15.7						
Fixed liabilities	33.4	27.4	31.8	34.9	33.8						
Long term borrowings	31.3	25.3	29.7	32.8	31.6						
Other fixed liabilities	2.1	2.2	2.2	2.2	2.2						
Net assets	81.4	86.0	84.8	85.2	85.9						
Total liabilities and equity	148.5	151.5	154.2	157.5	156.8						

Risk Reward – CMK (6958.T)

Further vehicle electrification and growth in new business areas awaited

PRICE TARGET ¥790

Our price target is ¥790, based on a DCF valuation (risk-free rate 2.6%; beta 1.3; risk premium 3.2%; WACC 5.1%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥60; our price target equates to 33.1x our F3/27E EPS of ¥23.8, and our forex rate assumption for the next quarter and thereafter is ¥155/¥.

RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥1,950	BASE CASE	¥790	BEAR CASE	¥410
Integrated ECU demand growth accelerates					
OP and EPS forecasts of ¥3.7bn/¥28.1 in F3/27, ¥5.3bn/¥43.5 in F3/28, ¥7.0bn/¥58.9 in F3/29, ¥8.9bn/¥78.5 in F3/30, and ¥12.1bn/¥109.4 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥140. Our fair value equates to 69.5x our F3/27E EPS of ¥28.1 and 1.7x BPS of ¥115.2.					
BU sales for Integrated ECU rise fully from F3/27					
OP and EPS forecasts of ¥3.2bn/¥23.8 in F3/27, ¥3.9bn/¥30.9 in F3/28, ¥4.5bn/¥36.5 in F3/29, ¥5.2bn/¥43.5 in F3/30, and ¥6.2bn/¥53.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥60. Our PT equates to 33.1x our F3/27E EPS of ¥23.8 and 0.7x BPS of ¥115.0.					
Weak automotive production, slow uptake for ADAS					
OP and EPS forecasts of ¥2.3bn/¥14.0 in F3/27, ¥2.5bn/¥18.2 in F3/28, ¥2.8bn/¥21.0 in F3/29, ¥2.8bn/¥21.0 in F3/30, and ¥2.7bn/¥21.0 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥30. Our fair value equates to 29.2x our F3/27E EPS of ¥14.0 and 0.4x BPS of ¥114.2.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Daishinku: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	38.6	39.6	40.9	43.4	46.2	Cash flow from operating activities	2.3	-1.8	4.0	5.6	6.2
Costs of Sales	29.3	30.4	31.2	33.0	34.6	EBIT	1.6	1.1	-0.1	1.1	1.9
Gross Profit	9.3	9.2	9.7	10.4	11.6	Depreciation	4.0	4.2	5.2	5.4	5.6
SG&A	8.4	8.1	8.7	8.9	9.3	Changes in Receivables	0.5	-0.2	0.1	-0.6	-0.7
Operating Income	0.9	1.1	1.0	1.5	2.3	Change in inventory	-1.9	-5.6	-1.7	-0.7	-0.7
Non-operating income/expenses	-0.5	-0.4	-0.7	-0.4	-0.4	Changes in Payables	1.0	0.1	0.1	0.3	0.3
Recurring Profit	0.4	0.7	0.3	1.1	1.9	Other	-2.8	-1.5	0.4	0.2	-0.1
Extraordinary income	1.2	0.3	-0.4	0.0	0.0	Cash flow from investing activities	-6.3	-3.1	-4.0	-4.3	-4.6
Profit before income taxes	1.6	1.1	-0.1	1.1	1.9	Purchase of PP&E	-6.9	-3.0	-4.0	-4.3	-4.6
Income taxes	0.6	0.5	0.0	0.3	0.6	Sales of PP&E	0.0	0.1	0.0	0.0	0.0
(Effective tax rate)	39%	42%	0%	27%	32%	Other	0.5	-0.2	0.0	0.0	0.0
Profit attributable to non-controlling interests	0.7	0.2	0.2	0.2	0.2	Cash flow from financing activities	-1.7	1.2	1.1	-3.7	-3.2
Net Profit	0.3	0.4	-0.3	0.6	1.1	Net Change in Borrowings	-0.1	2.7	2.5	-2.3	-1.8
Capex	1.0	4.8	4.0	4.3	4.6	Purchase of treasury shares	-0.3	-0.2	0.0	0.0	0.0
Depreciation & amortization	4.0	4.2	5.2	5.4	5.6	Cash dividend paid	-0.9	-0.9	-0.9	-0.9	-0.9
R&D	2.2	0.6	1.9	2.4	2.5	Other	-0.5	-0.4	-0.5	-0.5	-0.5
EBITDA	4.9	5.4	6.2	6.9	7.9	Effect of exchange rate changes	-0.1	1.2	-2.2	0.0	0.0
ROE	0.8%	1.1%	-0.8%	1.7%	3.1%	Changes in cash and cash equivalents	-5.9	-2.5	-1.0	-2.3	-1.5
No of average shares (mn)	32.1	31.8	31.8	31.8	31.8	Cash and cash equivalents at end of the period	18.5	16.0	14.9	12.6	11.1
EPS (JPY)	8.9	13.2	-9.4	18.9	34.6	FCF	-4.0	-4.9	0.0	1.3	1.6
BPS (JPY)	1,166.1	1,226.9	1,120.7	1,111.6	1,118.1	Sales by Application					
DPS (JPY)	28.0	28.0	28.0	28.0	28.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.2	150.7	155.0	155.0	155.0	Communication	10.2	10.2	10.7	11.8	13.0
Balance Sheet						Automotive	13.8	14.8	16.1	17.2	18.5
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Consumer	10.7	10.4	9.7	9.9	10.1
Total assets	89.9	98.8	97.0	94.9	93.8	Industrial	4.0	4.2	4.3	4.5	4.6
Current assets	48.9	53.4	53.7	52.6	52.5	(YoY)					
Cash and cash equivalents	18.7	16.3	14.9	12.6	11.1	Communication	-13%	0%	5%	10%	10%
Accounts receivable	9.1	10.2	10.1	10.7	11.4	Automotive	3%	8%	9%	7%	7%
Inventories	18.0	24.6	26.3	27.0	27.7	Consumer	15%	-3%	-6%	2%	2%
Other financial assets	3.1	2.3	2.3	2.3	2.3	Industrial	-19%	5%	2%	4%	4%
Fixed assets	41.0	45.4	43.4	42.3	41.3						
Tangible fixed assets	34.9	38.4	36.4	35.3	34.3						
Intangible fixed assets	1.0	0.8	0.8	0.8	0.8						
Other non-current assets	5.1	6.2	6.2	6.2	6.2						
Current liabilities	24.9	32.9	30.4	29.8	29.3						
Short term borrowings	16.9	22.6	20.9	19.9	18.9						
Accounts payable	3.4	4.1	4.2	4.5	4.8						
Other current liabilities	4.6	6.3	5.2	5.4	5.6						
Fixed liabilities	19.8	17.9	22.1	20.8	20.0						
Long term borrowings	16.3	14.1	18.3	17.0	16.2						
Other fixed liabilities	3.5	3.8	3.8	3.8	3.8						
Net assets	45.2	48.0	44.6	44.3	44.5						
Total liabilities and equity	89.9	98.8	97.0	94.9	93.8						

Risk Reward – Daishinku (6962.T)

Awaiting demand acceleration for new devices and production system upgrades

PRICE TARGET ¥840

Our price target is ¥840, based on a DCF valuation (risk-free rate 2.6%; beta 1.1; risk premium 3.2%; WACC 4.5%); beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward are that the firm maintains FCF/share of ¥80; our price target equates to 44.5x our F3/28E EPS of ¥18.9, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

UNDERWEIGHT THESIS

Sales weighting for photolithographic products has risen as demand for high-frequency products from clients has increased. We give the firm high marks for progressively ramping up capacity of TCXO, demand for which has been rising for use in communication modules or GPS applications. However, we believe Daishinku still needs time to build a network to expand sales of the Arkh Series, which has technological advantages in terms of both cost and productivity. We also think the firm will continue facing downward pressure on prices, esp. for consumer applications. Ongoing production issues could also become a bottleneck in the demand upcycle, and countermeasures should be closely monitored.

Risk Reward Themes

New Data Era: Negative

View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥1,900	BASE CASE	¥840	BEAR CASE	¥220
Telecom/auto growth speeds up					
OP and EPS forecasts of ¥1.3bn/¥3.1 in F3/27, ¥2.1bn/¥28.3 in F3/28, ¥3.3bn/¥53.5 in F3/29, ¥4.9bn/¥91.2 in F3/30, and ¥6.6bn/¥125.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥155. Our fair value equates to 67.1x our F3/28E EPS of ¥28.3 and 1.7x BPS of ¥1,127.0.					
Gold surge and higher costs limit recovery					
OP and EPS forecasts of ¥1.0bn/¥9.4 in F3/27, ¥1.5bn/¥18.9 in F3/28, ¥2.3bn/¥34.6 in F3/29, ¥2.9bn/¥47.2 in F3/30, and ¥3.6bn/¥62.9 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥80. Our PT equates to 44.5x our F3/28E EPS of ¥18.9 and 0.7x BPS of ¥1,120.7.					
Investment burden, slowing topline growth					
OP and EPS forecasts of ¥0.9bn/¥9.4 in F3/27, ¥1.1bn/¥9.4 in F3/28, ¥1.5bn/¥18.9 in F3/29, ¥1.6bn/¥22.0 in F3/30, and ¥1.7bn/¥25.2 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥30. Our fair value equates to 23.3x our F3/28E EPS of ¥9.4 and 0.2x BPS of ¥1,120.7.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Hamamatsu Photonics: Key Financial Data and Risk Reward

Income Statement						Cash Flow Statement					
(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e	(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e
Net Sales	204.0	212.1	232.8	245.8	265.7	Net Cash from Operations	38.1	37.8	37.1	48.2	48.5
COGS	100.1	110.7	120.9	129.6	142.4	Net Income	25.1	14.2	18.0	19.7	23.4
Gross Profit	103.9	101.4	111.8	116.2	123.3	Depreciation and Amortization	16.7	18.9	23.5	27.7	26.7
SG&A Expenses	71.8	85.2	89.3	90.7	92.5	Changes in Working Capital	-3.9	3.7	-5.4	-0.2	-2.6
Operating Profit	32.1	16.2	22.5	25.4	30.7	Other-net	0.0	1.0	1.0	1.0	1.0
Non-operating Income/Loss	2.4	2.6	2.5	2.5	2.5	Net Cash from Investing Activities	-73.7	-42.2	-42.9	-16.0	-16.0
Recurring Profit	34.5	18.8	25.0	27.9	33.2	Purchase of PP&E	-28.1	-35.8	-40.2	-15.0	-15.0
Extraordinary Income/Loss	0.9	1.9	0.3	0.3	0.3	Sale/Purchase of Investments	-0.5	-4.1	0.0	0.0	0.0
Pretax Profit	35.4	20.7	25.2	28.2	33.5	Other-net	-45.1	-2.2	-2.8	-1.0	-1.0
Income Tax	10.0	6.2	7.1	8.5	10.1	Free Cash Flow	-35.6	-4.4	-5.8	32.2	32.5
(Effective Tax Rate)	35%	35%	35%	35%	35%	Net Cash from Financing Activities	12.6	-2.8	-31.7	-27.7	-28.4
Minority Interest	0.3	0.4	0.1	0.0	0.0	Change in Interest-bearing Debt	24.5	29.0	0.0	0.0	0.0
Net Profit	25.1	14.2	18.0	19.7	23.4	Change in Treasury Stock	0.0	-20.0	-20.0	-15.0	-15.0
Capex	32.2	39.1	40.2	15.0	15.0	Payment of Cash Dividends	-11.8	-11.6	-11.7	-12.7	-13.4
Depreciation and Amortization	17.7	22.3	23.5	27.7	26.7	Other-net	-0.2	-0.3	0.0	0.0	0.0
R&D Expenses	13.6	18.4	18.8	19.9	21.3	Net Effect of Exchange Rate Changes	1.2	0.7	0.0	0.0	0.0
EBITDA	49.8	38.5	46.0	53.1	57.4	Net Change in Cash	-21.8	-6.5	-37.5	4.5	4.1
ROE	7.7%	4.4%	5.7%	6.5%	7.9%	Cash at End of Period	92.6	86.0	48.6	53.1	57.1
Average Share Outstanding (Diluted)	309.7	298.7	298.8	283.2	277.9	Segment Information					
EPS (Diluted, Yen)	81.2	47.4	61.7	69.5	84.2	(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e
BPS (Yen)	1,075.3	1,082.9	1,067.4	1,058.7	1,063.1	Net Sales	204.0	212.1	232.8	245.8	265.7
DPS (Yen)	38.0	38.0	40.0	45.0	45.0	Electron Tube	78.5	71.9	78.8	81.9	86.8
USD/JPY: Average Exchange Rate	149.4	149.2	155.2	155.0	155.0	Opto-semiconductor	78.9	79.5	89.9	98.0	104.8
YoY % Change						Imaging and Measurement Instruments	32.8	32.7	37.2	39.5	42.6
Net Sales	-8%	4%	10%	6%	8%	Others	3.0	5.7	4.7	2.5	4.8
Operating Profit	-43%	-50%	39%	13%	21%	YoY % Change					
Net Profit	-41%	-44%	27%	9%	19%	Electron Tube	-8%	4%	10%	6%	8%
As % of Sales						Opto-semiconductor	-6%	-8%	10%	4%	6%
Gross Profit	50.9%	47.8%	48.0%	47.3%	46.4%	Imaging and Measurement Instruments	-21%	1%	13%	9%	7%
Operating Profit	15.7%	7.6%	9.7%	10.4%	11.6%	Others	3%	0%	14%	6%	8%
Net Profit	12.3%	6.7%	7.7%	8.0%	8.8%	Operating Profit by Segment	180%	89%	-17%	-47%	93%
Capex	15.8%	18.4%	17.2%	6.1%	5.6%	Total Operating Profit	32.1	16.2	22.5	25.4	30.7
Depreciation and Amortization	8.7%	10.5%	10.1%	11.3%	10.0%	Electron Tube	23.8	19.0	22.5	24.6	26.6
Balance Sheet						Opto-semiconductor	17.9	12.6	15.5	14.6	17.7
(Y bn)	F9/24	F9/25	F9/26e	F9/27e	F9/28e	Imaging and Measurement Instruments	10.4	9.7	12.1	13.1	14.2
Total Assets	434.6	455.0	440.1	432.6	428.4	Others	0.7	0.9	0.0	0.3	0.3
Total Current Assets	239.0	236.4	204.9	210.1	217.6	Adjustments	-23.3	-21.6	-21.4	-22.0	-23.8
Cash and Cash Equivalents	97.0	90.6	53.1	57.6	61.6	Operating Margin	16%	8%	10%	10%	12%
Marketable Securities	2.9	6.5	6.5	6.5	6.5	Electron Tube	30%	28%	29%	30%	31%
Account Receivables	45.7	46.6	50.0	52.2	55.7	Opto-semiconductor	23%	16%	17%	15%	17%
Inventory	80.6	77.0	79.4	78.0	78.0	Imaging and Measurement Instruments	32%	30%	32%	33%	33%
Other Current Assets	12.8	15.8	15.8	15.8	15.8	Others	24%	15%	0%	10%	7%
Total Long Term Assets	195.6	218.6	235.2	222.5	210.8	Operating Profit by Segment					
Net PP&E	132.9	150.7	167.3	154.6	142.9	Total Operating Profit	32.1	16.2	22.5	25.4	30.7
Investments in Securities	35.7	34.6	34.6	34.6	34.6	Electron Tube	23.8	19.0	22.5	24.6	26.6
Other Assets	27.0	33.3	33.3	33.3	33.3	Opto-semiconductor	17.9	12.6	15.5	14.6	17.7
Total Current Liabilities	80.7	108.5	109.0	109.5	110.3	Imaging and Measurement Instruments	10.4	9.7	12.1	13.1	14.2
ST Debt and Current Portion of LT	28.4	55.5	55.5	55.5	55.5	Others	0.7	0.9	0.0	0.3	0.3
Accounts Payable	6.6	6.8	7.3	7.9	8.6	Adjustments	-23.3	-21.6	-21.4	-22.0	-23.8
Other Current Liabilities	45.8	46.2	46.2	46.2	46.2	Operating Margin	16%	8%	10%	10%	12%
Total Long Term Liabilities	20.9	23.0	23.0	23.0	23.0	Electron Tube	30%	28%	29%	30%	31%
Long Term Debt	8.5	10.6	10.6	10.6	10.6	Opto-semiconductor	23%	16%	17%	15%	17%
Other Long Term Liabilities	12.4	12.5	12.5	12.5	12.5	Imaging and Measurement Instruments	32%	30%	32%	33%	33%
Net Assets	333.0	323.5	308.1	299.9	295.4	Others	24%	15%	0%	10%	7%
Net Debt	-63.0	-31.0	6.5	2.0	-2.1						

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Risk Reward – Hamamatsu Photonics (6965.T)

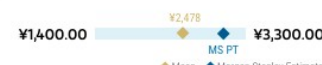
Stands out for high potential of its applied photonics products.

PRICE TARGET ¥3,000

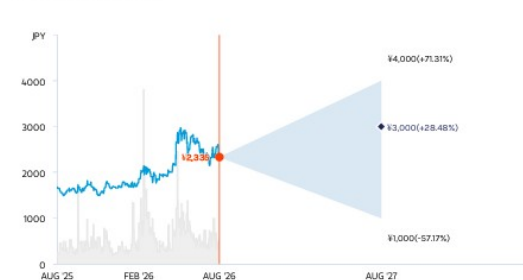
DCF model assumptions: 2.6% risk-free rate, 0.91 equity beta, 3.2% risk premium, yielding 5.5% WACC; zero growth from F9/36.

Consensus Price Target Distribution

Source: Refinitiv, Morgan Stanley Research



RISK REWARD CHART



Key: — Historical Stock Performance • Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

EQUAL-WEIGHT THESIS

■ Following the completion of customer inventory adjustments, we expect sales to bottom out not only in semiconductor manufacturing and inspection equipment, but also in medical and biotech applications, leading to a recovery in earnings.
■ While we expect a recovery from F9/26 following a trough in F9/25, it may take time to exceed the peak OP of ¥57.0bn in F9/22 given higher fixed costs, including depreciation and labor expenses.
■ Photonics applications, such as optoelectronic conversion and photonic integration, hold significant potential, including the prospect of yet-to-be-anticipated new products contributing to earnings.

Consensus Rating Distribution



MS Rating

Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥4,000	BASE CASE	¥3,000	BEAR CASE	¥1,000
Derived from DCF					
Earnings from SPE, FA, and automobile markets ahead of base case. OP ¥28.4bn in F9/26, ¥35.4bn in F9/27, and ¥77.1bn in F9/30. Same currency and DCF model assumptions as base case. Fair value equates to P/E 56.8 x F9/26 diluted EPS ¥70.4, 43.8 x F9/27e diluted EPS ¥91.3 and 20.2 x F9/27e diluted EPS ¥198.4.					
Earnings to bottom out in F9/25 and recover from F9/26, although it is likely to take time to reach a new record high. OP ¥22.5bn in F9/26, ¥25.4bn in F9/27, and ¥45.9bn in F9/30. Assumes ¥155/USD; DCF model assumptions: risk-free rate 2.6%, equity beta 0.91, risk premium 3.2%, yielding WACC 5.5%. Zero growth from F9/36. Base case fair value equates to P/E 48.6x F9/26 diluted EPS ¥61.7, 43.1x F9/27e diluted EPS ¥69.5 and 23.5x F9/30e diluted EPS ¥127.5.					
0.9x diluted book value ¥1,109 for F9/26 end					
Expect continued earnings decline in F9/26, with growth remaining modest from F9/27 onward. OP ¥15.5bn in F9/26, ¥16.5bn in F9/27, and ¥36.0bn in F9/30. Same currency assumptions as base case. ROE 3.5% in F9/26 and 3.7% in F9/27, lower than the 5.5% cost of capital; trades at 0.9x diluted book value for F9/26-end.					

Kyocera: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	2,014.5	2,070.2	1,951.5	1,977.2	2,102.9
COGS	1,455.3	1,462.6	1,316.5	1,319.6	1,452.6
Gross Profit	559.2	607.6	635.0	657.6	650.3
SG&A	531.9	489.5	502.4	509.1	471.5
Operating Profit	27.3	118.1	132.6	148.5	178.8
Non-operating Income/Loss	36.3	50.9	42.5	29.0	57.6
Pretax Profit	63.6	169.0	175.1	177.5	236.4
Income Tax	36.2	24.1	35.0	35.5	61.5
(Effective Tax Rate)	1%	0%	0%	0%	0%
Minority Interest	3.4	4.0	2.0	2.0	3.6
Net Profit	24.1	141.0	138.1	140.0	171.3
Capex	161.7	149.1	225.0	160.0	160.0
Depreciation and Amortization	111.7	110.9	120.0	125.0	120.0
R&D	104.3	115.7	120.0	125.0	125.0
EBITDA	139.0	229.1	252.6	273.5	298.8
ROE	0.7%	4.3%	4.2%	4.6%	5.8%
Average Share Outstanding (Diluted)	1,412.4	1,372.6	1,292.2	1,196.0	1,128.5
EPS (Diluted, Yen)	17.1	102.7	106.8	117.1	151.8
BPS (Yen)	2,195.0	2,203.5	2,254.0	2,284.4	2,349.5
DPS (Yen)	50.0	52.0	65.0	70.0	70.0
USD/JPY: Average Exchange Rate	151.8	150.8	155.0	155.0	155.0
EUR/JPY: Average Exchange Rate	163.8	174.8	180.0	180.0	180.0
YoY % Change					
Sales	1%	3%	-6%	1%	6%
Operating Profit	-71%	333%	12%	12%	20%
Net Profit	-76%	485%	-2%	1%	22%
As % of Sales					
Gross Profit	27.8%	29.4%	32.5%	33.3%	30.9%
Operating Profit	1.4%	5.7%	6.8%	7.5%	8.5%
Net Profit	1.2%	6.8%	7.1%	7.1%	8.1%
Capex	8.0%	7.2%	11.5%	8.1%	7.6%
Depreciation and Amortization	5.5%	5.4%	6.1%	6.3%	5.7%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	4,511.3	4,646.3	4,445.3	4,353.2	4,460.4
Total Current Assets	1,435.7	1,501.2	1,195.2	1,068.0	1,135.3
Cash and Cash Equivalents	444.7	455.9	209.9	87.5	92.5
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	411.2	416.6	367.1	371.9	395.6
Inventory	521.8	522.0	511.5	501.9	540.5
Other Current Assets	58.0	106.7	106.7	106.7	106.7
Total Long Term Assets	3,075.6	3,145.2	3,250.2	3,285.2	3,325.2
Net PP&E	651.9	685.5	790.5	825.5	865.5
Investments and Advances	1,770.3	1,804.0	1,804.0	1,804.0	1,804.0
Goodwill	282.2	274.0	274.0	274.0	274.0
Intangible Assets	142.1	121.2	121.2	121.2	121.2
Other Assets	229.1	260.5	260.5	260.5	260.5
Total Current Liabilities	491.7	525.0	514.0	514.5	533.0
ST Debt and Current Portion of LT Debt	45.8	56.1	56.1	56.1	56.1
Accounts Payable	207.0	194.8	183.8	184.2	202.8
Other Current Liabilities	238.8	274.2	274.2	274.2	274.2
Total Long Term Liabilities	776.4	753.9	753.9	853.9	953.9
Long Term Debt	202.6	189.0	189.0	289.0	389.0
Other Long Term Liabilities	573.8	565.0	565.0	565.0	565.0
Minority Interests	25.4	27.9	25.9	23.9	20.3
Total Equity	3,243.2	3,367.4	3,151.4	2,960.8	2,953.1
Net Debt (excl. KDDI Shares)	-385.3	-483.8	-307.2	-334.9	-489.9

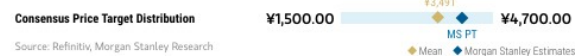
Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	237.9	226.2	305.1	258.2	244.0
Net Income	24.1	141.0	138.1	140.0	171.3
Depreciation and Amortization	160.9	158.2	120.0	125.0	120.0
Changes in Working Capital:	26.0	-41.4	49.0	5.2	-43.7
Other-net	26.8	-31.5	-2.0	-2.0	-3.6
Net Cash from Financing Activities	-150.5	74.5	-225.0	-160.0	-160.0
Purchase of PP&E	-154.7	-159.2	-225.0	-160.0	-160.0
Sale/Purchase of Investments	0.0	0.0	0.0	0.0	0.0
Other-net	4.2	233.7	0.0	0.0	0.0
Free cash flow	87.4	306.8	80.1	108.2	84.0
Net Cash from Financing Activities	-64.9	-312.0	-326.1	-230.6	-79.0
Change in Interest-bearing Debt	38.0	-6.2	0.0	100.0	100.0
Change in Treasury Stock	0.0	-200.0	-250.0	-250.0	-100.0
Payment of Cash Dividends	-73.3	-72.2	-76.1	-80.6	-79.0
Other-net	-29.6	-33.5	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-2.5	22.3	0.0	0.0	0.0
Net Change in Cash	20.0	11.1	-246.0	-122.4	5.0
Cash at End of Period	444.7	455.9	209.9	87.5	92.5
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	2,014.5	2,070.2	1,951.5	1,977.2	2,102.9
Core Components Business	567.1	653.4	654.0	688.7	736.9
Industrial & Automotive Components Unit	233.1	250.1	246.0	258.3	276.4
Semiconductor Components Unit	300.8	379.4	380.0	410.4	439.1
Others	33.3	23.9	28.0	20.0	21.4
Electronic Components Business	354.6	363.5	375.0	397.5	422.5
Solutions Business	1,111.0	1,070.9	932.5	901.0	954.2
Industrial Tools Unit	305.9	285.9	170.0	174.3	186.4
Document Solutions Unit/Information Eq	480.0	478.5	480.0	482.0	516.6
Communications Unit	225.5	219.2	190.0	194.8	208.4
Others	99.7	87.3	92.5	40.0	42.8
Others	17.7	17.7	17.7	17.7	17.7
Elimination	-35.4	-31.8	-26.0	-26.0	-27.8
Net OP	24.0	133.2	164.6	180.5	187.8
Core Components Business	-1.1	63.1	82.6	90.5	104.0
Industrial & Automotive Components Unit	25.0	18.7	30.0	35.0	40.2
Semiconductor Components Unit	-27.8	46.9	53.0	55.0	63.2
Others	1.7	-2.8	-0.4	0.5	0.5
Electronic Components Business	-0.8	7.3	22.0	25.0	30.6
Solutions Business	72.9	103.9	86.0	92.0	96.2
Others	0.0	0.0	0.0	0.0	0.0
Segment Margin	1.2%	6.4%	8.4%	9.1%	8.9%
Core Components Business	-0.2%	9.7%	12.6%	13.1%	14.1%
Industrial & Automotive Components Unit	10.7%	7.5%	12.2%	13.6%	14.6%
Semiconductor Components Unit	-9.3%	12.4%	13.9%	13.4%	14.4%
Electronic Components Business	-0.2%	2.0%	5.9%	6.3%	7.2%
Solutions Business	6.6%	9.7%	9.2%	10.2%	10.1%

Risk Reward – Kyocera (6971.T)

After balance sheet reform, sustainability of profit growth will be the key.

PRICE TARGET ¥4,000

Derived from the base case. DCF model assumptions: 2.6% risk-free rate, 0.91 equity beta, 3.2% risk premium, for WACC of 5.3%; zero growth from F3/36.



RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

BULL CASE ¥5,000 BASE CASE ¥4,000 BEAR CASE ¥1,500

Derived from DCF

Sales top the base case by 5% as parts demand recovers: OP ¥171.6bn in F3/27, ¥270.5bn in F3/28, ¥306.8bn in F3/31. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 42.2x on F3/27 diluted EPS of ¥171.6, 35.2x F3/28 diluted EPS of ¥207.5, 24.1x F3/31 diluted EPS of ¥306.8.

Derived from DCF

We aim to assess the trajectory of ROE improvement, driven by balance sheet enhancement through the sale of KDDI shares and shareholder returns, as well as by profit growth. OP ¥132.6bn in F3/27, ¥148.5bn in F3/28, ¥247.8bn in F3/31. Same FX assumptions as the base case. ROE of 34% in F3/27 and 30% in F3/28 remains below capital costs of 5.5%, prompting the stock to trade on P/B of 0.7x.

EQUAL-WEIGHT THESIS

- Needs to further strengthen core components business and restructure solution business.
- While the divestment of KDDI shares and enhanced shareholder returns should support ROE improvement, we also want to assess how much of the numerator is being driven by profit growth.
- We expect ROE to improve to 5.8% in F3/29 and 6.6% in F3/31, which is higher than cost of capital of 5.5%.

Consensus Rating Distribution



Source: Refinitiv, Morgan Stanley Research

Risk Reward Themes

Self-help: Positive
View descriptions of Risk Rewards Themes [here](#)

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart

Source: Company data, Morgan Stanley Research

Taiyo Yuden: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	411.3	475.1	523.8
COGS	269.9	273.4	301.0	338.0	375.2
Gross Profit	71.6	81.9	110.2	137.2	148.6
SG&A Expenses	61.1	61.9	65.0	66.9	68.4
Operating Profit	10.5	20.0	45.2	70.2	80.2
Non-operating Income/Loss	0.1	4.1	-1.2	-1.2	-1.2
Recurring Profit	10.5	24.1	44.0	69.0	79.0
Extraordinary Income/Loss	-2.2	-4.1	0.0	-2.0	-2.0
Profit Before Tax	8.4	20.1	44.0	67.0	77.0
Income Tax	6.0	5.3	13.2	20.1	23.1
(Effective Tax Rate)	72%	26%	30%	30%	30%
Minority Interest	0.0	0.0	0.0	0.0	0.0
Net Profit	2.3	14.8	30.8	46.9	53.9

Capex	64.2	40.4	40.0	60.0	60.0
Depreciation and Amortization	46.3	49.1	48.0	52.0	52.0
R&D Expenses	15.0	14.5	15.0	16.0	17.0
EBITDA	56.7	69.1	93.2	122.2	132.2
ROE	0.7%	4.5%	8.7%	12.3%	12.9%

Average Share Outstanding (Diluted)	144.3	137.7	137.7	137.7	137.7
EPS (Diluted, Yen)	16.1	107.6	224.0	340.9	391.7
BPS (Yen)	2,206.4	2,501.8	2,639.5	2,885.0	3,172.3
DPS (Yen)	90.0	90.0	100.0	110.0	120.0

USD/JPY: Average Exchange Rate	152.6	150.0	155.0	155.0	155.0
YoY % Change					
Net Sales	6%	4%	16%	16%	10%
Operating Profit	15%	91%	126%	55%	14%
Net Profit	-72%	536%	108%	52%	15%

Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	673.2	615.5	638.3	676.8	719.1

Total Current Assets	279.3	318.3	349.1	378.6	413.9
Cash and Cash Equivalents	78.2	100.1	133.8	140.5	157.6
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	80.5	86.4	94.8	108.2	117.8
Inventory	110.0	125.8	114.5	123.9	132.4
Other Current Assets	10.5	6.0	6.0	6.0	6.0
Total Long Term Assets	293.9	297.2	289.2	297.2	305.2
Net PP&E	286.8	290.0	282.0	290.0	298.0
Investments in Securities	0.0	0.0	0.0	0.0	0.0
Other Assets	7.0	7.3	7.3	7.3	7.3

Total Current Liabilities	79.6	93.8	97.6	101.3	105.0
ST Debt and Current Portion of LT Debt	19.7	35.2	35.2	35.2	35.2
Accounts Payable	27.2	26.5	30.3	34.0	37.8
Other Current Liabilities	32.8	32.1	32.1	32.1	32.1
Total Long Term Liabilities	174.4	177.4	177.4	177.4	177.4
Long Term Debt	144.7	136.0	136.0	136.0	136.0
Other Long Term Liabilities	29.7	41.4	41.4	41.4	41.4

Net Assets	319.2	344.4	363.4	397.2	436.7
Minority Interests	0.0	0.0	0.0	0.0	0.0
Stock Acquisition Rights	0.7	0.0	0.0	0.0	0.0

Net Debt	86.2	71.1	37.4	30.7	13.6
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Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	33.9	58.1	85.6	79.8	91.5
Net Income	2.3	14.8	30.8	46.9	53.9
Depreciation and Amortization	46.3	49.1	48.0	52.0	52.0
Changes in Working Capital	-13.5	-9.0	6.8	-19.1	-14.4
Other-net	-1.2	3.2	0.0	0.0	0.0
Net Cash from Investing Activities	-63.5	-25.7	-40.0	-60.0	-60.0
Purchase of PP&E	-62.6	-40.4	-40.0	-60.0	-60.0
Sale/Purchase of Investments	1.3	0.0	0.0	0.0	0.0
Other-net	-2.2	14.7	0.0	0.0	0.0
Free Cash Flow	-29.8	32.4	45.6	19.8	31.5

Net Cash from Financing Activities	3.0	-6.8	-11.9	-13.1	-14.4
Change in Interest-bearing Debt	15.7	6.5	0.0	0.0	0.0
Change in Treasury Stock	0.0	0.0	0.0	0.0	0.0
Payment of Cash Dividends	-11.2	-11.2	-11.9	-13.1	-14.4
Other-net	-1.5	-2.1	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	-0.9	4.9	0.0	0.0	0.0
Net Change in Cash	-27.4	30.5	33.7	6.7	17.2
Cash at End of Period	67.5	98.1	131.8	138.5	155.6

Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	341.4	355.3	411.3	475.1	523.8
Capacitors	232.1	251.8	304.8	362.7	406.2
Elina	0.0	0.0	0.0	0.0	0.0
Ferrite and Applied Products	61.5	64.3	68.2	73.0	78.1
Complex Devices	23.0	14.8	12.3	14.5	14.5
Others	24.8	24.5	26.0	25.0	25.0
Total Operating Profit	10.5	20.0	45.2	70.2	80.2

Operating Profit Margin	3.1%	5.6%	11.0%	14.8%	15.3%
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Risk Reward – Taiyo Yuden (6976.T)

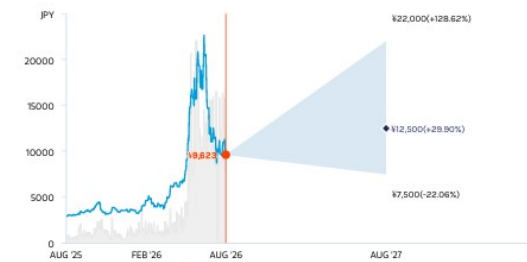
Even assuming continued earnings expansion, we view the stock as overvalued.

PRICE TARGET ¥12,500

Derived from the base case. DCF assumptions: 2.6% risk-free rate, 1.35 equity beta, 3.2% risk premium, for 6.5% WACC; zero growth from F3/36.



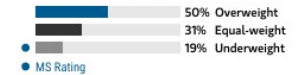
RISK REWARD CHART



UNDERWEIGHT THESIS

- While we expect continued earnings expansion, we believe the shares are overvalued.
- We expect continued earnings growth in the core MLCC business.
- We expect the contribution from premium MLCC products for AI/data center applications to remain limited.
- We expect sales growth for power inductors used in 5G smartphones and base stations to continue.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive
View descriptions of Risk Rewards Themes [here](#)

BULL CASE	¥22,000	BASE CASE	¥12,500	BEAR CASE	¥7,500
Derived from DCF		Derived from DCF		F3/27e diluted BPS ¥2,553 x P/B 2.9	
MLCC sales exceed the base case. OP ¥56.7bn in F3/27, ¥89.5bn in F3/28, ¥163.4bn in F3/31. Same FX & DCF model assumptions as the base case. Fair value equates to P/E 78.0x on F3/27e diluted EPS of ¥282.0, 50.1x on F3/28e diluted EPS of ¥439.0, and 27.0x on F3/31e diluted EPS of ¥814.5.		Earnings are expanding driven by growth in MLCC sales, but market expectations appear overly high: OP ¥45.2bn in F3/27, ¥70.2bn in F3/28, ¥100.2bn in F3/31. FX assumption: ¥155/US\$. Base case fair value equates to P/E 55.8x on F3/27e diluted EPS of ¥224.0, 36.7x on F3/28e diluted EPS of ¥340.9, and 25.3x on F3/31e diluted EPS of ¥493.3.		Stays competitive in core products, but slowing end demand drags profits down. OP ¥39.3bn in F3/27, OP ¥58.2bn in F3/28, ¥58.6bn in F3/31. Same FX and DCF assumptions as in the base case. ROE 74% in F3/27, 10.5% in F3/28, higher than cost of capital 6.9%, but with earnings lower than market expectation, P/B falls to 2.9x.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Murata Manufacturing: Key Financial Data and Risk Reward

Income Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,743.4	1,830.9	2,044.2	2,449.7	2,830.5
Cost of Goods Sold	1,025.7	1,056.0	1,129.7	1,316.9	1,543.3
Gross Profit	717.7	774.8	914.5	1,132.8	1,287.2
SG&A Expenses	438.0	493.0	482.6	508.9	532.9
Operating Profit	279.7	281.8	431.9	624.0	754.3
Non-Operating Income/Loss	24.7	26.8	11.0	12.0	12.0
Pretax profit	304.4	308.6	442.9	636.0	766.3
Income Tax	71.4	74.9	108.5	155.8	187.7
(Effective Tax Rate)	23%	24%	25%	25%	25%
Minority Interest	-0.8	-0.1	0.0	0.0	0.0
Net Profit	233.8	233.8	334.4	480.2	578.6
CapeX	180.5	247.8	250.0	250.0	250.0
Depreciation and Amortization	173.3	178.2	178.0	190.0	200.0
R&D	149.3	158.9	167.0	173.0	178.0
EBITDA	453.0	460.0	609.9	814.0	954.3
ROE	9.1%	8.8%	12.2%	16.7%	18.5%
Average Share Outstanding (Diluted)	1,869.3	1,831.4	1,801.9	1,765.7	1,738.5
EPS (Diluted, Yen)	125.08	127.65	185.56	271.93	332.78
BPS (Yen)	1,380.1	1,484.0	1,540.9	1,689.0	1,880.3
DPS (Yen)	57.0	65.0	70.0	75.0	82.0
USD/JPY: Average Exchange Rate	152.6	150.8	155.0	155.0	155.0
YoY % Change					
Net Sales	6%	5%	12%	20%	16%
Operating Profit	30%	1%	53%	44%	21%
Net Profit	29%	0%	43%	44%	20%
As % of Sales					
Gross Profit	41.2%	42.3%	44.7%	48.2%	45.5%
Operating Profit	16.0%	15.4%	21.1%	25.5%	26.6%
Net Profit	13.4%	12.8%	16.4%	19.6%	20.4%
CapeX	10.4%	13.5%	12.2%	10.2%	8.8%
Depreciation and Amortization	9.9%	9.7%	8.7%	7.8%	7.1%
Balance Sheet					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Total Assets	3,028.2	3,199.1	3,262.6	3,482.3	3,786.1
Total Current Assets	1,497.9	1,551.4	1,572.9	1,732.6	1,986.3
Cash and Cash Equivalents	625.1	653.7	586.1	601.5	702.7
Marketable Securities	0.0	0.0	0.0	0.0	0.0
Account Receivables	294.4	328.2	366.4	439.1	507.3
Inventory	482.8	520.5	541.3	612.9	697.2
Other Current Assets	95.5	79.1	79.1	79.1	79.1
Total Long Term Assets	1,530.3	1,617.7	1,689.7	1,749.7	1,799.7
Net PP&E	1,183.7	1,300.9	1,372.9	1,432.9	1,482.9
Investments in Securities	40.3	50.4	50.4	50.4	50.4
Other assets	306.3	266.4	266.4	266.4	266.4
Total Current Liabilities	276.5	316.5	322.1	336.2	353.3
ST Debt & Current Portion of LT Debt	0.8	1.7	1.7	1.7	1.7
Accounts Payable	66.6	79.8	85.4	99.5	116.6
Other Current Liabilities	208.1	234.9	234.9	234.9	234.9
Total Long Term Liabilities	171.7	164.8	164.8	164.8	164.8
Long Term Debt	1.7	1.5	1.5	1.5	1.5
Other Long Term Liabilities	170.0	163.3	163.3	163.3	163.3
Minority Interests	-0.8	-0.9	-0.9	-0.9	-0.9
Total Equity	2,580.0	2,717.8	2,776.7	2,982.3	3,268.9
Net Debt	-663.0	-700.8	-633.2	-648.7	-749.9

Cash Flow Statement					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Cash from Operations	451.9	425.2	458.9	540.0	643.1
Net Income	233.8	233.8	334.4	480.2	578.6
Depreciation and Amortization	173.3	178.2	178.0	190.0	200.0
Changes in Working Capital:	24.4	-18.0	-53.5	-130.2	-135.4
Other-net	20.3	31.3	0.0	0.0	0.0
Net Cash from Investing Activities	-208.1	-193.8	-250.0	-250.0	-250.0
Purchase of PP&E	-179.4	-243.0	-250.0	-250.0	-250.0
Sale/Purchase of Investments	3.4	2.2	0.0	0.0	0.0
Other-net	-32.1	46.9	0.0	0.0	0.0
Free cash flow	243.8	231.4	208.9	290.0	393.1
Net Cash from Financing Activities	-242.7	-221.8	-276.5	-274.6	-291.9
Change in Interest-bearing Debt	-50.4	0.8	0.0	0.0	0.0
Change in Treasury Stock	-80.0	-100.0	-150.0	-150.0	-150.0
Payment of Cash Dividends	-101.6	-110.7	-126.5	-124.6	-141.9
Other-net	-10.7	-11.9	0.0	0.0	0.0
Net Effect of Exchange Rate Changes	2.0	19.0	0.0	0.0	0.0
Net Change in Cash	3.1	28.6	-67.6	15.4	101.2
Segment Information					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales	1,793.3	1,830.9	2,044.2	2,449.7	2,830.5
Components Total	1,033.1	1,159.7	1,376.3	1,660.1	1,882.6
Capacitors	831.8	936.4	1,143.9	1,418.4	1,631.2
Inductors and EMI Filters	201.3	223.3	232.4	241.7	251.4
Devices/Modules	697.2	656.0	652.9	776.6	934.9
High Frequency Devices/Communications Module	443.6	394.8	377.9	453.5	544.2
Battery/Power supply	155.7	154.1	165.0	206.5	267.2
Functional Devices	97.8	107.1	110.0	116.6	123.6
Others	13.1	15.2	15.0	13.0	13.0
Total Operating Profit	279.7	281.8	431.9	624.0	754.3
Components Total	275.2	315.1	426.4	579.2	685.2
Devices/Modules	10.0	-26.5	11.5	50.7	75.0
Others	-5.4	-6.8	-6.0	-6.0	-6.0
Operating Profit Margin	15.6%	15.4%	21.1%	25.5%	26.6%
Components Total	26.6%	27.2%	31.0%	34.9%	36.4%
Devices/Modules	2.3%	-6.7%	3.0%	11.2%	13.8%
Sales by Application					
(Y bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Net Sales (excl. Non-product Sales)	1,793.3	1,830.9	2,044.2	2,449.7	2,830.5
AV	674.2	653.0	-	-	-
Communication	453.1	474.5	-	-	-
Computer and Peripheral	281.9	310.4	-	-	-
Automobile Electronics	150.4	142.7	-	-	-
Home Appliances and Others	183.7	250.3	-	-	-
As % of Total					
AV	38.7%	36%	-	-	-
Communication	26.0%	26%	-	-	-
Computer and Peripheral	16.2%	17%	-	-	-
Automobile Electronics	8.6%	8%	-	-	-
Home Appliances and Others	10.5%	14%	-	-	-

Risk Reward – Murata Manufacturing (6981.T) Top Pick

Expect AI/DC use value added MLCC sales and earnings to continue to expand

PRICE TARGET ¥12,500

Derived from our base case and a DCF model, with the following assumptions: 2.6% risk-free rate, 1.09 equity beta, 3.2% risk premium, yielding 6.1% WACC; zero growth from F3/36.



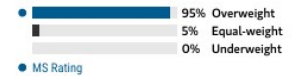
RISK REWARD CHART



OVERWEIGHT THESIS

- Expect AI/DC use value added MLCC sales and earnings expansion to continue.
- Expect a significant increase in the total capacitance required for MLCCs in AI/data center applications, alongside continued miniaturization and further increases in capacitance per unit.
- We expect Murata to remain competitive in MLCCs and other products over the long term.
- In F3/27 onward, we expect module earnings to improve as sales of RF modules rise.

Consensus Rating Distribution



Risk Reward Themes

New Data Era: Positive

Secular Growth: Positive

View descriptions of Risk Rewards Themes [here](#)

BULL CASE

¥15,500

Derived from DCF

Demand for AI/DC use MLCC is stronger than in the base case. OP assumptions: ¥494.9bn in F3/27, ¥707.0bn in F3/28, ¥1,205.7bn in F3/31. Same FX & DCF model assumptions as the base case. Bull case fair value equates to P/E of 73.1x on F3/27 diluted EPS of ¥212.0, 50.4x on F3/28 diluted EPS of ¥307.4, and 28.4x on F3/31 diluted EPS of ¥546.0.

BASE CASE

¥12,500

Derived from DCF

Earnings continue to grow and beat consensus over the medium/long term due to technological advantages & competitiveness, especially in MLCCs. Assumptions: OP ¥431.9bn in F3/27, ¥624.0bn in F3/28, ¥1,068.9bn in F3/31; FX ¥155/US\$, DCF model (see Valuation Methodology); zero growth from F3/36. Base case fair value equates to P/E 67.4x on F3/27 diluted EPS of ¥185.6, 46.0x on F3/28 diluted EPS of ¥271.9 and 25.8x on F3/31 diluted EPS of ¥484.6.

BEAR CASE

¥3,500

F3/27e BPS ¥1,490 x P/B 2.3

Murata remains competitive, but final demand for its components is lower than we envision. OP assumptions: ¥310.1bn in F3/27, ¥324.3bn in F3/28, ¥379.6bn in F3/31. Same FX assumptions as the base case. Assumes ROE moves from 8.8% in F3/26 to 9.0% in F3/27 and 9.2% in F3/28, higher than our capital cost estimate of 6.1%, but with earnings missing market expectations, P/B drops to 2.3x.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nichicon: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	175.6	168.7	167.0	197.2	207.1	Cash flow from operating activities	18.3	8.2	8.4	14.7	17.8
Costs of Sales	147.4	139.7	151.4	156.9	162.6	EBIT	8.5	7.2	9.7	12.9	15.6
Gross Profit	28.3	30.0	35.6	40.3	44.5	Depreciation	8.5	8.1	8.5	9.0	9.9
SG&A	23.1	23.5	26.2	27.9	29.4	Changes in Receivables	3.2	-1.2	-4.8	-2.8	-2.8
Operating Income	5.2	6.5	9.4	12.4	15.1	Change in inventory	3.7	1.8	-2.8	-0.8	-0.7
Non-operating income/expenses	2.3	1.9	0.3	0.5	0.5	Changes in Payables	-2.2	-6.0	2.4	1.4	1.4
Recurring Profit	7.5	8.3	9.7	12.9	15.6	Other	-3.3	-1.7	-4.5	-4.9	-5.6
Extraordinary income	1.0	-1.1	0.0	0.0	0.0	Cash flow from investing activities	-8.4	-6.6	-8.6	-10.4	-11.0
Profit before income taxes	8.5	7.2	9.7	12.9	15.6	Purchase of PP&E	-10.7	-7.1	-10.0	-11.8	-12.4
Income taxes	2.1	0.4	2.5	3.3	4.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	25%	5%	26%	26%	26%	Other	2.3	0.5	1.4	1.4	1.4
Profit attributable to non-controlling interests	0.5	0.5	0.5	0.5	0.5	Cash flow from financing activities	-14.3	-3.8	-1.6	-2.8	-3.8
Net Profit	5.9	6.3	6.7	9.1	11.1	Net Change in Borrowings	2.2	-0.8	0.9	-0.1	-1.0
Capex	11.1	7.6	10.0	11.8	12.4	Purchase of treasury shares	-1.6	0.0	0.0	0.0	0.0
Depreciation & amortization	8.5	8.1	8.5	9.0	9.9	Cash dividend paid	-2.5	-2.4	-2.6	-2.7	-2.8
R&D	0.0	0.0	0.0	0.0	0.0	Other	-12.5	-0.6	0.0	0.0	0.0
EBITDA	13.7	14.5	17.9	21.4	25.0	Effect of exchange rate changes	0.5	1.2	2.1	0.0	0.0
ROE	5.3%	5.5%	5.5%	7.1%	8.1%	Changes in cash and cash equivalents	-3.9	-1.1	0.2	1.5	3.0
						Cash and cash equivalents at end of the period	25.5	24.4	24.7	26.2	29.2
No. of average shares (mn)	68.3	67.1	67.2	67.2	67.2	FCF	10.0	1.6	-0.2	4.3	6.8
EPS (JPY)	86.0	94.0	99.8	135.5	165.3						
BPS (JPY)	1,644.3	1,780.9	1,873.2	1,968.7	2,092.0	Sales by Product					
DPS (JPY)	36.0	38.0	39.0	41.0	43.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
						Capacitor	99.2	102.7	117.3	124.6	131.5
USD/JPY (Average)	151.6	150.0	155.0	155.0	155.0	Aluminum capacitor	79.3	82.0	93.7	99.8	105.4
						Film capacitor	19.8	20.7	23.6	24.8	26.1
Balance Sheet						NECST	76.6	67.0	69.7	72.6	75.6
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	Applied system and equipment	7.3	7.2	8.6	9.0	9.5
Total assets	192.6	195.6	205.2	213.0	221.7	EV	5.6	3.5	4.0	4.3	4.5
Current assets	108.3	107.7	115.3	120.5	126.9	Power source	26.3	21.7	18.6	18.7	18.7
Cash and cash equivalents	25.5	24.5	24.7	26.2	29.2	Storage	35.6	32.4	36.3	38.5	40.8
Accounts receivable	44.8	47.4	52.2	55.1	57.9	Others	1.9	2.2	2.2	2.2	2.2
Inventories	33.2	32.3	35.1	35.9	36.6	(YoY)					
Other financial assets	4.7	3.5	3.3	3.3	3.3	Capacitor	-7%	4%	14%	6%	6%
Fixed assets	84.3	87.9	89.9	92.5	94.8	Aluminum capacitor	-9%	3%	14%	6%	6%
Tangible fixed assets	53.1	52.5	54.7	57.5	60.0	Film capacitor	-19%	6%	24%	18%	15%
Intangible fixed assets	1.9	2.1	1.9	1.7	1.6	NECST	2%	-13%	4%	4%	4%
Other non-current assets	29.3	33.2	33.2	33.2	33.2	Applied system and equipment	6%	-1%	20%	5%	5%
Current liabilities	51.6	47.2	50.1	51.4	52.5	EV	-31%	-37%	15%	5%	5%
Short term borrowings	9.2	10.9	11.3	11.1	10.8	Power source	11%	-17%	-14%	0%	0%
Accounts payable	28.5	23.3	25.7	27.1	28.4	Storage	4%	-9%	12%	6%	6%
Other current liabilities	13.9	13.0	13.1	13.2	13.3	Others	-20%	15%	0%	0%	0%
Fixed liabilities	27.2	24.8	25.4	25.5	24.8	OP by Product					
Long term borrowings	15.8	13.3	13.8	13.9	13.2	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Other fixed liabilities	11.5	11.6	11.6	11.6	11.6	Capacitor	1.6	4.6	7.2	9.6	12.0
Net assets	113.8	123.5	129.7	136.1	144.4	Aluminum capacitor	0.5	3.1	4.2	6.3	8.5
						Film capacitor	1.0	1.5	2.9	3.3	3.5
Total liabilities and equity	192.6	195.6	205.2	213.0	221.7	NECST	3.7	1.9	2.2	2.8	3.1
						Applied system and equipment	1.1	0.8	1.1	1.2	1.3
						EV	-0.3	-0.9	-0.7	-0.7	-0.7
						Power source	0.5	0.1	0.3	0.3	0.4
						Storage	2.2	1.5	1.6	1.9	2.1
						Others/Adjustment	0.2	0.2	0.0	0.0	0.0
						(OPM)					
						Capacitor	1.6%	4.5%	6.1%	7.7%	9.1%
						Aluminum capacitor	0.7%	3.8%	4.5%	6.3%	8.0%
						Film capacitor	5.1%	7.2%	12.4%	13.2%	13.5%
						NECST	4.8%	2.8%	3.2%	3.9%	4.1%
						Applied system and equipment	14.4%	11.3%	12.6%	13.6%	14.0%
						EV	-5.4%	-21.3%	-17.6%	-16.3%	-16.0%
						Power source	1.9%	0.3%	1.4%	1.8%	2.1%
						Storage	6.2%	4.7%	4.4%	5.0%	5.2%

Risk Reward – Nichicon (6996.T)

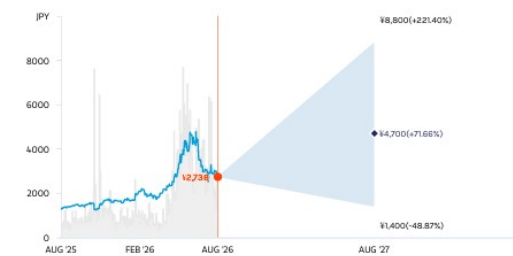
Strong growth led by AI server capacitors and household batteries

PRICE TARGET ¥4,700

Our price target is ¥4,700, based on a DCF valuation (risk-free rate 2.6%; beta 1.0; risk premium 3.2%; WACC 4.7%). Beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and our assumptions for F3/32 onward is that the firm maintains FCF/share of ¥275. Our price target equates to 47.1x our F3/27e EPS of ¥99.8, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.



RISK REWARD CHART



BULL CASE

Continuing strong growth for AEC & NECST sales

OP and EPS forecasts of ¥9.7bn/¥102.7 in F3/27, ¥16.4bn/¥180.2 in F3/28, ¥22.7bn/¥250.2 in F3/29, ¥29.8bn/¥327.6 in F3/30, and ¥39.1bn/¥430.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥530. Our fair value here equates to 85.7x estimated F3/27 EPS of ¥102.7 and 4.7x BPS of ¥1,875.4.

BASE CASE

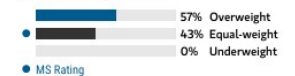
Value-added products drive earnings expansion

OP and EPS forecasts of ¥9.4bn/¥99.8 in F3/27, ¥12.4bn/¥135.5 in F3/28, ¥15.1bn/¥165.3 in F3/29, ¥18.1bn/¥198.0 in F3/30, and ¥22.2bn/¥244.2 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥275. Our fair value equates to 47.1x our F3/27e EPS of ¥99.8 and 2.5x BPS of ¥1,872.4.

EQUAL-WEIGHT THESIS

The equity story is centered on increased sales of high value-added, high-capacity capacitors and the stable growth of the NECST business. From F3/27, we expect the capacitor business to expand, driven primarily by large-size products for AI servers and film capacitors for xEV applications, while competition in general-purpose products is likely to remain intense amid rivalry with Chinese players. We expect NECST business to deliver stable growth, supported by the wider adoption of residential energy storage systems.

Consensus Rating Distribution



Risk Reward Themes

Electric Vehicles: Positive
New Data Era: Positive

View descriptions of Risk Rewards Themes [here](#)

BEAR CASE

Slowing sales growth, rising costs take a toll

OP and EPS forecasts of ¥3.7bn/¥37.2 in F3/27, ¥5.0bn/¥53.6 in F3/28, ¥5.5bn/¥59.6 in F3/29, ¥5.4bn/¥59.6 in F3/30, and ¥6.1bn/¥67.0 in F3/31. FCF/share is maintained at ¥75 from F3/32. Our fair value here equates to 37.6x estimated F3/27 EPS of ¥37.2 and 0.8x BPS of ¥1,809.9.

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

Nippon Chemi-con: Key Financial Data and Risk Reward

Income Statement						Cashflow Statement					
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	122.7	138.8	154.0	158.3	165.0	Cash flow from operating activities	-0.5	7.6	4.8	11.0	11.5
Costs of Sales	99.0	112.4	124.5	126.2	130.0	EBIT	0.6	3.5	5.0	7.4	9.5
Gross Profit	23.7	24.4	29.5	32.1	35.0	Depreciation	7.1	7.3	7.0	6.8	6.8
SG&A	20.0	21.1	22.5	23.0	23.8	Changes in Receivables	2.7	-4.3	-3.7	-0.9	-1.4
Operating Income	3.7	3.4	7.0	9.1	11.2	Change in inventory	-3.8	3.1	-3.5	-0.5	-1.0
Non-operating income/expenses	-2.2	-1.3	-2.0	-1.7	-1.7	Changes in Payables	-1.0	-0.5	1.1	0.3	0.4
Recurring Profit	1.6	2.1	5.0	7.4	9.5	Other	-6.1	-1.5	-1.1	-2.0	-2.8
Extraordinary income	-1.0	1.4	0.0	0.0	0.0	Cash flow from investing activities	-9.8	-6.3	-5.7	-6.3	-6.6
Profit before income taxes	0.6	3.5	5.0	7.4	9.5	Purchase of PP&E	-9.3	-4.4	-5.5	-6.1	-6.3
Income taxes	0.5	1.1	1.7	2.5	3.2	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	88%	31%	34%	34%	34%	Other	-0.4	-0.9	-0.2	-0.2	-0.3
Profit attributable to non-controlling interests	0.0	0.1	0.0	0.0	0.0	Cash flow from financing activities	-11.9	-6.2	-0.3	-1.7	-1.8
Net Profit	0.0	2.4	3.3	4.9	6.3	Net Change In Borrowings	-10.6	-4.6	2.8	-0.6	-0.7
Capex	7.6	5.9	5.7	6.3	6.6	Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Depreciation & amortization	7.1	7.3	7.0	6.8	6.8	Cash dividend paid	-0.2	-0.6	-1.1	-1.1	-1.1
R&D	0.0	0.0	0.0	0.0	0.0	Other	-1.2	-1.1	-2.0	0.0	0.0
EBITDA	10.8	10.7	14.0	15.9	18.0	Effect of exchange rate changes	0.7	1.3	3.5	0.0	0.0
ROE	0.1%	4.0%	5.1%	7.2%	8.7%	Changes in cash and cash equivalents	-21.4	-2.6	2.3	3.0	3.1
No of average shares (mn)	39.9	34.4	39.5	39.5	39.5	Cash and cash equivalents at end of the period	23.9	21.3	23.6	26.6	29.7
EPS (JPY)	1.0	68.8	83.5	124.0	159.4	FCF	-10.2	2.3	-0.9	4.7	4.9
BPS (JPY)	2,631.2	2,594.1	2,415.6	2,554.8	2,745.2	Sales by Product					
DPS (JPY)	0.0	20.0	20.0	20.0	20.0	(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e
USD/JPY (Average)	152.6	150.8	155.0	155.0	155.0	Capacitor	118.0	131.8	149.9	154.2	160.9
Balance Sheet						Aluminum capacitor	106.1	120.0	138.3	142.6	149.3
(JPY bn)	F3/25	F3/26	F3/27e	F3/28e	F3/29e	DLCAP	3.8	3.6	3.2	3.2	3.2
Total assets	162.7	166.0	173.7	177.2	182.1	Other capacitor	2.5	2.6	2.4	2.4	2.4
Current assets	86.6	87.6	97.0	101.5	107.1	Capacitor material	5.7	5.6	5.9	5.9	5.9
Cash and cash equivalents	23.9	21.3	23.6	26.6	29.7	Mechanical part	3.4	3.2	2.6	2.6	2.6
Accounts receivable	28.9	29.2	32.9	33.8	35.2	Other	1.2	1.8	1.4	1.4	1.4
Inventories	33.1	31.2	34.7	35.3	36.3	(YoY)					
Other financial assets	0.8	5.8	5.8	5.8	5.8	Capacitor	-19%	12%	14%	3%	4%
Fixed assets	76.1	78.4	76.6	75.7	75.1	Aluminum capacitor	-20%	13%	15%	3%	5%
Tangible fixed assets	48.3	47.5	46.2	45.7	45.5	DLCAP	-28%	-4%	-11%	0%	0%
Intangible fixed assets	2.3	2.6	2.1	1.6	1.2	Other capacitor	-26%	4%	-5%	0%	0%
Other non-current assets	25.4	28.3	28.3	28.3	28.3	Capacitor material	24%	-1%	5%	0%	0%
Current liabilities	60.6	58.5	61.4	61.2	61.7	Mechanical part	-2%	-8%	-16%	0%	0%
Short term borrowings	13.1	8.7	9.8	10.1	10.5	Other	-21%	81%	-22%	0%	0%
Accounts payable	36.7	34.6	36.4	36.0	36.0						
Other current liabilities	10.8	15.1	15.2	15.2	15.2						
Fixed liabilities	45.4	44.4	45.5	45.3	44.6						
Long term borrowings	38.1	35.9	37.0	36.8	36.1						
Other fixed liabilities	7.3	8.5	8.5	8.5	8.5						
Net assets	56.7	63.1	66.8	70.6	75.8						
Total liabilities and equity	162.7	166.0	173.7	177.2	182.1						

Risk Reward – Nippon Chemi-Con (6997.T)

Near/mid-term targets present high hurdles, risk of dilution too

PRICE TARGET ¥3,500

Our price target is ¥3,500, based on a DCF valuation (risk-free rate 2.6%; beta 1.1; risk premium 3.2%; WACC 4.2%). Beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and we assume for F3/32 onward that the firm maintains FCF/share of ¥245. Our price target equates to 41.9x our F3/27e EPS of ¥83.5, and our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



Source: Refinitiv, Morgan Stanley Research

BULL CASE	¥7,100	BASE CASE	¥3,500	BEAR CASE	¥1,500
Strong demand growth for AI/DC aluminum capacitors					
OP and EPS forecasts of ¥7.9bn/¥113.9 in F3/27, ¥11.1bn/¥169.6 in F3/28, ¥14.3bn/¥222.7 in F3/29, ¥18.2bn/¥291.0 in F3/30, and ¥23.4bn/¥379.6 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥450. Our fair value here equates to 62.3x estimated F3/27 EPS of ¥113.9 and 2.9x BPS of ¥2,459.4.					
Recovery continues, but cost & dilution concerns					
OP and EPS forecasts of ¥7.0bn/¥83.5 in F3/27, ¥9.1bn/¥124.0 in F3/28, ¥11.2bn/¥159.4 in F3/29, ¥13.0bn/¥189.8 in F3/30, and ¥15.2bn/¥227.8 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥245. Our PT equates to 41.9x our F3/27e EPS of ¥83.5 and 14x BPS of ¥2,415.6.					
Earnings remain subdued at the trough					
OP and EPS forecasts of ¥6.3bn/¥88.6 in F3/27, ¥6.9bn/¥98.7 in F3/28, ¥7.3bn/¥106.3 in F3/29, ¥7.5bn/¥111.4 in F3/30, and ¥7.6bn/¥116.4 in F3/31. Our calculations also assume that from F3/32 onward the firm maintains FCF/share at ¥120. Our fair value here equates to 16.9x estimated F3/27 EPS of ¥88.6 and 0.6x BPS of ¥2,422.9.					

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

KOA: Key Financial Data and Risk Reward

Income Statement (JPY bn)						Cashflow Statement (JPY bn)					
	F3/25	F3/26	F3/27e	F3/28e	F3/29e		F3/25	F3/26	F3/27e	F3/28e	F3/29e
Sales	64.1	72.3	78.9	85.6	88.6	Cash flow from operating activities	8.1	9.1	10.0	9.4	10.7
Costs of Sales	46.0	50.4	55.8	57.7	59.9	EBIT	1.3	5.3	6.7	4.6	6.5
Gross Profit	18.2	21.9	23.1	25.9	28.7	Depreciation	5.9	7.1	7.2	7.3	7.4
SG&A	17.0	18.2	19.9	20.7	21.7	Changes in Receivables	0.3	-1.6	-1.5	-1.0	-1.1
Operating Income	1.2	3.6	3.2	5.2	7.0	Change in inventory	0.7	0.5	-1.2	-0.4	-0.4
Non-operating income/expenses	0.1	1.6	0.5	-0.6	-0.5	Changes in Payables	-0.5	-0.9	0.3	0.2	0.2
Recurring Profit	1.2	5.2	3.7	4.6	6.5	Other	0.5	-1.3	-1.5	-1.3	-1.9
Extraordinary income	0.0	0.0	3.0	0.0	0.0	Cash flow from investing activities	23.9	-7.1	-4.1	-5.9	-6.2
Profit before income taxes	1.3	5.3	6.7	4.6	6.5	Purchase of PP&E	-25.1	-7.5	-4.0	-5.8	-6.1
Income taxes	1.0	1.3	1.7	1.4	2.0	Sales of PP&E	0.0	0.0	0.0	0.0	0.0
(Effective tax rate)	79%	25%	25%	30%	31%	Other	1.1	0.4	-0.1	-0.1	-0.1
Profit attributable to non-controlling interests	0.0	0.0	0.0	0.0	0.0	Cash flow from financing activities	11.3	-0.5	-6.1	-2.7	-3.2
Net Profit	0.3	4.0	5.0	3.2	4.5	Net Change in Borrowings	13.2	0.7	-3.7	-1.4	-1.9
						Purchase of treasury shares	0.0	0.0	0.0	0.0	0.0
Capex	24.2	5.9	4.1	5.9	6.2	Cash dividend paid	-1.8	-1.1	-1.4	-1.3	-1.3
Depreciation & amortization	5.9	7.1	7.2	7.3	7.4	Other	-0.1	-0.1	0.0	0.0	0.0
R&D	3.6	3.6	3.5	3.7	3.9	Effect of exchange rate changes	0.2	1.1	-6.9	0.0	0.0
EBITDA	7.0	10.7	10.4	12.5	14.4	Changes in cash and cash equivalents	-4.4	2.6	-6.2	0.8	1.3
ROE	0.3%	4.7%	5.8%	3.7%	5.1%	Cash and cash equivalents at end of the period	26.6	29.0	22.8	23.6	24.9
No. of average shares (mn)	37.1	37.1	37.1	37.1	37.1	FCF	-15.8	2.0	5.9	3.5	4.5
EPS (JPY)	7.0	106.4	134.6	86.2	121.2	Sales by Customer Region (JPY bn)					
BPS (JPY)	2,104.4	2,385.1	2,296.4	2,348.1	2,434.2		F3/25	F3/26	F3/27e	F3/28e	F3/29e
DPS (JPY)	40.0	32.0	39.0	30.0	40.0	Japan	16.5	18.3	19.0	19.7	20.5
USD/JPY (Average)	152.6	151.1	155.0	155.0	155.0	North America	11.0	11.5	12.1	12.5	12.9
						Asia	24.5	28.7	32.5	35.6	39.1
						Europe	12.1	13.8	15.3	15.7	16.1
						(YoY)					
						Japan	-9%	11%	4%	4%	4%
						North America	-3%	5%	5%	3%	3%
						Asia	5%	17%	13%	10%	10%
						Europe	2%	14%	11%	2%	2%
						Sales by Application (JPY bn)					
							F3/25	F3/26	F3/27e	F3/28e	F3/29e
						Automotive	34.7	37.4	-	-	-
						Industrial	7.1	8.5	-	-	-
						Home appliance	3.1	3.2	-	-	-
						Communication	4.5	5.6	-	-	-
						Power supply	3.2	4.0	-	-	-
						Others	11.7	13.5	-	-	-
						(YoY)					
						Automotive	4%	8%	-	-	-
						Industrial	-21%	20%	-	-	-
						Home appliance	-1%	5%	-	-	-
						Communication	12%	26%	-	-	-
						Power supply	10%	26%	-	-	-
						Others	-9%	16%	-	-	-
Balance Sheet (JPY bn)											
	F3/25	F3/26	F3/27e	F3/28e	F3/29e						
Total assets	141.4	151.6	144.8	145.5	147.0						
Current assets	57.5	62.7	59.2	61.5	64.3						
Cash and cash equivalents	26.6	29.0	22.8	23.6	24.9						
Accounts receivable	13.8	16.0	17.4	18.5	19.6						
Inventories	13.4	13.5	14.7	15.1	15.5						
Other financial assets	3.6	4.3	4.3	4.3	4.3						
Fixed assets	83.9	88.8	85.6	84.0	82.7						
Tangible fixed assets	71.5	74.5	71.3	69.8	68.5						
Intangible fixed assets	3.2	3.0	2.9	2.9	2.9						
Other non-current assets	9.2	11.3	11.3	11.3	11.3						
Current liabilities	15.9	24.8	22.4	21.4	20.4						
Short term borrowings	3.3	9.9	7.1	6.0	4.7						
Accounts payable	5.0	3.2	3.5	3.7	4.0						
Other current liabilities	7.6	11.7	11.7	11.7	11.7						
Fixed liabilities	47.3	38.1	37.1	36.9	36.3						
Long term borrowings	41.2	35.5	34.5	34.3	33.7						
Other fixed liabilities	6.1	2.6	2.6	2.6	2.6						
Net assets	78.1	88.6	85.3	87.2	90.4						
Total liabilities and equity	141.4	151.6	144.8	145.5	147.0						

Risk Reward – KOA (6999.T)

Earnings recovery continues centered on AI/DCs but is milder than MTP targets

PRICE TARGET ¥3,000

Our price target is ¥3,000, based on a DCF valuation (risk-free rate 2.6%; beta 1.2; risk premium 3.2%; WACC 5.0%). Beyond our 3-year base earnings models, we also build forecasts for F3/30–31 into our DCF-based valuation, and we assume for F3/32 onward that the firm maintains FCF/share of ¥190. Our price target equates to 22.3x our F3/27e EPS of ¥134.6. Our forex rate assumption for the next quarter and thereafter is ¥155/\$.

RISK REWARD CHART



EQUAL-WEIGHT THESIS

The equity story revolves around ongoing negative effects of fixed costs associated with adding to production capacity. We think this is being undertaken amid recognition of the emergence of competing Taiwanese makers in the automotive resistor market. We also think that effects of rising fixed costs alongside investment to boost production will make attaining of earnings targets in the current mid-term plan difficult. In the short term, we also expect the firm to be hit by high prices for precious metals. On the other hand, we expect earnings to head into a mild recovery from F3/28 as demand for value-added products such as shunt resistant devices for AI/data centers is growing.

Risk Reward Themes

New Data Era: Positive

Pricing Power: Negative

View descriptions of Risk Rewards Themes [here](#)

	¥4,800	BASE CASE	¥3,000	BEAR CASE	¥1,300
BULL CASE					
Achieves medium-term earnings targets		Mild growth led by AI/DC resistors continues		Earnings remain sluggish as sales growth slows	
OP and EPS forecasts of ¥5.2bn/¥175.0 in F3/27, ¥8.4bn/¥145.4 in F3/28, ¥10.7bn/¥188.5 in F3/29, ¥12.4bn/¥220.8 in F3/30, and ¥15.0bn/¥269.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥320. Our fair value here equates to 27.4x estimated F3/27 EPS of ¥175.0 and 2.1x BPS of ¥2,336.8.		OP and EPS forecasts of ¥3.2bn/¥134.6 in F3/27, ¥5.2bn/¥86.2 in F3/28, ¥7.0bn/¥121.2 in F3/29, ¥8.0bn/¥140.0 in F3/30, and ¥9.3bn/¥164.3 in F3/31. Our calculations also assume that from F3/32 onward the firm can maintain FCF/share at ¥190. Our PT equates to 22.3x our F3/27e EPS of ¥134.6 and 1.3x BPS of ¥2,296.4.		OP and EPS forecasts of ¥1.7bn/¥105.0 in F3/27, ¥3.0bn/¥45.8 in F3/28, ¥3.7bn/¥59.2 in F3/29, ¥4.1bn/¥70.0 in F3/30, and ¥4.1bn/¥70.0 in F3/31. Our calculations also assume that from F3/32 onward the firm maintains FCF/share at ¥70. Our fair value here equates to 12.4x estimated F3/27 EPS of ¥105.0 and 0.6x BPS of ¥2,266.8.	

Note: e = Morgan Stanley Research estimates; Stock price as of the Aug 20 close for Risk Reward Chart
Source: Company data, Morgan Stanley Research

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(as of July 31, 2026)

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Stock Rating Category	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

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COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/21/2026)
Shoji Sato		
ALPS ALPINE (6770.T)	O (03/17/2026)	¥2,194
Hamamatsu Photonics (6965.T)	E (06/17/2026)	¥2,367
Ibiden (4062.T)	O (08/04/2026)	¥19,190
Kyocera (6971.T)	E (06/25/2020)	¥3,608
Mabuchi Motor (6592.T)	E (11/03/2022)	¥1,771
Minebea Mitsumi (6479.T)	E (10/23/2025)	¥3,365
Murata Manufacturing (6981.T)	O (11/26/2025)	¥7,091
Nidec (6594.T)	NR (09/05/2025)	¥2,627
Niterra (5334.T)	O (01/17/2024)	¥8,803
Taiyo Yuden (6976.T)	U (06/17/2026)	¥9,300

TDK (6762.T)	O (08/02/2022)	¥3,041
Sota Harashima		
CMK (6958.T)	E (02/28/2025)	¥722
Daishinku (6962.T)	U (06/17/2026)	¥785
Hirose Electric (6806.T)	O (07/10/2024)	¥25,665
IRISO Electronics (6908.T)	E (08/02/2022)	¥2,737
Japan Aviation Electronics Industry (6807.T)	E (01/17/2024)	¥2,363
KOA (6999.T)	E (06/17/2026)	¥2,191
Meiko Electronics (6787.T)	E (04/03/2026)	¥17,160
Nichicon (6996.T)	E (11/10/2021)	¥2,697
Nihon Dempa Kogyo (6779.T)	E (03/07/2024)	¥2,732
Nippon Chemi-Con (6997.T)	U (09/20/2024)	¥2,747

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